Main Routine

EDC2 - v2.00 added PdM with QualityRx, TrendRx and Server Interface
EDC2 - v2.01 added Event DtOwner and DtType Mask Configuration; removed Shared Andon/AVI Events
EDC3 - v3.00 added Stuck Download Detection for bad configuration, expanded Server Status block
v3.50 - L980 Controller; Gateway Configuration Download; TransNum Handshake

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

1

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	801
Dest	Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806549717
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806549717

2

Reset

Reset Event Configuration
(use before Download All to defrag the Event table for faster searches)

Gateway Reset CFGResetEventText

Toggle to Reset Configuration ManualReset

Gateway Reset CFGResetEventText (U)

Toggle to Reset Configuration ManualReset (U)

AutoConfig (L)

JSR	
Routine Name	_810_Reset

Main Routine

Configuration

3

AutoConfig

Toggle to Load User Configuration ManualConfig

s:FS

AutoConfig U

Toggle to Load User Configuration ManualConfig U

Configuration

JSR

Routine Name _811_Config

Reset for Configuration and Status Data Errors

4

Manual Reset for Configuration Status Errors ManualResetER

Manual Reset for Configuration Status Errors ManualResetER U

Manual Reset of Gateway Status Errors EDC_StatusErrorReset

Manual Reset of Gateway Status Errors EDC_StatusErrorReset U

Server Interface Status Reset Errors EDC_Status01.ServerCfg.Update.ResetER L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...

Status EDC_Status01.ServerCfg

Server Interface Status Reset Errors EDC_Status01.ServerPdm.Update.ResetER L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...

Status EDC_Status01.ServerPdm

Server Interface Status Reset Errors EDC_Status01.ServerStat.Update.ResetER L

Update Status State for Transactions

aoiEDC_StatusUpdateServerStatus

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...

Status EDC_Status01.ServerStat

Server Interface Status Reset Errors EDC_Status01.ServerHealth.Update.ResetER L

Update Health State for Transactions

aoiEDC_StatusUpdateServerHealth

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerHealth ...

Status EDC_Status01.ServerHealth

Server Interface Status Reset Errors EDC_Status01.ServerMask.Update.ResetER L

Update Mask State for Transactions

aoiEDC_StatusUpdateServerMask

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...

Status EDC_Status01.ServerMask

Main Routine

Line Configuration Request

Configuration Request to Server via Transaction Manager

5 JSR Routine Name _820_CfgRequest

PdM Configuration Request

Configuration Request to Server via Transaction Manager

6 JSR Routine Name _845_PdmRequest

Status Request
DN005 Health Status Alarm Callout

Status Request to Server via Transaction Manager

Detect and Callout CIS Message Errors

Detect and Callout Health Status Errors

7 JSR Routine Name _860_StatRequest | JSR Routine Name _870_StatCisErrCallout | JSR Routine Name _871_StatHealthErrCallout

EQ Health Request
DN014 Health Displays

EQ Health Request to Server via Transaction Manager

8 JSR Routine Name _880_HealthRequest

Event Mask Configuration Request

Configuration Request to Server via Transaction Manager

9 JSR Routine Name _890_MaskRequest

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

10

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	802
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	4
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	20
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.LOGi
	30
Dest	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	9003
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

11

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

12

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
59
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.8
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	-20378558

Local Program Scan Time & Fault Routine Program Fault Occurrences

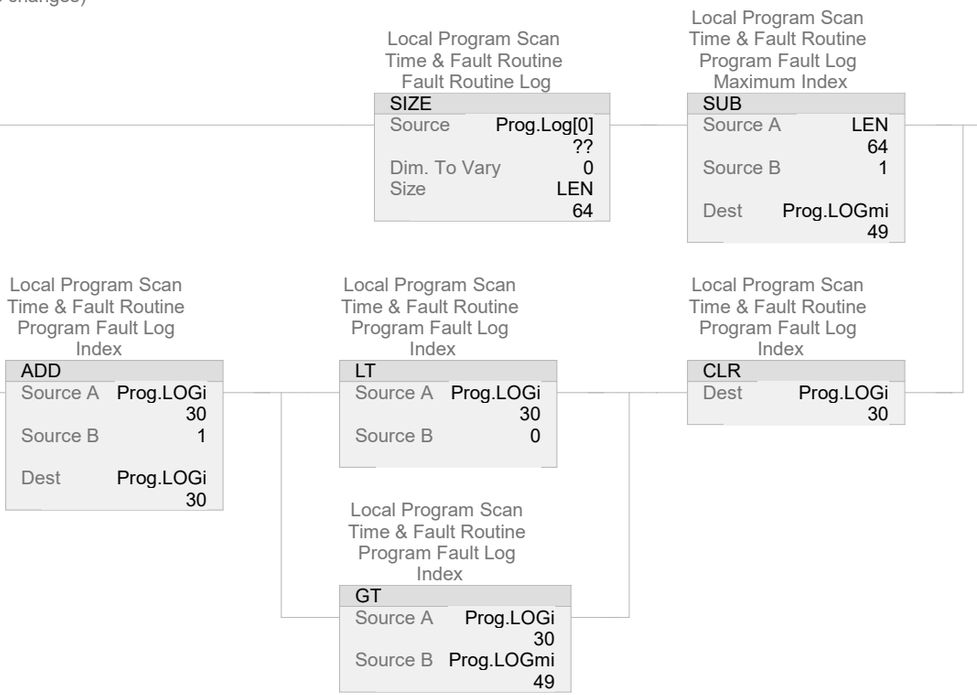
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

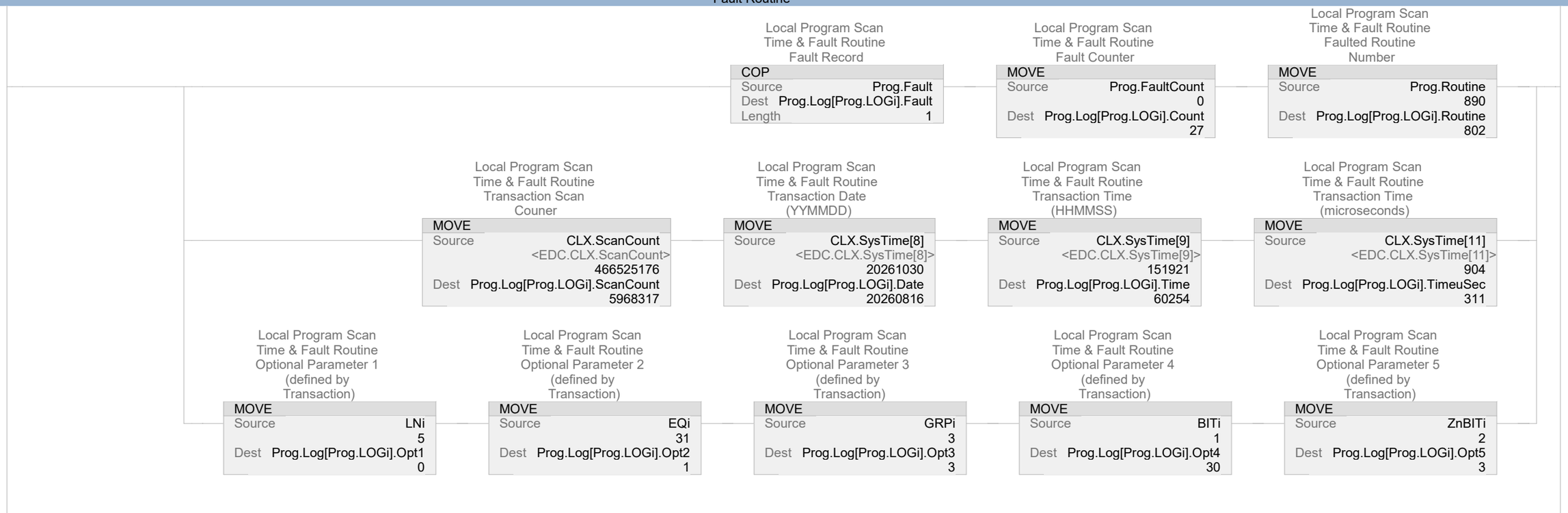
Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

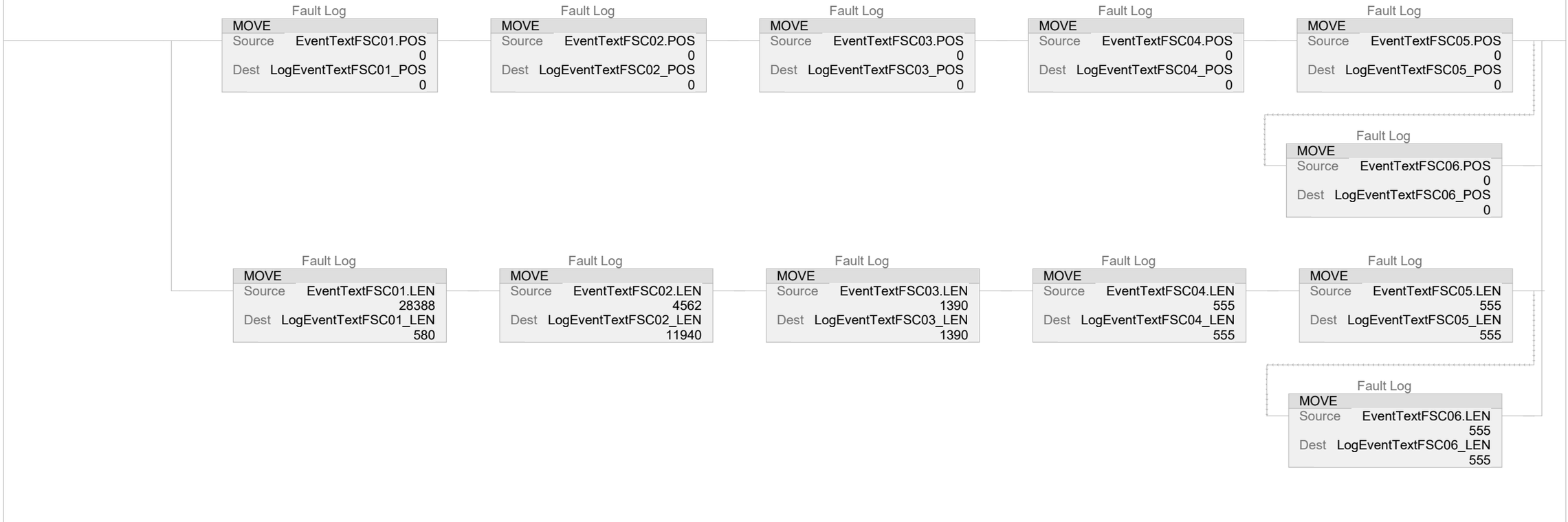
3



Fault Routine

Trap FSC Parameters

4



Fault Routine

Trap Mask Download Error

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

EQ
Source A Prog.Routine 890
Source B 896

Fault Log

ADD
Source A LogMaskFaultCount 4
Source B 1
Dest LogMaskFaultCount 4

Fault Log

MOVE
Source MaskEQi 26
Dest LogMaskEQi 13

Fault Log

MOVE
Source MaskLNi 9
Dest LogMaskLNi 1

Fault Log

MOVE
Source MaskEvLKi 164
Dest LogMaskEvLKi 91

Fault Log

MOVE
Source LOOPi 21
Dest LogLoopi 21

Fault Log

MOVE
Source TOOLi 915
Dest LogTOOLi 915

Fault Log

MOVE
Source TRNDi 4
Dest LogTRNDi 4

Fault Log

MOVE
Source MSKi 8
Dest LogMSKi 8

Fault Log

COP
Source ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Dest LogServerMask_DataRspLast
Length 1

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

6

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log

Prog.LogFault

(L)

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

7

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 802
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Fault Routine

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

8

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type
0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code
0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow
-20378558

(End)

Server Interface Reset
(called before Full Configuration Download)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 810
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Clear Event Config Faults

2

Server Interface Status State = Red - Equip Event Array Full ER
EDC_Status01.ServerCfg.Update.EquipEventArrayFull
(U)

Reset Equipment Event Text, Fragment Statistics and Errors
(before Download All to defrag the Event table for faster searches)

3

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt01[0]
Length EDC_HMI.CFG.EvTxt01.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt02[0]
Length EDC_HMI.CFG.EvTxt02.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt03[0]
Length EDC_HMI.CFG.EvTxt03.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt04[0]
Length EDC_HMI.CFG.EvTxt04.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt05[0]
Length EDC_HMI.CFG.EvTxt05.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt06[0]
Length EDC_HMI.CFG.EvTxt06.n

Event Controls Event Control

FLL
Source 0
Dest EDC_Event.CTL
Length 1

CLR
Dest EventEmptyLastFD 45063

Manual Reset for Configuration Status Errors
ManualResetER
(L)

v2.0 Clear Event Masks

4

DT Type = Alarm Event Mask

FLL	
Source	0
Dest	EDC_EventMaskAlarm
Length	1

DT Type = Alert Event Mask

FLL	
Source	0
Dest	EDC_EventMaskAlert
Length	1

DT Type = Fault Event Mask

FLL	
Source	-1
Dest	EDC_EventMaskFault
Length	1

DT Type = Ignore Event Mask

FLL	
Source	-1
Dest	EDC_EventMaskIgnore
Length	1

DT Owner = Maintenance Event Mask

FLL	
Source	-1
Dest	EDC_EventMaskMaint
Length	1

No Display Event Mask

FLL	
Source	0
Dest	EDC_EventMaskNoDisp
Length	1

May 2026 - clear Event KPI Option Masks
and Equipment KPI Option Packet Config/Status

5

KPI Option 1 Event Mask

FLL	
Source	0
Dest	EDC_EventMaskOpt01
Length	1

KPI Option 2 Event Mask

FLL	
Source	0
Dest	EDC_EventMaskOpt02
Length	1

KPI Option 3 Event Mask

FLL	
Source	0
Dest	EDC_EventMaskOpt03
Length	1

KPI Option 4 Event Mask

FLL	
Source	0
Dest	EDC_EventMaskOpt04
Length	1

KPI Option 5 Event Mask

FLL	
Source	-1
Dest	EDC_EventMaskOpt05
Length	1

Event KPI Options

FLL	
Source	0
Dest	EDC_EventOpt
Length	1

Clear Carrier Scanners

6

Carrier Tracking from EQ Quality Tools EDC Tracking Scanner Configuration

FLL	
Source	0
Dest	EDC_Track.ScanCfg[0]
Length	EDC_Track.Cfg.SCN.n

Reset Line, Group, Equip
Clear Line Loop Index

7

Line Index

CLR	
Dest	LNi 5

Line Loop

lbl_LineLoop
LBL

Equipment Index
CLR
Dest EQi
31

Reset Line Config
Reset Carrier Tracking Line Config

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].CFG
Length 1

Carrier Tracking from EQ Quality Tools Tracking Conveyor
FLL
Source 0
Dest EDC_Track.Conv[LNi].Cfg
Length 1

Reset Group Config

Line Index
LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.GRP.mi
<EDC.CFG.GRP.mi>
3

MOVE
Source LNi
5
Dest GRPi
3

Equipment Data Collection Offline Group Configuration
FLL
Source 0
Dest EDC.Offline.CFG[GRPi]
Length 1

11

Equipment Loop
Clear Equipment Event Text Search range
DN006 - support multiple Offline Lines
DN009 -add Equip.CTL to reset (otherwise deleted Equipment auto-configures as new Equipment)
v2.0 Clear Event Lookup Index
EDC2 WL09 - save CTL.*Msg to avoid Duplicate Message Errors during full download

lbl_EquipLoop
LBL

Equipment Data Collection - Line Equipment Control
COP
Source EDC.Line[LNi].Equip[EQi].CTL
Dest TempCTL
Length 1

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].Equip[EQi].CFG
Length 1

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].Equip[EQi].CFGoffline
Length 1

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].Equip[EQi].CTL
Length 1

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTsi
23544

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTei
23779

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi
144

12

EDC2 WL09 - restore CTL.*Msg to avoid Duplicate Message Errors during full download

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.CalloutMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.CalloutMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.FaultMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.FaultMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.QualityMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.QualityMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.TrendMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.TrendMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.VersionMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.VersionMsg
Length 1

Increment Equipment Index

13

Equipment Index
ADD
Source A EQi 31
Source B 1
Dest EQi 31

Equipment Index
LE
Source A EQi 31
Source B CFG.EQ.mi <EDC.CFG.EQ.mi> 99

lbl_EquipLoop
JMP

Increment Line Index

14

Line Index
ADD
Source A LNi 5
Source B 1
Dest LNi 5

Line Index
LE
Source A LNi 5
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

lbl_LineLoop
JMP

Reset Diagnostic Counts

15

Server Response Command diagnostic count
SIZE
Source RspCmd[0] 44710
Dim. To Vary 0
Size SIZE 100

Server Response Command diagnostic count
FLL
Source 0
Dest RspCmd[0]
Length SIZE

(End)

Configuration

HMI Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 811
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Fast and Slow Request Times
Fast is used when Last Transaction Response had data

2

Server Configuration from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerCfg.Ctl.ReqFastTime <EDC.ServerCfg.Ctl.ReqFastTime> 0

Server Configuration from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerCfg.Ctl.ReqSlowTime <EDC.ServerCfg.Ctl.ReqSlowTime> 10000

Fast and Slow Request Times
Fast is used when Last Transaction Response had data

3

Server PdM Configuration from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerPdm.Ctl.ReqFastTime <EDC.ServerPdm.Ctl.ReqFastTime> 0

Server PdM Configuration from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerPdm.Ctl.ReqSlowTime <EDC.ServerPdm.Ctl.ReqSlowTime> 10000

Configuration

4

Fast and Slow Status Request Times
Fast is used when Last Transaction Response not last row of table

Server Status from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerStat.Ctl.ReqFastTime
<EDC.ServerStat.Ctl.ReqFastTime>
0

Server Status from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerStat.Ctl.ReqSlowTime
<EDC.ServerStat.Ctl.ReqSlowTime>
10000

5

Fast and Slow Health Request Times
Fast is used when Last Transaction Response has new data

(Slow time is relatively fast -- with 200+ Equipment, guaranteed a refresh in ~3 minutes even if data changes aren't detected)

Server EQ Health from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerHealth.Ctl.ReqFastTime
<EDC.ServerHealth.Ctl.ReqFastTime>
0

Server EQ Health from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 1000
Dest ServerHealth.Ctl.ReqSlowTime
<EDC.ServerHealth.Ctl.ReqSlowTime>
1000

6

Fast and Slow Request Times
Fast is used when Last Transaction Response had data

Server Ignore Mask Configuration from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerMask.Ctl.ReqFastTime
<EDC.ServerMask.Ctl.ReqFastTime>
0

Server Ignore Mask Configuration from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerMask.Ctl.ReqSlowTime
<EDC.ServerMask.Ctl.ReqSlowTime>
10000

Configuration

Reset Empty Event Last Found

7

CLR
Dest EventEmptyLastFD
45063

Configuration

Event Mask Equipment, Packet, Data and Bit Arrays
Configure Array Indexes

*** NOTE: Event CTL and EventMast Equipment Arrays should be same SIZE --- use the smaller array to set array limits ***

8

Event Controls Event Control Equipment Array

SIZE
Source EDC_Event.CTL.Equip[0] ??
Dim. To Vary 0
Size EDC_Event.CFG.EQ.n 3000

DT Type = Alarm Event Mask Event Mask Equipment Blocks

SIZE
Source EDC_EventMaskAlarm.Equip[0] ??
Dim. To Vary 0
Size LEN 64

Temp Result for Array Size

LT
Source A LEN 64
Source B EDC_Event.CFG.EQ.n 3000

Event Controls Array Size

MOVE
Source LEN 64
Dest EDC_Event.CFG.EQ.n 3000

Event Controls Array Max Index

SUB
Source A EDC_Event.CFG.EQ.n 3000
Source B 1
Dest EDC_Event.CFG.EQ.mi 2999

DT Type = Alarm Event Mask Event Mask Equipment Packet of 8 DINTs

SIZE
Source EDC_EventMaskAlarm.Equip[0].Packet[0] ??
Dim. To Vary 0
Size EDC_Event.CFG.PKT.n 8

Event Controls Array Max Index

SUB
Source A EDC_Event.CFG.PKT.n 8
Source B 1
Dest EDC_Event.CFG.PKT.mi 7

DT Type = Alarm Event Mask Event Packet Data

SIZE
Source EDC_EventMaskAlarm.Equip[0].Packet[0].Data[0] 16#0000_0000
Dim. To Vary 0
Size EDC_Event.CFG.DAT.n 8

Event Controls Array Max Index

SUB
Source A EDC_Event.CFG.DAT.n 8
Source B 1
Dest EDC_Event.CFG.DAT.mi 7

Event Controls Array Size

MOVE
Source 32
Dest EDC_Event.CFG.BIT.n 32

Event Controls Array Max Index

SUB
Source A EDC_Event.CFG.BIT.n 32
Source B 1
Dest EDC_Event.CFG.BIT.mi 31

Configuration



Oview Display Zone/Mask Initialization

Overview Display Text
Create Zone/Mask text string for Overview Display
("A, B, C") for display on Event Table Zoom

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 818
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Overview Configuration for Rank
(Overview Mask defaults to all Zones)

2

MOVE
Source EDC_HMI.CFG.Oview.RankNum 1
Dest RankNum 1

MOVE
Source EDC_HMI.CFG.Oview.RankChar 4
Dest RankChar 4

Oview Display Zone/Mask Initialization

Overview shows all Zones - create ZoneList showing range of first and last Zones
(i.e. "A-T" for 20 Zones)

3

ASCII
Zone Letter

FLL	
Source	0
Dest	ZoneList
Length	1

ADD	
Source A	ascii.A <EDC.ascii.A> 'A'
Source B	EDC_HMI.CFG.ZN.si 0
Dest	ZoneLetter 'H'

MOVE	
Source	ZoneLetter 'H'
Dest	ZoneList.DATA[ZoneList.LEN] '$00'

ADD	
Source A	ZoneList.LEN 3
Source B	1
Dest	ZoneList.LEN 3

MOVE	
Source	ascii.Hyphen <EDC.ascii.Hyphen> '-'
Dest	ZoneList.DATA[ZoneList.LEN] '$00'

ADD	
Source A	ZoneList.LEN 3
Source B	1
Dest	ZoneList.LEN 3

ASCII
Zone Letter

ADD	
Source A	ascii.A <EDC.ascii.A> 'A'
Source B	EDC_HMI.CFG.ZN.mi 20
Dest	ZoneLetter 'H'

MOVE	
Source	ZoneLetter 'H'
Dest	ZoneList.DATA[ZoneList.LEN] '$00'

ADD	
Source A	ZoneList.LEN 3
Source B	1
Dest	ZoneList.LEN 3

Convert RankNum and RankChar to RankList
(i.e. "4C")

4

FLL	
Source	0
Dest	RankList
Length	1

MOVE	
Source	2
Dest	RankList.LEN 2

ADD	
Source A	RankNum 1
Source B	48
Dest	RankList.DATA[0] '1'

ADD	
Source A	RankChar 4
Source B	64
Dest	RankList.DATA[1] 'D'

Oview Display Zone/Mask Initialization

Save ZoneList and RankList to HMI Display

5

HMI Zone List ("A,B,C")

COP	
Source	ZoneList
Dest	EDC_HMI.Oview.ZoneList
Length	1

HMI Rank List ("A1,A2, B1,C1")

COP	
Source	RankList
Dest	EDC_HMI.Oview.RankList
Length	1

(End)

Zone Display Zone/Mask Initialization

Zone Display Text
Create Zone/Mask text string for Zone Display
("A, B, C") for display on n Zone Tables

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 819
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Zone Configuration for Mask and Rank

2

MOVE
Source EDC_HMI.CFG.Zone[ZNi].ZoneMask 16#0000_0000
Dest ZoneMask 16#0000_0000

MOVE
Source EDC_HMI.CFG.Zone[ZNi].RankNum 1
Dest RankNum 1

MOVE
Source EDC_HMI.CFG.Zone[ZNi].RankChar 4
Dest RankChar 4

Intialize Zone Loop

3

AddComma
U

FLL
Source 0
Dest ZoneList
Length 1

CLR
Dest LOOPi 21

Zone Display Zone/Mask Initialization

Convert ZoneMask to ZoneList
(i.e. 00-0000-0000-0000-0011 = "AB")

add each Zone letter to Zone List
eliminate commas since there are 20 Zones

4

lbl_ZoneLoop
LBL

ZoneMask.[LOOPi]

AddComma
AFI

MOVE
Source ascii.Comma
<EDC.ascii.Comma>
','
Dest ZoneList.DATA[ZoneList.LEN]
'$00'

ADD
Source A ZoneList.LEN
3
Source B 1
Dest ZoneList.LEN
3

ASCII
Zone Letter

ADD
Source A ascii.A
<EDC.ascii.A>
'A'
Source B LOOPi
21
Dest ZoneLetter
'H'

MOVE
Source ZoneLetter
'H'
Dest ZoneList.DATA[ZoneList.LEN]
'$00'

ADD
Source A ZoneList.LEN
3
Source B 1
Dest ZoneList.LEN
3

AddComma
L

Increment Loop

5

ADD
Source A LOOPi
21
Source B 1
Dest LOOPi
21

LE
Source A LOOPi
21
Source B EDC_HMI.CFG.ZN.mi
20

lbl_ZoneLoop
JMP

If No Zones assigned, set list to N/A

6

EQ
Source A ZoneList.LEN
3
Source B 0

COP
Source strNA
Dest ZoneList
Length 1

Zone Display Zone/Mask Initialization

Convert RankNum and RankChar to RankList
(i.e. "4C")

7

FLL
Source 0
Dest RankList
Length 1

MOVE
Source 2
Dest RankList.LEN
2

ADD
Source A RankNum
1
Source B 48
Dest RankList.DATA[0]
'1'

ADD
Source A RankChar
4
Source B 64
Dest RankList.DATA[1]
'D'

Save ZoneList and RankList to HMI Display

8

HMI Maintenance Zone Displays

COP
Source ZoneList
Dest EDC_HMI.Zone[ZNi].ZoneList
Length 1

HMI Maintenance Zone Displays

COP
Source RankList
Dest EDC_HMI.Zone[ZNi].RankList
Length 1

(End)

Configuration Request to Server via Transaction Manager

Configuration Data Request to SQL Server via Transaction Manager

- Logix Requests Data (-1 Command)
- Server Responds with Data (1-18 Command), No Data (0 Command) or No Response (-1)

- if Data, Logix processes Data and Sends Ack (Echo Server Command)
- Server uses Ack to set Download Timestamp for Config Record
- Server Responds with next Data or NoData

- if No Data, Logix waits and then Requests Data (-1 Command)

0 NOP

Request Commands to TM/SQL

-1 = New Data Request (delay between requests)
1-18 = Ack Data Response and New Data Request (no delay between requests)
99 - Full Download Request

1 NOP

Response Commands from Server

-1 = Data Error
0 = No Data to Download
1 = Line Configuration
2 = Offline Groups Configuration
3 = HMI Overview Display Configuration
4 = HMI Zone Display Configuration
5 = Equipment Configuration
6 = Schedule Hours Configuration
7 = Gateway Configuration
9 = Carrier Scanner Configuration
16 = Equipment Event Configuration - PreDefined
17 = Equipment Event Configuration - EqDefined
18 = Shop Configuration (Reset and Download All)

2 NOP

Configuration Request to Server via Transaction Manager

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

3

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	820
Dest	Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806549717
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data

4

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Response Duplicate Error
ServerCfg.Ctl.RspDupER
<EDC.ServerCfg.Ctl.RspDupER>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerCfg.Ctl.ReqFastTime
<EDC.ServerCfg.Ctl.ReqFastTime>
0
Dest ServerCfg.Ctl.ReqNextTMR.PRE
<EDC.ServerCfg.Ctl.ReqNextTMR.PRE>
10000

Server Configuration from Transaction Manager Request Next Record Timer

TON
Timer ServerCfg.Ctl.ReqNextTMR
<EDC.ServerCfg.Ctl.ReqNextTMR>
Preset 10000
Accum 7766
EN
DN

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Response Duplicate Error
ServerCfg.Ctl.RspDupER
<EDC.ServerCfg.Ctl.RspDupER>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerCfg.Ctl.ReqSlowTime
<EDC.ServerCfg.Ctl.ReqSlowTime>
10000
Dest ServerCfg.Ctl.ReqNextTMR.PRE
<EDC.ServerCfg.Ctl.ReqNextTMR.PRE>
10000

Request Next
Ready for next Transaction Manager Trigger

5

Server Configuration from Transaction Manager Request Next Record Timer
ServerCfg.Ctl.ReqNextTMR.DN
<EDC.ServerCfg.Ctl.ReqNextTMR.DN>

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>
L

Server Configuration from Transaction Manager Request Record Complete
ServerCfg.Ctl.ReqComplete
<EDC.ServerCfg.Ctl.ReqComplete>
U

Configuration Request to Server via Transaction Manager

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

Mar 24, 2024 -- added ReqInProgressTimeout
instead of resending last Request, Init a new request
(was locking up on repeated Gateway Download requests because SQL response >1 minute)

6

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Configuration Request to Server via Transaction Manager - Data Init

JSR
Routine Name _821_CfgRequestInit

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Configuration Request to Server via Transaction Manager

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

7

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B 1
Dest ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B 1

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721

Configuration Request to Server via Transaction Manager

8

Request Next
Log Transaction & Performance

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4000
Input Par	ServerCfg.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerCfg.DataReq.Command
	<EDC.ServerCfg.DataReq.Command>
Input Par	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
Input Par	ServerCfg.Ctl.TransactionNumber
	<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR

Dest	TempPerfMsg.TransTMR.ACC
	110

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4002
Input Par	ServerCfg.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerCfg.DataReq.Command
	<EDC.ServerCfg.DataReq.Command>
Input Par	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
Input Par	ServerCfg.Ctl.TransactionNumber
	<EDC.ServerCfg.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans

aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerCfg.Msg.EN

Configuration Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

9

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B ServerCfg.Ctl.Interface
<EDC.ServerCfg.Ctl.Interface>
8721

Server Configuration from Transaction Manager Request Record Complete
ServerCfg.Ctl.ReqComplete
<EDC.ServerCfg.Ctl.ReqComplete>
L

Configuration Request to Server via Transaction Manager

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
Also problems with asynchronous data from TM - handshake can update before data values.
2-Check method is fast and avoids errors for occasional slow update of data.
Wait 1 scan before 1st check to eliminate 99% of missing data issues.
2nd check after 500msec if Check 1 fails.
(anything >2 seconds will generate a TM Transaction error based on Tag Retrieval rate)

10

Server Configuration from Transaction Manager Request Record Complete
ServerCfg.Ctl.ReqComplete
<EDC.ServerCfg.Ctl.ReqComplete>

Server Configuration from Transaction Manager Request Record First Check Timer

MOVE
Source 10
Dest ServerCfg.Ctl.ReqCompleteCheckTMR.PRE
<EDC.ServerCfg.Ctl.ReqCompleteCheckTMR.PRE>
10

Server Configuration from Transaction Manager Request Record First Check Timer

TON
Timer ServerCfg.Ctl.ReqCompleteCheckTMR
<EDC.ServerCfg.Ctl.ReqCompleteCheckTMR>
Preset 10
Accum 16
EN
DN

Server Configuration from Transaction Manager Request Record First Check Timer
ServerCfg.Ctl.ReqCompleteCheckTMR.DN
<EDC.ServerCfg.Ctl.ReqCompleteCheckTMR.DN>

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Complete Delay Timer

MOVE
Source 500
Dest ServerCfg.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerCfg.Ctl.ReqCompleteDelayTMR.PRE>
500

Server Configuration from Transaction Manager Request Complete Delay Timer

TON
Timer ServerCfg.Ctl.ReqCompleteDelayTMR
<EDC.ServerCfg.Ctl.ReqCompleteDelayTMR>
Preset 500
Accum 507
EN
DN

Server Configuration from Transaction Manager Request Complete Delay Timer
ServerCfg.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerCfg.Ctl.ReqCompleteDelayTMR.DN>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Configuration Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforced in data update.)

11

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record First Check 1-shot
ServerCfg.Ctl.ReqCompleteCheckONS
<EDC.ServerCfg.Ctl.ReqCompleteCheckONS>
ONS

Server Configuration from Transaction Manager Request Complete 1st Check Count

ADD
Source A ServerCfg.Ctl.ReqComp1stChkCnt
<EDC.ServerCfg.Ctl.ReqComp1stChkCnt>
1663573
Source B 1
Dest ServerCfg.Ctl.ReqComp1stChkCnt
<EDC.ServerCfg.Ctl.ReqComp1stChkCnt>
1663573

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Request Record Complete 1-shot
ServerCfg.Ctl.ReqCompleteONS
<EDC.ServerCfg.Ctl.ReqCompleteONS>
ONS

Server Configuration from Transaction Manager Request Complete 2nd Check Count

ADD
Source A ServerCfg.Ctl.ReqComp2ndChkCnt
<EDC.ServerCfg.Ctl.ReqComp2ndChkCnt>
2506
Source B 1
Dest ServerCfg.Ctl.ReqComp2ndChkCnt
<EDC.ServerCfg.Ctl.ReqComp2ndChkCnt>
2506

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4001
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataReq.Command
<EDC.ServerCfg.DataReq.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerCfg.Update.MsgRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Equipment Data

Configuration Request to Server via Transaction Manager

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerCfg.Msg.DN

Configuration Response from Server via Transaction Manager

JSR
Routine Name _830_CfgResponse

Request Complete Delay Reset Request

12

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>
U

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

13

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Request in Progress Timer

MOVE
Source 12000
Dest ServerCfg.Ctl.ReqInProgressTMR.PRE
<EDC.ServerCfg.Ctl.ReqInProgressTMR.PRE>
12000

Server Configuration from Transaction Manager Request in Progress Timer

TON
Timer ServerCfg.Ctl.ReqInProgressTMR
<EDC.ServerCfg.Ctl.ReqInProgressTMR>
Preset 12000
Accum 0
EN
DN

Server Configuration from Transaction Manager Request in Progress Timer
ServerCfg.Ctl.ReqInProgressTMR.DN
<EDC.ServerCfg.Ctl.ReqInProgressTMR.DN>

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

14

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4004
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataReq.Command
<EDC.ServerCfg.DataReq.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Msg Rx ER
EDC_Status01.ServerCfg.Update.MsgRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerCfg.Msg.ER

Server Configuration from Transaction Manager Request in Progress Timer
ServerCfg.Ctl.ReqInProgressTMR
<EDC.ServerCfg.Ctl.ReqInProgressTMR>
RES

(End)

Configuration Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Server to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 821
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Create Data Init Packet
(FLL -1 corrupts FLOAT values and causes TM Transaction Failure; overwrite with MOV)

2

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager - Data
FLL
Source -1
Dest ServerCfgInit
Length 1

Server Configuration from Transaction Manager - Data Name
FLL
Source 0
Dest ServerCfgInit.Name
Length 1

Server Configuration from Transaction Manager - Data Short Name
FLL
Source 0
Dest ServerCfgInit.ShortName
Length 1

Server Configuration from Transaction Manager - Data Equipment Event Text
FLL
Source 0
Dest ServerCfgInit.EqEventText
Length 1

Server Configuration from Transaction Manager - Data Name
MOVE
Source 1
Dest ServerCfgInit.Name.DATA[0] '$01'

Server Configuration from Transaction Manager - Data Short Name
MOVE
Source 1
Dest ServerCfgInit.ShortName.DATA[0] '$01'

Server Configuration from Transaction Manager - Data Equipment Event Text
MOVE
Source 1
Dest ServerCfgInit.EqEventText.DATA[0] '$01'

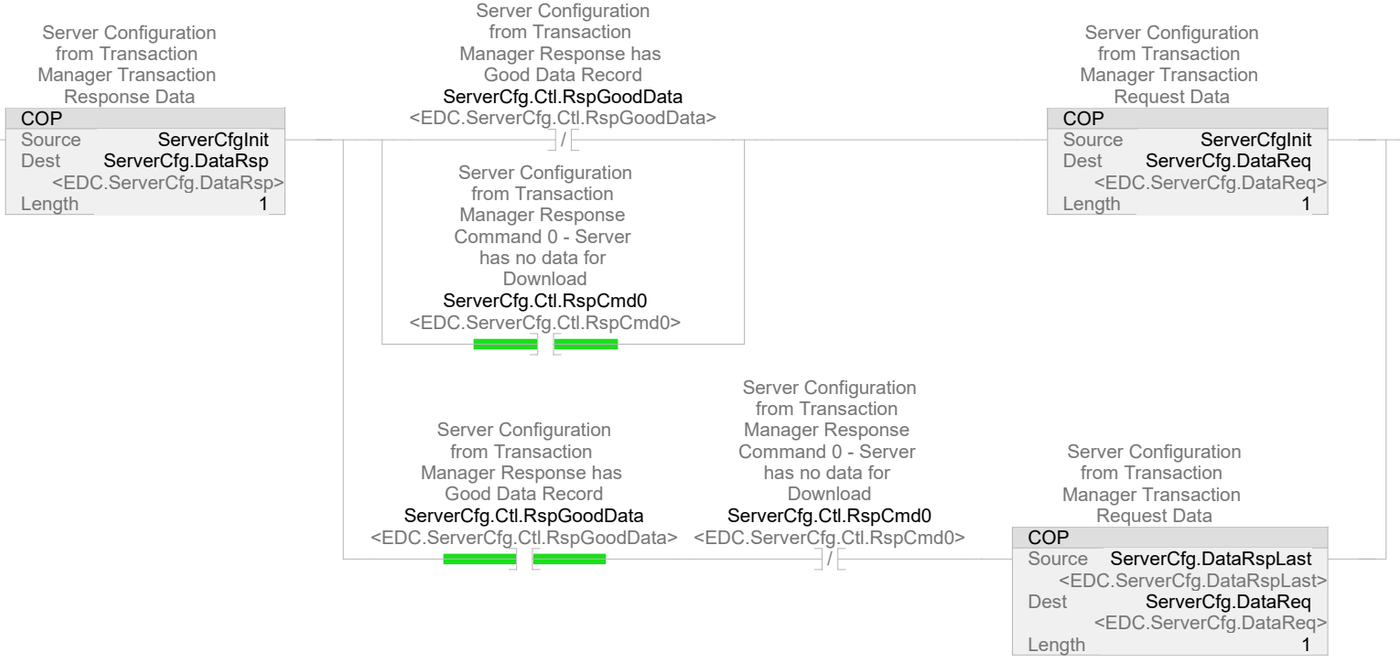
Server Configuration from Transaction Manager - Data Target OA - Green Status
MOVE
Source -1
Dest ServerCfgInit.TargetOAGrn -1.0

Server Configuration from Transaction Manager - Data Target OA - Yellow Status
MOVE
Source -1
Dest ServerCfgInit.TargetOAYel -1.0

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Clear Response to ensure new data on next DB Response
DHF017 Jan 30, 2018 -- Clear Server Request if no Ack for Good Message Response
Copy Response to Request for Good Data Ack to DB

3



Configuration Request to Server via Transaction Manager - Data Init

Config Download ALL Request - Command 99
Automatic Sunday morning Defragment of Event Text

DHF012 - Fixed Nov/Dec bug in aoiEDC_DayofWeek_v1_2
DN009 - perform Sunday Config Download ALL regardless of Status Event Defrag count
(this also removes any newly disabled Equipment from the Overview)

4

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

CLx Module Status Hour (HH)

EQ	
Source A	CLX.TransTime.Hour <EDC.CLX.TransTime.Hour>
	15
Source B	6

CLx Module Status Minute (MM)

EQ	
Source A	CLX.TransTime.Min <EDC.CLX.TransTime.Min>
	19
Source B	0

CLx Module Status CLx Wallclock Day of Week (0=Saturday, 6=Friday)

EQ	
Source A	CLX.DayOfWeek <EDC.CLX.DayOfWeek>
	6
Source B	1

Server Configuration from Transaction Manager Request/Response Command

MOVE	
Source	99
Dest	ServerCfg.DataReq.Command <EDC.ServerCfg.DataReq.Command>
	-1

Initiate Full Download (manual set)
ManualFullDownload

Request Full Configuration Download (from Gway Reset)
CFG_FullDownload

Initiate Full Download (manual set)
ManualFullDownload
U

Request Full Configuration Download (from Gway Reset)
CFG_FullDownload
U

Store Last Request

5

Server Configuration from Transaction Manager Transaction Last Request Data

COP	
Source	ServerCfg.DataReq <EDC.ServerCfg.DataReq>
Dest	ServerCfg.DataReqLast <EDC.ServerCfg.DataReqLast>
Length	1

(End)

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 830
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Detect Duplicate Config Data Response
(this indicates the database is stuck on a bad config item)

2

Server Configuration from Transaction Manager Request/Response Command

NE
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 0

Server Configuration from Transaction Manager Request/Response Command

NE
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B -1

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B ServerCfg.DataRsp.Command
<EDC.ServerCfg.DataRsp.Command>
0

Server Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.ShopKey
<EDC.ServerCfg.DataRspLast.ShopKey>
-1
Source B ServerCfg.DataRsp.ShopKey
<EDC.ServerCfg.DataRsp.ShopKey>
-1

Server Configuration from Transaction Manager Line Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.LineKey
<EDC.ServerCfg.DataRspLast.LineKey>
-1
Source B ServerCfg.DataRsp.LineKey
<EDC.ServerCfg.DataRsp.LineKey>
-1

Server Configuration from Transaction Manager Equip Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.EqKey
<EDC.ServerCfg.DataRspLast.EqKey>
-1
Source B ServerCfg.DataRsp.EqKey
<EDC.ServerCfg.DataRsp.EqKey>
-1

Server Configuration from Transaction Manager Equipment ID

EQ
Source A ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Source B ServerCfg.DataRsp.EqID
<EDC.ServerCfg.DataRsp.EqID>
-1

Server Configuration from Transaction Manager Line ID (0=Offline)

EQ
Source A ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Source B ServerCfg.DataRsp.LineID
<EDC.ServerCfg.DataRsp.LineID>
-1

Server Configuration from Transaction Manager Pitch ID (0-350)

EQ
Source A ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
Source B ServerCfg.DataRsp.PitchID
<EDC.ServerCfg.DataRsp.PitchID>
-1

Server Configuration from Transaction Manager Position ID (0-5)

EQ
Source A ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
Source B ServerCfg.DataRsp.PositionID
<EDC.ServerCfg.DataRsp.PositionID>
-1

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

EQ
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Source B ServerCfg.DataRsp.ZoneDispID
<EDC.ServerCfg.DataRsp.ZoneDispID>
-1

Server Configuration from Transaction Manager Offline Group ID (1-4)

EQ
Source A ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID>
-1
Source B ServerCfg.DataRsp.OfflineGroupID
<EDC.ServerCfg.DataRsp.OfflineGroupID>
-1

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

EQ
Source A ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Source B ServerCfg.DataRsp.EqEventID
<EDC.ServerCfg.DataRsp.EqEventID>
-1

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

EQ
Source A ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Source B ServerCfg.DataRsp.EqEventID
<EDC.ServerCfg.DataRsp.EqEventID>
-1

Server Configuration from Transaction Manager Response Duplicate
ServerCfg.Ctl.RspDup
<EDC.ServerCfg.Ctl.RspDup>

Configuration Response from Server via Transaction Manager

3

Duplicate Message Count and Detection
(this indicates the database is stuck on a bad config item)

Server Configuration from Transaction Manager Response Duplicate
ServerCfg.Ctl.RspDup
<EDC.ServerCfg.Ctl.RspDup>

ADD
Source A RspDuplicateCount 0
Source B 1
Dest RspDuplicateCount 0

GT
Source A RspDuplicateCount 0
Source B RspDuplicateMax 53

MOVE
Source RspDuplicateCount 0
Dest RspDuplicateMax 53

GT
Source A RspDuplicateCount 0
Source B 5

Server Configuration from Transaction Manager Response Duplicate Error
ServerCfg.Ctl.RspDupER
<EDC.ServerCfg.Ctl.RspDupER>

Server Configuration from Transaction Manager Response Duplicate
ServerCfg.Ctl.RspDup
<EDC.ServerCfg.Ctl.RspDup>

CLR
Dest RspDuplicateCount 0

4

Move Response to working tag and Initialize Response Good Data
DN011 Separate Req & Rsp Tags

Moving DataRsp to DataLastRsp prevents SQL/TM from changing data while Logix is processing.

Server Configuration from Transaction Manager Transaction Last Response Data

COP
Source ServerCfg.DataRsp <EDC.ServerCfg.DataRsp>
Dest ServerCfg.DataRspLast <EDC.ServerCfg.DataRspLast>
Length 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Configuration Response from Server via Transaction Manager

Check Response Command for Bad Data
Valid Commands 0, 1-7, 9, 16-18

5

Server Configuration from Transaction Manager Request/Response Command

LT
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 0

Server Configuration from Transaction Manager Request/Response Command

LIMIT
Low Limit 8
Test ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
High Limit 8

Server Configuration from Transaction Manager Request/Response Command

LIMIT
Low Limit 10
Test ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
High Limit 15

Server Configuration from Transaction Manager Request/Response Command

GT
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 18

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID> 2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4012
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC> 0
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans (...)
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerCfg.Msg.KA

lbl_RspComp
JMP

Configuration Response from Server via Transaction Manager

If Response Command received, process Response

6

RspCommandRcvd

Server has no new Configuration data to download

7

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Response Command diagnostic count

ADD
Source A RspCmd[0]
44710
Source B 1
Dest RspCmd[0]
44710

Process Response Command

8

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 1

Configuration Response from Server via Transaction Manager

JSR
Routine Name _831_CfgResponseLine

Server Response Command diagnostic count

ADD
Source A RspCmd[1]
29
Source B 1
Dest RspCmd[1]
29

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 2

Configuration Response from Server via Transaction Manager

JSR
Routine Name _832_CfgResponseOfflineGroup

Server Response Command diagnostic count

ADD
Source A RspCmd[2]
4
Source B 1
Dest RspCmd[2]
4

Configuration Response from Server via Transaction Manager

Process Response Command

9

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 3

Configuration Response from Server via Transaction Manager

JSR
Routine Name _833_CfgResponseOviewDisplay

Server Response Command diagnostic count

ADD
Source A RspCmd[3]
1
Source B 1
Dest RspCmd[3]
1

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 4

Configuration Response from Server via Transaction Manager

JSR
Routine Name _834_CfgResponseZoneDisplay

Server Response Command diagnostic count

ADD
Source A RspCmd[4]
21
Source B 1
Dest RspCmd[4]
21

Configuration Response from Server via Transaction Manager

Process Response Command

10

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 5

Configuration Response from Server via Transaction Manager

JSR
Routine Name _835_CfgResponseEquipment

Server Response Command diagnostic count

ADD
Source A RspCmd[5] 300
Source B 1
Dest RspCmd[5] 300

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 6

Configuration Response from Server via Transaction Manager

JSR
Routine Name _840_CfgResponseSchedule

Server Response Command diagnostic count

ADD
Source A RspCmd[6] 23
Source B 1
Dest RspCmd[6] 23

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 7

Configuration Response from Server via Transaction Manager

JSR
Routine Name _841_CfgResponseGateway

Server Response Command diagnostic count

ADD
Source A RspCmd[7] 1
Source B 1
Dest RspCmd[7] 1

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 9

Configuration Response from Server via Transaction Manager

JSR
Routine Name _839_CfgResponseCarrierScanner

Server Response Command diagnostic count

ADD
Source A RspCmd[9] 5
Source B 1
Dest RspCmd[9] 5

Configuration Response from Server via Transaction Manager

Process Response Command

11

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 16

Server Response Command diagnostic count

ADD
Source A RspCmd[16]
32
Source B 1
Dest RspCmd[16]
32

Server Configuration from Transaction Manager Download Started - Event Received
ServerCfg.Ctl.DownloadEventRcvd
<EDC.ServerCfg.Ctl.DownloadEventRcvd>

Configuration Response from Server via Transaction Manager

JSR
Routine Name _836_CfgResponseEquipmentEventTxt

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 17

Server Response Command diagnostic count

ADD
Source A RspCmd[17]
96886
Source B 1
Dest RspCmd[17]
96886

Process Response Command

12

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 18

Configuration Response from Server via Transaction Manager

JSR
Routine Name _838_CfgResponseShop

Server Response Command diagnostic count

ADD
Source A RspCmd[18]
2
Source B 1
Dest RspCmd[18]
2

Command Response Complete

13

lbl_RspComp
LBL

NOP

Configuration Response from Server via Transaction Manager

Store Last Good, Bad or Incomplete Response w/ Diagnostic Counts
Incomplete - Command Num received with incomplete data packet on on 1st Check
Bad - Incomplete data packet on 2nd Check

14

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Transaction Last Good Response Data

COP
Source ServerCfg.DataRspLast
<EDC.ServerCfg.DataRspLast>
Dest ServerCfg.DataRspLastGood
<EDC.ServerCfg.DataRspLastGood>
Length 1

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Response Good on 1st Check

ADD
Source A ServerCfg.Ctl.RspGood1stChkCnt
<EDC.ServerCfg.Ctl.RspGood1stChkCnt>
1715275
Source B 1
Dest ServerCfg.Ctl.RspGood1stChkCnt
<EDC.ServerCfg.Ctl.RspGood1stChkCnt>
1715275

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Response Good on 2nd Check

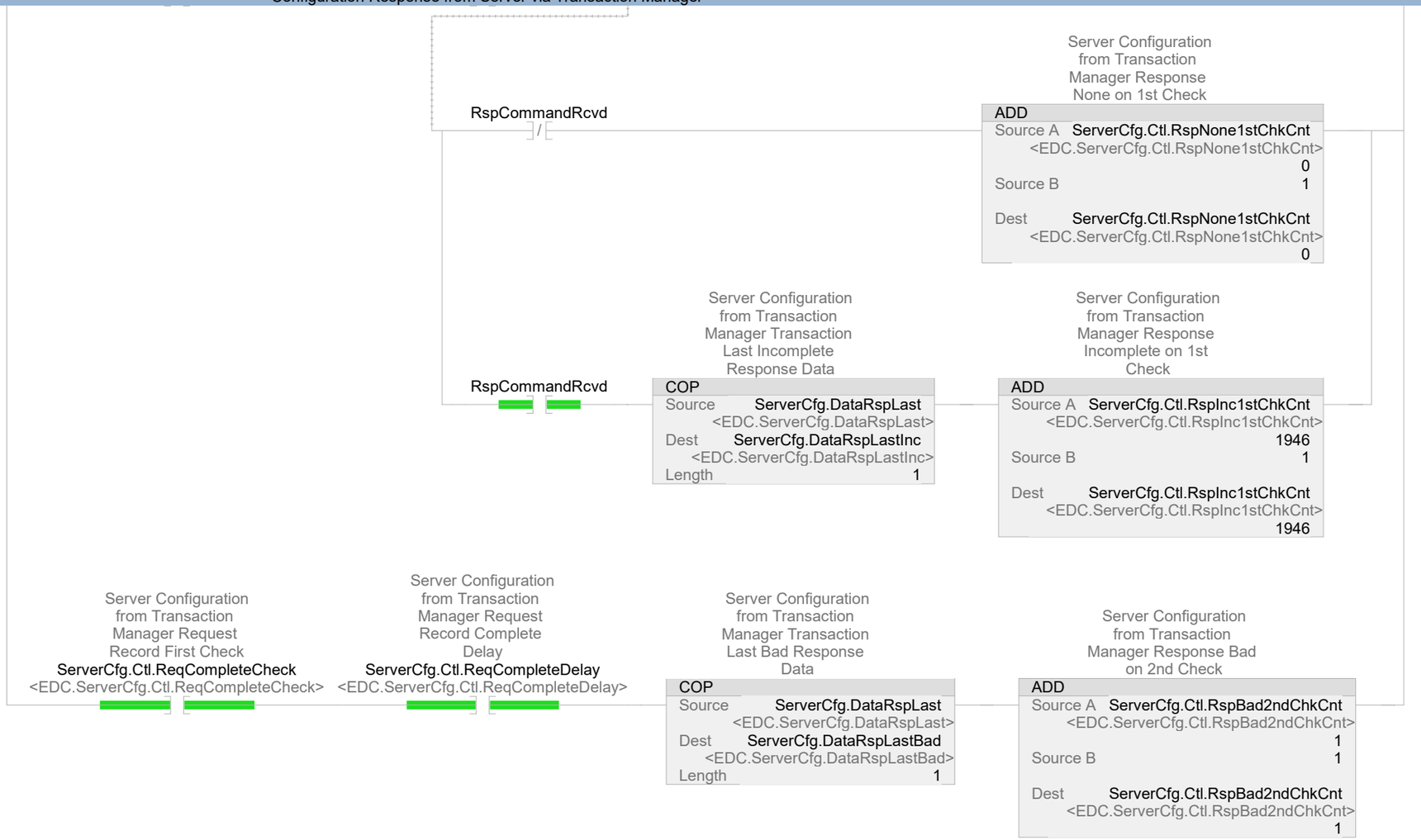
ADD
Source A ServerCfg.Ctl.RspGood2ndChkCnt
<EDC.ServerCfg.Ctl.RspGood2ndChkCnt>
3211
Source B 1
Dest ServerCfg.Ctl.RspGood2ndChkCnt
<EDC.ServerCfg.Ctl.RspGood2ndChkCnt>
3211

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Configuration Response from Server via Transaction Manager



Configuration Response from Server via Transaction Manager

Download ALL configuration is complete

15

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>
U

Server Configuration from Transaction Manager Response Download All Complete
ServerCfg.Ctl.DownloadAllComplete
<EDC.ServerCfg.Ctl.DownloadAllComplete>
L

Server Configuration from Transaction Manager Response Download All Complete 1-shot
ServerCfg.Ctl.DownloadAllCompleteONS
<EDC.ServerCfg.Ctl.DownloadAllCompleteONS>

HMI Refresh HMI Events after Config All
EDC_HMI.CTL.DownloadAllRefresh
L

EDC2 WL31 -- ensure BlockEqDriver doesn't get stuck on in a failed download
(better to collect data from Disabled than not collect from Enabled)

EDC2 WL31 Equipment Complete -- once download reaches the Events, all Equipment should have a good configuration
EDC2 WL31 Block EqDriver -- don't service EqDriver Messages until Equipment Configuration is complete and EQ Disable status can be re-established

16

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>
/

Server Configuration from Transaction Manager Download Started - Event Received
ServerCfg.Ctl.DownloadEventRcvd
<EDC.ServerCfg.Ctl.DownloadEventRcvd>

Server Configuration from Transaction Manager Download Started - Equipment Config Complete
ServerCfg.Ctl.DownloadEquipComplete
<EDC.ServerCfg.Ctl.DownloadEquipComplete>
L

Server Configuration from Transaction Manager Download Started - Block Equipment Drivers until Equipment Config Complete
ServerCfg.Ctl.DownloadBlockEqDriver
<EDC.ServerCfg.Ctl.DownloadBlockEqDriver>
U

Configuration Response from Server via Transaction Manager

DHF013 - Nov 27, 2017 -- Periodic Refresh of Event Text during Download ALL configuration

17

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

Download All Periodic Refresh of Event Text

MUL
Source A 3
Source B 60000
Dest DownloadAllPeriodicRefreshTMR.PRE 180000

Download All Periodic Refresh of Event Text

TON
Timer DownloadAllPeriodicRefreshTMR
Preset 180000
Accum 0
EN
DN

Download All Periodic Refresh of Event Text
DownloadAllPeriodicRefreshTMR.DN

Download All Periodic Refresh of Event Text
DownloadAllPeriodicRefreshTMR
RES

HMI Refresh HMI Events after Config All
EDC_HMI.CTL.DownloadAllRefresh
L

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 1 - Line
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 831
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data
v3.5 only check LineID lower limit to determine Good Data (upper limit is only used to determine if LineID can be saved in array)

2

Server Configuration from Transaction Manager Line ID (0=Offline)

GE
Source A ServerCfg.DataRspLast.LineID <EDC.ServerCfg.DataRspLast.LineID> -1
Source B 0

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0] <EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Short Name

NE
Source A ServerCfg.DataRspLast.ShortName.DATA[0] <EDC.ServerCfg.DataRspLast.ShortName.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Target OA - Green Status

GE
Source A ServerCfg.DataRspLast.TargetOAGrn <EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Source B 0

Server Configuration from Transaction Manager Target OA - Yellow Status

GE
Source A ServerCfg.DataRspLast.TargetOAYel <EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData <EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4013
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerCfg.Update.LineRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

v3.5 Check LIneID for valid Index
if SQL is configured for more Lines than Gateway, Gateway will ack the invalid LineID without saving
(the alternative is to continuously retry the invalid LineID index)

4

Server Configuration from Transaction Manager Line ID (0=Offline)

GT
Source A ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Source B CFG.LN.mi
<EDC.CFG.LN.mi>
28

lbl_ExitGoodStatus
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

5

Line Index

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID> -1
Dest LNi
5

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.Line[LNi].CFG.Name
Length 1

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.Line[LNi].CFG.ShortName
Length 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAGrn
<EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Dest EDC.Line[LNi].CFG.TargetOAGrn
96.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAYel
<EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Dest EDC.Line[LNi].CFG.TargetOAYel
94.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID> -1
Dest EDC.Line[LNi].CFG.Takt
56300

Check if Line is already configured for Offline

6

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline

Line was configured for Offline
OffLineOn

Configuration Response from Server via Transaction Manager

DN006 - support multiple Offline Lines
If Line Name contains 'Offline' latch the Offline configuration

July 31, 2023 - added 'Other' to Offline search for eBW
Feb 12, 2024 - added UPPER for case-insensitive search of OFFLINE & OTHER

7

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline
U

Server Configuration from Transaction Manager Name

UPPER
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
''
Dest strNameUPPER
'OFFLINE2'

Line Name contains 'Offline'

CLR
Dest OfflineFD
1

FIND
Source strNameUPPER
'OFFLINE2'
Search strOFFLINE
'OFFLINE'
Start 1
Result OfflineFD
1

Line Name contains 'Offline'

LIMIT
Low Limit 1
Test OfflineFD
1
High Limit strNameUPPER.LEN
8

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline
L

Line Name contains 'Other'

CLR
Dest OtherFD
0

FIND
Source strNameUPPER
'OFFLINE2'
Search strOTHER
'OTHER'
Start 1
Result OtherFD
0

Line Name contains 'Other'

LIMIT
Low Limit 1
Test OtherFD
0
High Limit strNameUPPER.LEN
8

Line Index

LT
Source A LNi
5
Source B EDC.CFG.OfflineFirstLine
0

Equipment Data Collection First Offline Line (for multiline support during Processing)

MOVE
Source LNi
5
Dest EDC.CFG.OfflineFirstLine
0

Line Index

GT
Source A LNi
5
Source B EDC.CFG.OfflineLastLine
28

Equipment Data Collection Last Offline Line (for multiline support during Processing)

MOVE
Source LNi
5
Dest EDC.CFG.OfflineLastLine
28

Configuration Response from Server via Transaction Manager

when the Line's Offline status is On or stayed Off,
the OfflineFirstLine and OfflineLastLine are configured in previous rung
and can skip the update logic

Line was configured for Offline
OffLineOn
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline

lbl_SkipOffline
JMP

8

when a Line's Offline status was On and turned Off,
must search for the new OfflineFirstLine and OfflineLastLine values

initialize Line index and OfflineFirstLine/OfflineLastLine values

CLR
Dest index
22

Equipment Data Collection First Offline Line (for multiline support during Processing)

CLR
Dest EDC.CFG.OfflineFirstLine
0

Equipment Data Collection Last Offline Line (for multiline support during Processing)

CLR
Dest EDC.CFG.OfflineLastLine
28

9

set OfflineFirstLine and OfflineLastLine values for Offline Lines

lbl_Offline
LBL

Equipment Data Collection Lines and Equipment Array
EDC.Line[index].CFG.Offline

LT
Source A index
22
Source B EDC.CFG.OfflineFirstLine
0

Equipment Data Collection First Offline Line (for multiline support during Processing)

MOVE
Source index
22
Dest EDC.CFG.OfflineFirstLine
0

GT
Source A index
22
Source B EDC.CFG.OfflineLastLine
28

Equipment Data Collection Last Offline Line (for multiline support during Processing)

MOVE
Source index
22
Dest EDC.CFG.OfflineLastLine
28

10

Configuration Response from Server via Transaction Manager

Increment Loop Index

11

ADD
Source A index 22
Source B 1
Dest index 22

LE
Source A index 22
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

lbl_Offline
JMP

12

lbl_SkipOffline
LBL

NOP

Data OK

Server Interface Status State = Red- Line Rx Data OK

Update Status State for Transactions

13

lbl_ExitGoodStatus
LBL

EDC_Status01.ServerCfg.Update.LineRxDataOK
L

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Exit Routine

14

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 2 - Offline Group
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 832
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Offline Group ID (1-4)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.OfflineGroupID <EDC.ServerCfg.DataRspLast.OfflineGroupID> -1
High Limit CFG.GRP.mi <EDC.CFG.GRP.mi> 3

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0] <EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Short Name

NE
Source A ServerCfg.DataRspLast.ShortName.DATA[0] <EDC.ServerCfg.DataRspLast.ShortName.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Target OA - Green Status

GE
Source A ServerCfg.DataRspLast.TargetOAGrn <EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Source B 0

Server Configuration from Transaction Manager Target OA - Yellow Status

GE
Source A ServerCfg.DataRspLast.TargetOAYel <EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData <EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4014
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-Offline Group Rx Data ER
EDC_Status01.ServerCfg.Update.OfflineRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

4

MOVE
Source ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID> -1
Dest GRPi
3

Equipment Data Collection Offline Group Configuration

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.Offline.CFG[GRPi].Name
Length 1

HMI Offline Data Array by Group Index (1-Trim, 2-Chassis, 3-Final)

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_HMI.Oview.Offline[GRPi].Name
Length 1

Equipment Data Collection Offline Group Configuration

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.Offline.CFG[GRPi].ShortName
Length 1

Equipment Data Collection Offline Group Configuration

MOVE
Source ServerCfg.DataRspLast.TargetOAGrn
<EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Dest EDC.Offline.CFG[GRPi].TargetOAGrn
96.0

Equipment Data Collection Offline Group Configuration

MOVE
Source ServerCfg.DataRspLast.TargetOAYel
<EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Dest EDC.Offline.CFG[GRPi].TargetOAYel
94.0

Data OK

Server Interface Status State = Red-Offline Group Rx Data OK
EDC_Status01.ServerCfg.Update.OfflineRxDataOK

5

L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Exit Routine

lbl_Exit

6

LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 3 - Overview Display
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 833
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

GE
Source A ServerCfg.DataRspLast.RankNum <EDC.ServerCfg.DataRspLast.RankNum> -1
Source B 0

Server Configuration from Transaction Manager Maintenance Rank Assignments (A-D = Values 1-4)

GE
Source A ServerCfg.DataRspLast.RankChar <EDC.ServerCfg.DataRspLast.RankChar> -1
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4015
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- HMI Oview Display Rx Data ER
EDC_Status01.ServerCfg.Update.HmiOviewRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Process Data and Store

4

HMI Maintenance Risk Rank Assignments (4-1)

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Dest EDC_HMI.CFG.Oview.RankNum
1

HMI Maintenance Rank Assignments (A-D = Values 1-4)

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar>
-1
Dest EDC_HMI.CFG.Oview.RankChar
4

Initialize Zone/Rank Text for Overview

5

Oview Display Zone/Mask Initialization

JSR
Routine Name _818_InitOview

Configuration Response from Server via Transaction Manager

Data OK

Server Interface
Status State = Red-
HMI Oview Display Rx
Data OK
EDC_Status01.ServerCfg.Update.HmiOviewRxDataOK

Update Status State
for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

6 L

Exit Routine

lbl_Exit
7 LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 4 - Zone Display
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 834
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data
DHF034 June 2020 -- Remove Zone check, all values 0000-0000 thru 1111-1111 are legal
(when Zone H is checked, the value appears as a negative in the SINT signed integer)
Feb 2, 2023 - add Group Name

2

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
High Limit EDC_HMI.CFG.ZN.mi 20

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0]
<EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Maintenance Zone Mask (A-T)

GE
Source A ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
Source B 0

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

GE
Source A ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum> -1
Source B 0

Server Configuration from Transaction Manager Maintenance Rank Assignments (A-D = Values 1-4)

GE
Source A ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar> -1
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4016
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- HMI Zone Display Rx Data ER
EDC_Status01.ServerCfg.Update.HmiZoneRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig (...)
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store
Feb 2023 - added Group Name

4

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
Dest ZNi 20

HMI Zone Display Config

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_HMI.CFG.Zone[ZNi].Name
Length 1

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
Dest EDC_HMI.CFG.Zone[ZNi].ZoneMask 16#0000_0000

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum> -1
Dest EDC_HMI.CFG.Zone[ZNi].RankNum 1

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar> -1
Dest EDC_HMI.CFG.Zone[ZNi].RankChar 4

Data OK

5

Server Interface Status State = Red- HMI Zone Display Rx Data OK

EDC_Status01.ServerCfg.Update.HmiZoneRxDataOK
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Initialize Zone/Rank Text for Zone w/ ZNi

6

Zone Display Zone/Mask Initialization

JSR
Routine Name _819_InitZone

Exit Routine

7

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 5 - Equipment
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 835
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Convert EQID to LNi and EQi

2

Equipment Index

MOD
Source A ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1
Source B 100
Dest EQi 31

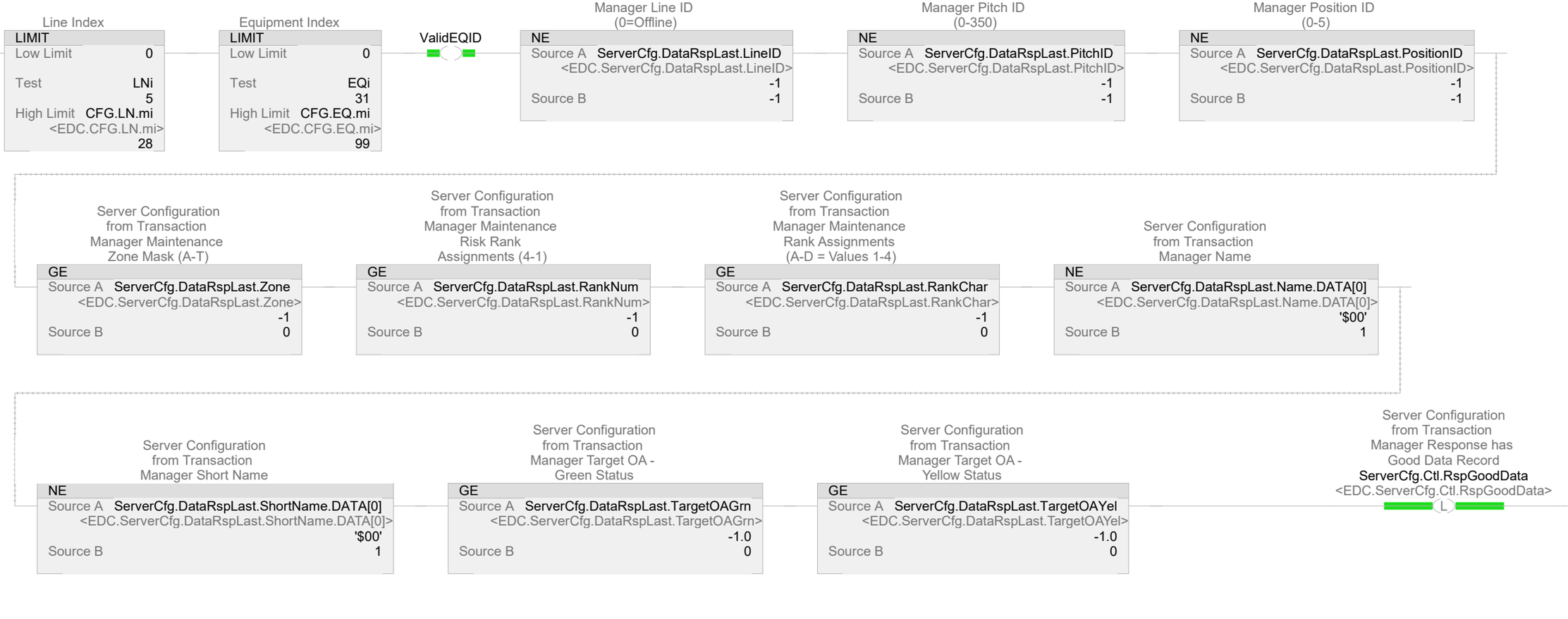
Line Index

CPT
Dest LNi 5
Expression (ServerCfg.DataRspLast.EqID - EQi) / 100

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

3



Configuration Response from Server via Transaction Manager

DN009 May 5, 2020 -- Acknowlege LNi and EQi for EQID 2900-9999
(Configuration allows EQID 1-9999 - values outside array limits are for decommissioned Equipment)

EDC2 - ack any EQID outside configured LNi, EQi
(unused EQID can be assigned to unconfigured LNi, EQi)

4

ValidEQID
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

lbl_DataOk
JMP

Data Error - Indexes out of Range
Log Transaction & Performance

5

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4017
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Rx Data ER
EDC_Status01.ServerCfg.Update.EquipRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

6

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Line 5

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Pitch 23

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Position 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.RankNum 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.RankChar 3

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.Line[LNi].Equip[EQi].CFG.Name
Length 1

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.Line[LNi].Equip[EQi].CFG.ShortName
Length 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAGrn
<EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Dest EDC.Line[LNi].Equip[EQi].CFG.TargetOAGrn 98.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAYel
<EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Dest EDC.Line[LNi].Equip[EQi].CFG.TargetOAYel 96.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Takt 56300

Convert Zone Number to BIt
Zone 1-20 to Bits 0-19
(allows easy compare to Zone Masks)

7

Equipment Data Collection Lines and Equipment Array

CLR
Dest EDC.Line[LNi].Equip[EQi].CFG.Zone
2#0000_0000_0000_0000_0000_0000_0100_0000

Server Configuration from Transaction Manager Maintenance Zone Mask (A-T)

LIMIT
Low Limit 1
Test ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
High Limit 20

SUB
Source A ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
Source B 1
Dest ZnBITi 2

Equipment Data Collection Lines and Equipment Array

EDC.Line[LNi].Equip[EQi].CFG.Zone.[ZnBITi]
(L)

Configuration Response from Server via Transaction Manager

DN006 - support multiple Offline Lines

8

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.Group
0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupDisplayRow
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayRow>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.DisplayRow
1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupDisplayCol
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayCol>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.DisplayCol
1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupLine
<EDC.ServerCfg.DataRspLast.OfflineGroupLine>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.Line
-1

Configuration Response from Server via Transaction Manager

v3.5 refresh Events if any config changes to Equipment
(don't wait for next Callout/Health update)

9

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.Update

L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.UpdateOview

L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom

Equipment Data Collection Lines and Equipment Array

FLL

Source	-1
Dest	EDC.Line[LNi].Equip[EQi].EventCTL.UpdateZone
Length	1

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager

DHF006 - New Configuration to support Disabled Equipment
Defined - exists in Configuration database
Disabled - temporarily or permanently out of commission - clear Callout table (this will clear Events)
CTL.DisableFaultMsg & CTL.DisableFaultMsgBulk sends Fault Messages for all Packets once to clear all Events in DB
CTL.DisableEvents refreshes Event Display once to clear any NoDisplay Events
v3.5 Add Data Refresh Request for re-Enabled Equipment

Bit 0 - Disable
Bit 1 - Disable Event Display
Bit 2 - Decommission

10

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Defined
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DisableFaultMsg
L

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].CTL.DisablePKTi
0

Server Configuration from Transaction Manager Command-specific Record ID for TransLog
MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4033
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DisableEvents
L

Callout Table (LNi, EQi)
FLL
Source 0
Dest EDC_EqCallout[LNi,EQi]
Length 1

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DataRefreshReq

Configuration Response from Server via Transaction Manager

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4032
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable
U

EDC2 - v2.01 added Equip.CFG.DisableEvents to replace PreDefined.15 from EqDriver v1.x
Bit 0 - Disable
Bit 1 - Disable Event Display
Bit 2 - Decommission

11

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DisableEvents
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents
U

Configuration Response from Server via Transaction Manager

v3.5 added Equip.CFG.Decom
Bit 0 - Disable
Bit 1 - Disable Event Display
Bit 2 - Decommission

12

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom
U

Check for good Event Looklup Index for EDC_Event EvLKi

13

Equipment-Event Index for EDC_Event Lookup
CLR
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array
LIMIT
Low Limit 0
Test EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
High Limit EDC_Event.CFG.EQ.mi 2999

Equipment-Event Index for EDC_Event Lookup
MOVE
Source EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
Dest EvLKi 164

Event Lookup Index within array limits
EvLKi_IndexOK

Equipment-Event Index for EDC_Event matches

14

Event Lookup Index EQID Match
EvLKi_EQIDMatch
U

Event Lookup Index within array limits
EvLKi_IndexOK

Event Controls Equipment Fragment Statistics Array
EQ
Source A EDC_Event.CTL.Equip[EvLKi].EQID 926
Source B ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Configuration Response from Server via Transaction Manager

Search for EQID Index if not found
Store result to avoid future searches

15

Event Lookup Index EQID Match: EvLKi_EQIDMatch

Fragment Statistics Equipment Search: EventEqFSC (RES)

Fragment Statistics Equipment Search: MOVE
Source 0
Dest EventEqFSC.POS 0

Fragment Statistics Equipment Search: MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEqFSC.LEN 3000

Fragment Statistics Equipment Search: FSC
Control EventEqFSC
Length 3000
Position 0
Mode ALL
Expression EDC_Event.CTL.Equip[EventEqFSC.POS].EQID=ServerCfg.DataRspLast.EqID
EN DN ER

Fragment Statistics Equipment Search: EventEqFSC.FD

Equipment-Event Index for EDC_Event Lookup: MOVE
Source EventEqFSC.POS 0
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array: MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Lookup Index EQID Match: EvLKi_EQIDMatch (L)

Event Lookup Index Found: EvLKi_Search

Search for Empty Index if new Equipment
Store result to avoid future searches

16

Event Lookup Index EQID Match: EvLKi_EQIDMatch

Fragment Statistics Empty Search: EventEmptyFSC (RES)

Fragment Statistics Empty Search: MOVE
Source 0
Dest EventEmptyFSC.POS 271

Fragment Statistics Empty Search: MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEmptyFSC.LEN 3000

Fragment Statistics Empty Search: FSC
Control EventEmptyFSC
Length 3000
Position 271
Mode ALL
Expression EDC_Event.CTL.Equip[EventEmptyFSC.POS].EQID = 0
EN DN ER

Fragment Statistics Empty Search: EventEmptyFSC.FD

Equipment-Event Index for EDC_Event Lookup: MOVE
Source EventEmptyFSC.POS 271
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array: MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Controls Equipment Fragment Statistics Array: MOVE
Source ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1
Dest EDC_Event.CTL.Equip[EvLKi].EQID 926

Event Controls Equipment with Event Text Qty: ADD
Source A EDC_Event.CTL.EqQty 272
Source B 1
Dest EDC_Event.CTL.EqQty 272

Event Lookup Index EQID Match: EvLKi_EQIDMatch (L)

Event Lookup Index New: EvLKi_New

Configuration Response from Server via Transaction Manager

New or Newly-Enabled EQID Data Refresh - Initialize Event Masks to DtOwner = Maintenance, DtType = Fault and Ignore = pass data
Initialize Mask Config words (new data for EqDriver)

*** Ignore Mask is inverted from Configuration -- when OFF, the Mask bit is 1 to pass data thru MEQ ***
v3.5 Add Data Refresh Request for re-Enabled Equipment, but only clear masks for New Equipment
(Disabled Equipment Events & Masks are updated so Events work correctly if re-Enabled)

17

Event Lookup Index New: EvLKi_New

Event Lookup Index EQID Match: EvLKi_EQIDMatch

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CTL.DataRefreshReq

Temp Result for Array Size

MUL
Source A: EDC_Event.CFG.PKT.n8
Source B: EDC_Event.CFG.DAT.n8
Dest: LEN 64

Event Lookup Index New: EvLKi_New

DT Owner = Maintenance Event Mask Equipment Mask Equip Array

FLL
Source: -1
Dest: EDC_EventMaskMaint.Equip[EvLKi].Packet[0]
Length: LEN

DT Type = Fault Event Mask Equipment Mask Equip Array

FLL
Source: -1
Dest: EDC_EventMaskFault.Equip[EvLKi].Packet[0]
Length: LEN

DT Type = Alarm Event Mask Equipment Mask Equip Array

FLL
Source: 0
Dest: EDC_EventMaskAlarm.Equip[EvLKi].Packet[0]
Length: LEN

DT Type = Alert Event Mask Equipment Mask Equip Array

FLL
Source: 0
Dest: EDC_EventMaskAlert.Equip[EvLKi].Packet[0]
Length: LEN

DT Type = Ignore Event Mask Equipment Mask Equip Array

FLL
Source: -1
Dest: EDC_EventMaskIgnore.Equip[EvLKi].Packet[0]
Length: LEN

No Display Event Mask Equipment Mask Equip Array

FLL
Source: 0
Dest: EDC_EventMaskNoDisp.Equip[EvLKi].Packet[0]
Length: LEN

Event Controls Equipment Fragment Statistics Array

MOVE
Source: -1
Dest: EDC_Event.CTL.Equip[EvLKi].MaintMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source: -1
Dest: EDC_Event.CTL.Equip[EvLKi].FaultMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source: -1
Dest: EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source: -1
Dest: EDC_Event.CTL.Equip[EvLKi].AlertMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source: -1
Dest: EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source: -1
Dest: EDC_Event.CTL.Equip[EvLKi].DataRefreshReq 2#0000_0000

Configuration Response from Server via Transaction Manager

Equipment Not Found and Statistics Array is Full
(shouldn't happen - requires Manual Intervention and Manual Reset of Error)

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

Log Transactions

Event Lookup Index EQID Match
EvLKi_EQIDMatch

18

```
MOVE
Source  ServerCfg.DataRspLast.EqEventID
        <EDC.ServerCfg.DataRspLast.EqEventID>
        -1
Dest    ServerCfg.Ctl.CommandID
        <EDC.ServerCfg.Ctl.CommandID>
        2317
```

```
JSR
Routine Name  _901_TransLogSBR
Input Par     4025
Input Par     ServerCfg.Ctl.ReqInProgressTMR.ACC
              <EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par     ServerCfg.DataRspLast.Command
              <EDC.ServerCfg.DataRspLast.Command>
Input Par     ServerCfg.Ctl.CommandID
              <EDC.ServerCfg.Ctl.CommandID>
Input Par     ServerCfg.Ctl.TransactionNumber
              <EDC.ServerCfg.Ctl.TransactionNumber>
```

Server Interface Status State = Red - Equip Event Fragmentation Statistics Array Full ER

Update Status State for Transactions

EDC_Status01.ServerCfg.Update.EventStatsArrayFull (L)

```
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS...  aoiEDC_StatusUpdateServerConfig
Status                   EDC_Status01.ServerCfg
```

Data OK
DN009 May 5, 2020 -- Ack data for any EQID 1-9999

Server Interface Status State = Red - Equip Rx Data OK

Update Status State for Transactions

lbl_DataOk
LBL

19

EDC_Status01.ServerCfg.Update.EquipRxDataOK (L)

```
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS...  aoiEDC_StatusUpdateServerConfig
Status                   EDC_Status01.ServerCfg
```

Exit Routine

lbl_Exit
LBL

20

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 16 - Equipment Event
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing
EDC2 - v2.01 added Event DtOwner and DtType Mask Configuration; removed Shared Andon/AVI Events

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 836
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Convert EventID into sub-indexes
EqEventID -- WRDi & BITi (Word and Bit Indexes for EqEvent)
WRDi -- PKTi & DATI (8 Packets with 8 Data for Masks)

(Jan 16, 2017 -- EQID removed from ServerCfg.Data.EqEventID and sent in separate ServerCfg.Data.EQID)
Combine EQID and EqEventID into EventID
EQID = -1 for PreDefined Faults

2

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest EQID
531

Event Mask Bit Index (0-31)

MOD
Source A ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Source B 100
Dest BITi
1

Event Word Index (0-64)

CPT
Dest WRDi
59
Expression (ServerCfg.DataRspLast.EqEventID - BITi) / 100

Event Mask Data Index (0-7)

MOD
Source A WRDi
59
Source B 8
Dest DATi
3

Event Mask Packet Index (0-7)

CPT
Dest PKTi
7
Expression (WRDi - DATi) / 8

Pre-Defined Event

3

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 16

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
High Limit 31

Line Index

MOVE
Source 0
Dest LNi
5

Equipment Index

MOVE
Source 0
Dest EQi
31

EQ EventID for Event Table storage (EEEEeeee)

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest EqEventID
9265901

PreDefined Event
EventTextPreDef

Configuration Response from Server via Transaction Manager

Convert EQID into sub-indexes
EQID = LNi & EQi

DN009 May 5, 2020 -- Acknowlege LNi and EQi for EQID 2900-9999
(Configuration allows EQID 1-9999 - values outside array limits are for decommissioned Equipment)

4

PreDefined Event
EventTextPreDef
/

Equipment Index
MOD
Source A EQID
531
Source B
100
Dest EQi
31

Line Index
CPT
Dest LNi
5
Expression (EQID - EQi) / 100

Line Index
LIMIT
Low Limit CFG.LN.si
<EDC.CFG.LN.si>
0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

Valid Line Index
ValidLNi

Valid Line Index
ValidLNi
/

Line Index
LIMIT
Low Limit CFG.LN.n
<EDC.CFG.LN.n>
29
Test LNi
5
High Limit
99

Decommissioned Line
DecomLNi

Equipment Index
LIMIT
Low Limit
0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

Valid Equipment Index
ValidEQi

Valid Line Index
ValidLNi
/

Line Index
MOVE
Source
0
Dest LNi
5

Configuration Response from Server via Transaction Manager

Equipment-Defined Event

5

PreDefined Event EventTextPreDef

Valid Line Index ValidLNi

Valid Equipment Index ValidEQi

EqDefined Event EventTextEqDef

Search Text Tag01 EventTextTag01 (U)

Search Text Tag02 EventTextTag02 (U)

Search Text Tag03 EventTextTag03 (U)

Search Text Tag04 EventTextTag04 (U)

Search Text Tag05 EventTextTag05 (U)

Search Text Tag06 EventTextTag06 (U)

Event Text Tag01 Search EventTextFSC01 (RES)

Event Text Tag02 Search EventTextFSC02 (RES)

Event Text Tag03 Search EventTextFSC03 (RES)

Event Text Tag04 Search EventTextFSC04 (RES)

Event Text Tag05 Search EventTextFSC05 (RES)

Event Text Tag06 Search EventTextFSC06 (RES)

EQ EventID for Event Table storage (EEEEeeee)

CPT

Dest	EqEventID 9265901
Expression	(EQID * 10000) + ServerCfg.DataRspLast.EqEventID

Configuration Response from Server via Transaction Manager

Good Data Packet
EDC2 - v2.01 added DtOwner (RankNum), DtType (RankChar), NoDisplay (Cor
May 2026 - added Summary Data KPI Option ID for EqDefined Events

6

PreDefined Event
EventTextPreDef

EqDefined Event
EventTextEqDef

EQ EventID for Event Table storage (EEEEeeee)

LIMIT
Low Limit 0
Test EqEventID
9265901
High Limit 99999999

Server Configuration from Transaction Manager Equipment Event Text

NE
Source A ServerCfg.DataRspLast.EqEventText.DATA[0]
<EDC.ServerCfg.DataRspLast.EqEventText.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

NE
Source A ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Source B -1

Server Configuration from Transaction Manager Maintenance Rank Assignments (A-D = Values 1-4)

NE
Source A ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar>
-1
Source B -1

Server Configuration from Transaction Manager Config Status Bits

NE
Source A ServerCfg.DataRspLast.ConfigStatus
<EDC.ServerCfg.DataRspLast.ConfigStatus>
-1
Source B -1

Server Configuration from Transaction Manager Offline Group Display Row

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.OfflineGroupDisplayRow
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayRow>
-1
High Limit 5

PreDefined Event
EventTextPreDef

0=Production
1=Maintenance

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Dest DtOwner
1

Server Configuration from Transaction Manager Equipment Event Text

EQ
Source A ServerCfg.DataRspLast.EqEventText.DATA[0]
<EDC.ServerCfg.DataRspLast.EqEventText.DATA[0]>
'$00'
Source B 1

Configuration Response from Server via Transaction Manager

gStatus)

Server Configuration
from Transaction
Manager Response has
Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

0=Not Used
1=Fault
2=Alarm
3=Alert
4=Info
5=Ignore

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar>
-1
Dest DtType
4

0 = not defined
1-5 = Option 1-5

MOVE
Source ServerCfg.DataRspLast.OfflineGroupDisplayRow
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayRow>
-1
Dest KpiOptionID
0

MOVE
Source ServerCfg.DataRspLast.ConfigStatus
<EDC.ServerCfg.DataRspLast.ConfigStatus>
-1
Dest NoDisplay
0

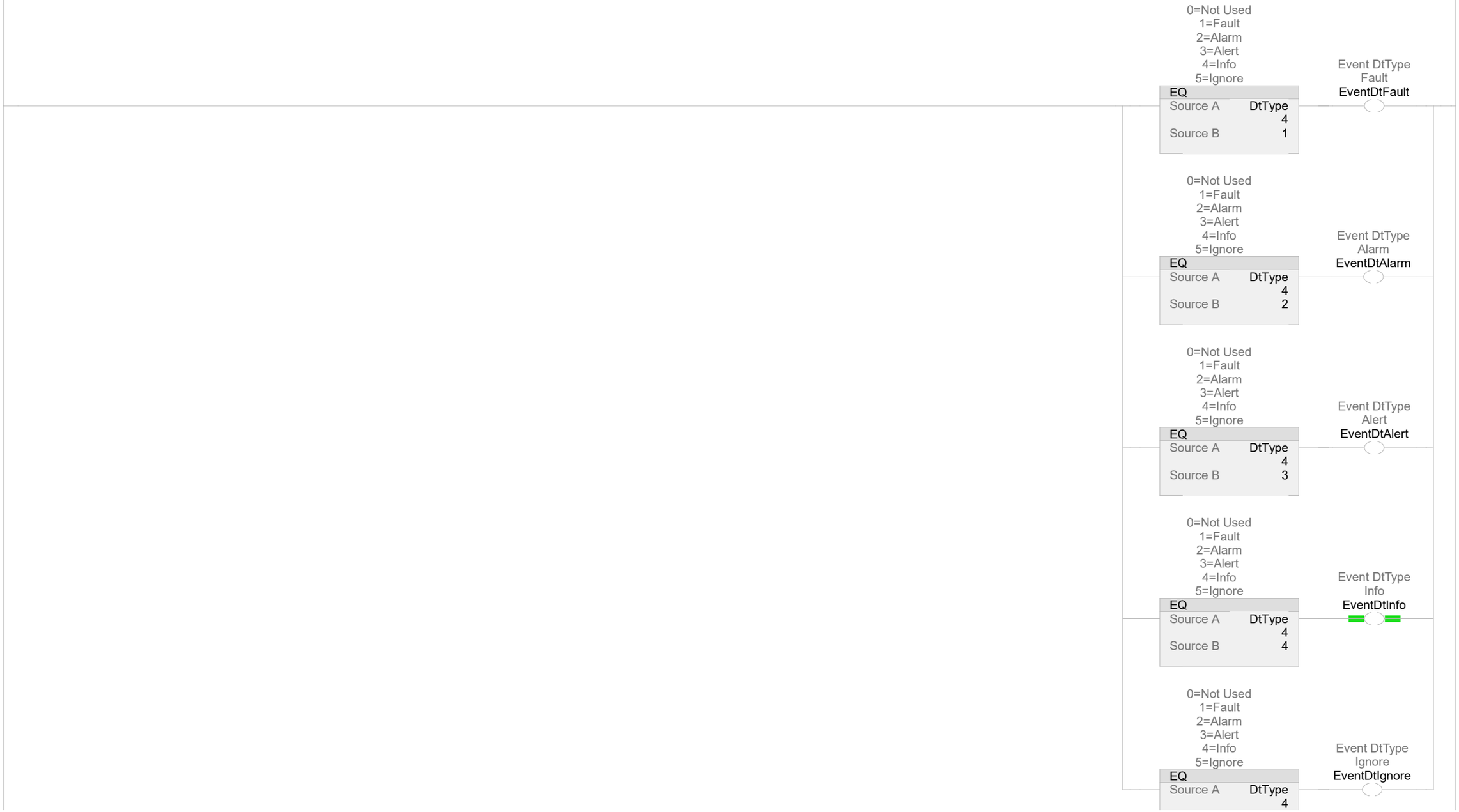
Fault Log

COP
Source ServerCfg.DataRspLast
<EDC.ServerCfg.DataRspLast>
Dest LogServerCfg_DataRespLast
Length 1

Configuration Response from Server via Transaction Manager

EDC2 - v2.01 Downtime Types

7



Configuration Response from Server via Transaction Manager

Source B 5

EDC2 - v2.01 Downtime Owner and No Display Option

8

0=Production
1=Maintenance
EQ
Source A DtOwner
1
Source B 1

Event DtOwner
Maintenance
EventDtMaint

EQ
Source A NoDisplay
0
Source B 1

Event NoDisplay
EventNoDisplay

If DtType = Info, Ignore or set for NoDisplay, don't save the Event Text - just update the Masks

9

PreDefined
Event
EventTextPreDef
/

Event DtType
Info
EventDtInfo

Event DtType
Ignore
EventDtIgnore

Event NoDisplay
EventNoDisplay

lbl_DataOk
JMP

Configuration Response from Server via Transaction Manager

DN009 May 5, 2020 -- Acknowlege LNi and EQi for EQID 2900-9999
(Configuration allows EQID 1-9999 - values outside array limits are for decommissioned Equipment)

(this shouldn't happen - Server should timestamp these complete without sending)

10

Decommissioned Line
DecomLNi

Valid Line Index
ValidLNi
/

Valid Equipment Index
ValidEQi

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Decommissioned Equipment
DecomEQi

lbl_DataOk
JMP

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

11

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	LNi
	5
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	EQi
	31
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	EqEventID
	9265901
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE	
Source	ServerCfg.DataRspLast.EqEventID
	<EDC.ServerCfg.DataRspLast.EqEventID>
	-1
Dest	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
	2317

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4023
Input Par	ServerCfg.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerCfg.DataRspLast.Command
	<EDC.ServerCfg.DataRspLast.Command>
Input Par	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
Input Par	ServerCfg.Ctl.TransactionNumber
	<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Event Rx Data ER
EDC_Status01.ServerCfg.Update.EquipEventRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig	
aoiEDC_StatusUpdateS...	aoiEDC_StatusUpdateServerConfig ...
Status	EDC_Status01.ServerCfg

lbl_Exit
JMP

Get Event Text Start/End Index for this Equipment

12

Event Text Start Index

MOVE	
Source	EDC.Line[LNi].Equip[EQi].EventCTL.TXTsi
	23544
Dest	TXTsi
	28256

Event Text End Index

MOVE	
Source	EDC.Line[LNi].Equip[EQi].EventCTL.TXTei
	23779
Dest	TXTei
	28387

Configuration Response from Server via Transaction Manager

Tag01
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

13

EqDefined Event: EventTextEqDef

Event Text Start Index

LIMIT	
Low Limit	EDC_HMI.CFG.EvTxt01.si 0
Test	TXTsi 28256
High Limit	EDC_HMI.CFG.EvTxt01.ei 32499

Search Text Tag01: EventTextTag01 (L)

Event Text Tag01 Search

MOVE	
Source	TXTsi 28256
Dest	EventTextFSC01.POS 0

Event Text Tag01 Search

MOVE	
Source	EDC_HMI.CFG.EvTxt01.n 32500
Dest	EventTextFSC01.LEN 28388

Event Text End Index

LIMIT	
Low Limit	EDC_HMI.CFG.EvTxt01.si 0
Test	TXTei 28387
High Limit	EDC_HMI.CFG.EvTxt01.ei 32499

Event Text Tag01 Search

ADD	
Source A	TXTei 28387
Source B	1
Dest	EventTextFSC01.LEN 28388

Tag01 Event Text Search

14

EqDefined Event: EventTextEqDef

Search Text Tag01: EventTextTag01

Event Text Tag01 Search

FSC	
Control	EventTextFSC01
Length	28388
Position	0
Mode	ALL
Expression	EDC_EventTxt01[EventTextFSC01.POS].EqEventID=EqEventID

EN DN ER

Configuration Response from Server via Transaction Manager

15

Tag02
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD

LIMIT (Event Text Start Index)
- Low Limit: EDC_HMI.CFG.EvTxt01.si 0
- Test: TXTsi 28256
- High Limit: EDC_HMI.CFG.EvTxt02.ei 64999

GT (Event Text End Index)
- Source A: TXTei 28387
- Source B: EDC_HMI.CFG.EvTxt02.si 32500

Search Text Tag02: EventTextTag02 (L)

SUB (Event Text Tag02 Search)
- Source A: TXTsi 28256
- Source B: EDC_HMI.CFG.EvTxt02.si 32500
- Dest: EventTextFSC02.POS 0

LT (Event Text Tag02 Search)
- Source A: EventTextFSC02.POS 0
- Source B: 0

MOVE (Event Text Tag02 Search)
- Source: 0
- Dest: EventTextFSC02.POS 0

MOVE (Event Text Tag02 Search)
- Source: EDC_HMI.CFG.EvTxt02.n 32500
- Dest: EventTextFSC02.LEN 4562

LIMIT (Event Text End Index)
- Low Limit: EDC_HMI.CFG.EvTxt02.si 32500
- Test: TXTei 28387
- High Limit: EDC_HMI.CFG.EvTxt02.ei 64999

CPT (Event Text Tag02 Search)
- Dest: EventTextFSC02.LEN 4562
- Expression: TXTei-EDC_HMI.CFG.EvTxt02.si+1

16

Tag02 Event Text Search

EqDefined Event: EventTextEqDef
Search Text Tag02: EventTextTag02

FSC (Event Text Tag02 Search)
- Control: EventTextFSC02
- Length: 4562
- Position: 0
- Mode: ALL
- Expression: EDC_EventTxt02[EventTextFSC02.POS].EqEventID=EqEventID

EN DN ER

Configuration Response from Server via Transaction Manager

Tag03
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

17

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD

Event Text Start Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt01.si 0
Test: TXTsi 28256
High Limit: EDC_HMI.CFG.EvTxt03.ei 97499

Event Text End Index
GT
Source A: TXTei 28387
Source B: EDC_HMI.CFG.EvTxt03.si 65000

Search Text Tag03: EventTextTag03 (L)

Event Text Tag03 Search
SUB
Source A: TXTsi 28256
Source B: EDC_HMI.CFG.EvTxt03.si 65000
Dest: EventTextFSC03.POS 0

Event Text Tag03 Search
LT
Source A: EventTextFSC03.POS 0
Source B: 0

Event Text Tag03 Search
MOVE
Source: 0
Dest: EventTextFSC03.POS 0

Event Text Tag03 Search
MOVE
Source: EDC_HMI.CFG.EvTxt03.n 32500
Dest: EventTextFSC03.LEN 1390

Event Text End Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt03.si 65000
Test: TXTei 28387
High Limit: EDC_HMI.CFG.EvTxt03.ei 97499

Event Text Tag03 Search
CPT
Dest: EventTextFSC03.LEN 1390
Expression: TXTei-EDC_HMI.CFG.EvTxt03.si+1

Tag03 Event Text Search

18

EqDefined Event: EventTextEqDef
Search Text Tag03: EventTextTag03

Event Text Tag03 Search
FSC
Control: EventTextFSC03
Length: 1390
Position: 0
Mode: ALL
Expression: EDC_EventTxt03[EventTextFSC03.POS].EqEventID=EqEventID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Tag04
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

19

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD
Event Text Tag03 Search: EventTextFSC03.FD

Event Text Start Index — LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt01.si 0
- Test: TXTsi 28256
- High Limit: EDC_HMI.CFG.EvTxt04.ei 129999

Event Text End Index — GT
- Source A: TXTei 28387
- Source B: EDC_HMI.CFG.EvTxt04.si 97500

Search Text Tag04: EventTextTag04 (L)

Event Text Tag04 Search — SUB
- Source A: TXTsi 28256
- Source B: EDC_HMI.CFG.EvTxt04.si 97500
- Dest: EventTextFSC04.POS 0

Event Text Tag04 Search — LT
- Source A: EventTextFSC04.POS 0
- Source B: 0

Event Text Tag04 Search — MOVE
- Source: 0
- Dest: EventTextFSC04.POS 0

Event Text Tag04 Search — MOVE
- Source: EDC_HMI.CFG.EvTxt04.n 32500
- Dest: EventTextFSC04.LEN 555

Event Text End Index — LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt04.si 97500
- Test: TXTei 28387
- High Limit: EDC_HMI.CFG.EvTxt04.ei 129999

Event Text Tag04 Search — CPT
- Dest: EventTextFSC04.LEN 555
- Expression: TXTei-EDC_HMI.CFG.EvTxt04.si+1

Tag04 Event Text Search

20

EqDefined Event: EventTextEqDef
Search Text Tag04: EventTextTag04

Event Text Tag04 Search — FSC
- Control: EventTextFSC04
- Length: 555
- Position: 0
- Mode: ALL
- Expression: EDC_EventTxt04[EventTextFSC04.POS].EqEventID=EqEventID

(EN) (DN) (ER)

Configuration Response from Server via Transaction Manager

Tag05
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

21

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD
Event Text Tag03 Search: EventTextFSC03.FD
Event Text Tag04 Search: EventTextFSC04.FD

Event Text Start Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt01.si 0
Test: TXTsi 28256
High Limit: EDC_HMI.CFG.EvTxt05.ei 162499

Event Text End Index
GT
Source A: TXTei 28387
Source B: EDC_HMI.CFG.EvTxt05.si 130000

Search Text Tag05: EventTextTag05 (L)

Event Text Tag05 Search
SUB
Source A: TXTsi 28256
Source B: EDC_HMI.CFG.EvTxt05.si 130000
Dest: EventTextFSC05.POS 0

Event Text Tag05 Search
LT
Source A: EventTextFSC05.POS 0
Source B: 0

Event Text Tag05 Search
MOVE
Source: 0
Dest: EventTextFSC05.POS 0

Event Text Tag05 Search
MOVE
Source: EDC_HMI.CFG.EvTxt05.n 32500
Dest: EventTextFSC05.LEN 555

Event Text End Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt05.si 130000
Test: TXTei 28387
High Limit: EDC_HMI.CFG.EvTxt05.ei 162499

Event Text Tag05 Search
CPT
Dest: EventTextFSC05.LEN 555
Expression: TXTei-EDC_HMI.CFG.EvTxt05.si+1

Tag05 Event Text Search

22

EqDefined Event: EventTextEqDef
Search Text Tag05: EventTextTag05

Event Text Tag05 Search
FSC
Control: EventTextFSC05
Length: 555
Position: 0
Mode: ALL
Expression: EDC_EventTxt05[EventTextFSC05.POS].EqEventID=EqEventID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Tag06
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

23

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD
Event Text Tag03 Search: EventTextFSC03.FD
Event Text Tag04 Search: EventTextFSC04.FD
Event Text Tag05 Search: EventTextFSC05.FD

Event Text Start Index — LIMIT
Low Limit	EDC_HMI.CFG.EvTxt01.si 0
Test	TXTsi 28256
High Limit	EDC_HMI.CFG.EvTxt06.ei 194999

Event Text End Index — GT
Source A	TXTei 28387
Source B	EDC_HMI.CFG.EvTxt06.si 162500

Search Text Tag06 EventTextTag06 (L)

Event Text Tag06 Search — SUB
Source A	TXTsi 28256
Source B	EDC_HMI.CFG.EvTxt06.si 162500
Dest	EventTextFSC06.POS 0

Event Text Tag06 Search — LT
Source A	EventTextFSC06.POS 0
Source B	0

Event Text Tag06 Search — MOVE
Source	0
Dest	EventTextFSC06.POS 0

Event Text Tag06 Search — MOVE
Source	EDC_HMI.CFG.EvTxt06.n 32500
Dest	EventTextFSC06.LEN 555

Event Text End Index — LIMIT
Low Limit	EDC_HMI.CFG.EvTxt06.si 162500
Test	TXTei 28387
High Limit	EDC_HMI.CFG.EvTxt06.ei 194999

Event Text Tag06 Search — CPT
Dest	EventTextFSC06.LEN 555
Expression	TXTei-EDC_HMI.CFG.EvTxt06.si+1

Tag06 Event Text Search

24

EqDefined Event: EventTextEqDef
Search Text Tag06: EventTextTag06

Event Text Tag06 Search — FSC
Control	EventTextFSC06
Length	555
Position	0
Mode	ALL
Expression	EDC_EventTxt06[EventTextFSC06.POS].EqEventID=EqEventID

EN DN ER

Configuration Response from Server via Transaction Manager

Tag01
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

25

EqDefined Event EventTextEqDef
Event Text Tag01 Empty Search EventTextEmptyFSC01 RES
Event Text Tag01 Search EventTextFSC01.FD /
Event Text Tag02 Search EventTextFSC02.FD /
Event Text Tag03 Search EventTextFSC03.FD /
Event Text Tag04 Search EventTextFSC04.FD /
Event Text Tag05 Search EventTextFSC05.FD /
Event Text Tag06 Search EventTextFSC06.FD /

LT
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt01.ei 32499

Event Text Tag01 Empty Search
MOVE
Source EventEmptyLastFD 45063
Dest EventTextEmptyFSC01.POS 0

Event Text Tag01 Empty Search
MOVE
Source EDC_HMI.CFG.EvTxt01.n 32500
Dest EventTextEmptyFSC01.LEN 32500

Event Text Tag01 Empty Search
LT
Source A EventTextEmptyFSC01.POS 0
Source B 32

Event Text Tag01 Empty Search
MOVE
Source 32
Dest EventTextEmptyFSC01.POS 0

Event Text Tag01 Empty Search
FSC
Control EventTextEmptyFSC01
Length 32500
Position 0
Mode ALL
Expression EDC_EventTxt01[EventTextEmptyFSC01.POS].EqEventID = 0
EN
DN
ER

Event Text Tag01 Empty Search EventTextEmptyFSC01.FD

Event Text Tag01 Empty Search
MOVE
Source EventTextEmptyFSC01.POS 0
Dest EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

Tag02
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

26

EqDefined Event EventTextEqDef
Event Text Tag02 Empty Search EventTextEmptyFSC02 (RES)
Event Text Tag01 Search EventTextFSC01.FD (/)
Event Text Tag02 Search EventTextFSC02.FD (/)
Event Text Tag03 Search EventTextFSC03.FD (/)
Event Text Tag04 Search EventTextFSC04.FD (/)
Event Text Tag05 Search EventTextFSC05.FD (/)
Event Text Tag06 Search EventTextFSC06.FD (/)
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD (/)

LT
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt02.ei 64999

Event Text Tag02 Empty Search
SUB
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt02.si 32500
Dest EventTextEmptyFSC02.POS 0

Event Text Tag02 Empty Search
MOVE
Source EDC_HMI.CFG.EvTxt02.n 32500
Dest EventTextEmptyFSC02.LEN 32500

Event Text Tag02 Empty Search
LT
Source A EventTextEmptyFSC02.POS 0
Source B 0

Event Text Tag02 Empty Search
MOVE
Source 0
Dest EventTextEmptyFSC02.POS 0

Event Text Tag02 Empty Search
FSC
Control EventTextEmptyFSC02
Length 32500
Position 0
Mode ALL
Expression EDC_EventTxt02[EventTextEmptyFSC02.POS].EqEventID = 0
EN
DN
ER

Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

ADD
Source A EventTextEmptyFSC02.POS 0
Source B EDC_HMI.CFG.EvTxt02.si 32500
Dest EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

Tag03
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

27

EqDefined Event EventTextEqDef
Event Text Tag03 Empty Search EventTextEmptyFSC03 (RES)
Event Text Tag01 Search EventTextFSC01.FD (/)
Event Text Tag02 Search EventTextFSC02.FD (/)
Event Text Tag03 Search EventTextFSC03.FD (/)
Event Text Tag04 Search EventTextFSC04.FD (/)
Event Text Tag05 Search EventTextFSC05.FD (/)
Event Text Tag06 Search EventTextFSC06.FD (/)
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD (/)
Event Text Tag02 Empty Search EventTextEmptyFSC02.FD (/)

LT
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt03.ei 97499

Event Text Tag03 Empty Search
SUB
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt03.si 65000
Dest: EventTextEmptyFSC03.POS 0

Event Text Tag03 Empty Search
MOVE
Source: EDC_HMI.CFG.EvTxt03.n 32500
Dest: EventTextEmptyFSC03.LEN 25000

Event Text Tag03 Empty Search
LT
Source A: EventTextEmptyFSC03.POS 0
Source B: 0

Event Text Tag03 Empty Search
MOVE
Source: 0
Dest: EventTextEmptyFSC03.POS 0

Event Text Tag03 Empty Search
FSC
Control: EventTextEmptyFSC03
Length: 25000
Position: 0
Mode: ALL
Expression: EDC_EventTxt03[EventTextEmptyFSC03.POS].EqEventID = 0
EN DN ER

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD

ADD
Source A: EventTextEmptyFSC03.POS 0
Source B: EDC_HMI.CFG.EvTxt03.si 65000
Dest: EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

28

Tag04
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

EqDefined Event EventTextEqDef
Event Text Tag04 Empty Search EventTextEmptyFSC04 (RES)
Event Text Tag01 Search EventTextFSC01.FD
Event Text Tag02 Search EventTextFSC02.FD
Event Text Tag03 Search EventTextFSC03.FD
Event Text Tag04 Search EventTextFSC04.FD
Event Text Tag05 Search EventTextFSC05.FD
Event Text Tag06 Search EventTextFSC06.FD
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD
Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD

LT
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt04.ei 129999

Event Text Tag04 Empty Search

SUB
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt04.si 97500
Dest: EventTextEmptyFSC04.POS 0

Event Text Tag04 Empty Search

MOVE
Source: EDC_HMI.CFG.EvTxt04.n 32500
Dest: EventTextEmptyFSC04.LEN 25000

Event Text Tag04 Empty Search

LT
Source A: EventTextEmptyFSC04.POS 0
Source B: 0

Event Text Tag04 Empty Search

MOVE
Source: 0
Dest: EventTextEmptyFSC04.POS 0

Event Text Tag04 Empty Search

FSC
Control: EventTextEmptyFSC04
Length: 25000
Position: 0
Mode: ALL
Expression: EDC_EventTxt04[EventTextEmptyFSC04.POS].EqEventID=0
(EN) (DN) (ER)

Event Text Tag04 Empty Search EventTextEmptyFSC04.FD

ADD
Source A: EventTextEmptyFSC04.POS 0
Source B: EDC_HMI.CFG.EvTxt04.si 97500
Dest: EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

29

Tag05
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

EqDefined Event EventTextEqDef
Event Text Tag05 Empty Search EventTextEmptyFSC05 (RES)
Event Text Tag01 Search EventTextFSC01.FD
Event Text Tag02 Search EventTextFSC02.FD
Event Text Tag03 Search EventTextFSC03.FD
Event Text Tag04 Search EventTextFSC04.FD
Event Text Tag05 Search EventTextFSC05.FD
Event Text Tag06 Search EventTextFSC06.FD
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD
Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD
Event Text Tag04 Empty Search EventTextEmptyFSC04.FD

LT
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt05.ei 162499

Event Text Tag05 Empty Search
SUB
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt05.si 130000
Dest: EventTextEmptyFSC05.POS 0

Event Text Tag05 Empty Search
MOVE
Source: EDC_HMI.CFG.EvTxt05.n 32500
Dest: EventTextEmptyFSC05.LEN 25000

Event Text Tag05 Empty Search
LT
Source A: EventTextEmptyFSC05.POS 0
Source B: 0

Event Text Tag05 Empty Search
MOVE
Source: 0
Dest: EventTextEmptyFSC05.POS 0

Event Text Tag05 Empty Search
FSC
Control: EventTextEmptyFSC05
Length: 25000
Position: 0
Mode: ALL
Expression: EDC_EventTxt05[EventTextEmptyFSC05.POS].EqEventID=0
EN DN ER

Event Text Tag05 Empty Search EventTextEmptyFSC05.FD

ADD
Source A: EventTextEmptyFSC05.POS 0
Source B: EDC_HMI.CFG.EvTxt05.si 130000
Dest: EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

Tag06
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

30

EqDefined Event EventTextEqDef | Event Text Tag06 Empty Search EventTextEmptyFSC06 (RES) | Event Text Tag01 Search EventTextFSC01.FD | Event Text Tag02 Search EventTextFSC02.FD | Event Text Tag03 Search EventTextFSC03.FD | Event Text Tag04 Search EventTextFSC04.FD | Event Text Tag05 Search EventTextFSC05.FD | Event Text Tag06 Search EventTextFSC06.FD | Event Text Tag01 Empty Search EventTextEmptyFSC01.FD | Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD | Event Text Tag04 Empty Search EventTextEmptyFSC04.FD | Event Text Tag05 Empty Search EventTextEmptyFSC05.FD

LT
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt06.ei 194999

Event Text Tag06 Empty Search
SUB
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt06.si 162500
Dest EventTextEmptyFSC06.POS 0

Event Text Tag06 Empty Search
MOVE
Source EDC_HMI.CFG.EvTxt06.n 32500
Dest EventTextEmptyFSC06.LEN 25000

Event Text Tag06 Empty Search
LT
Source A EventTextEmptyFSC06.POS 0
Source B 0

Event Text Tag06 Empty Search
MOVE
Source 0
Dest EventTextEmptyFSC06.POS 0

Event Text Tag06 Empty Search
FSC
Control EventTextEmptyFSC06
Length 25000
Position 0
Mode ALL
Expression EDC_EventTxt06[EventTextEmptyFSC06.POS].EqEventID=0
EN DN ER

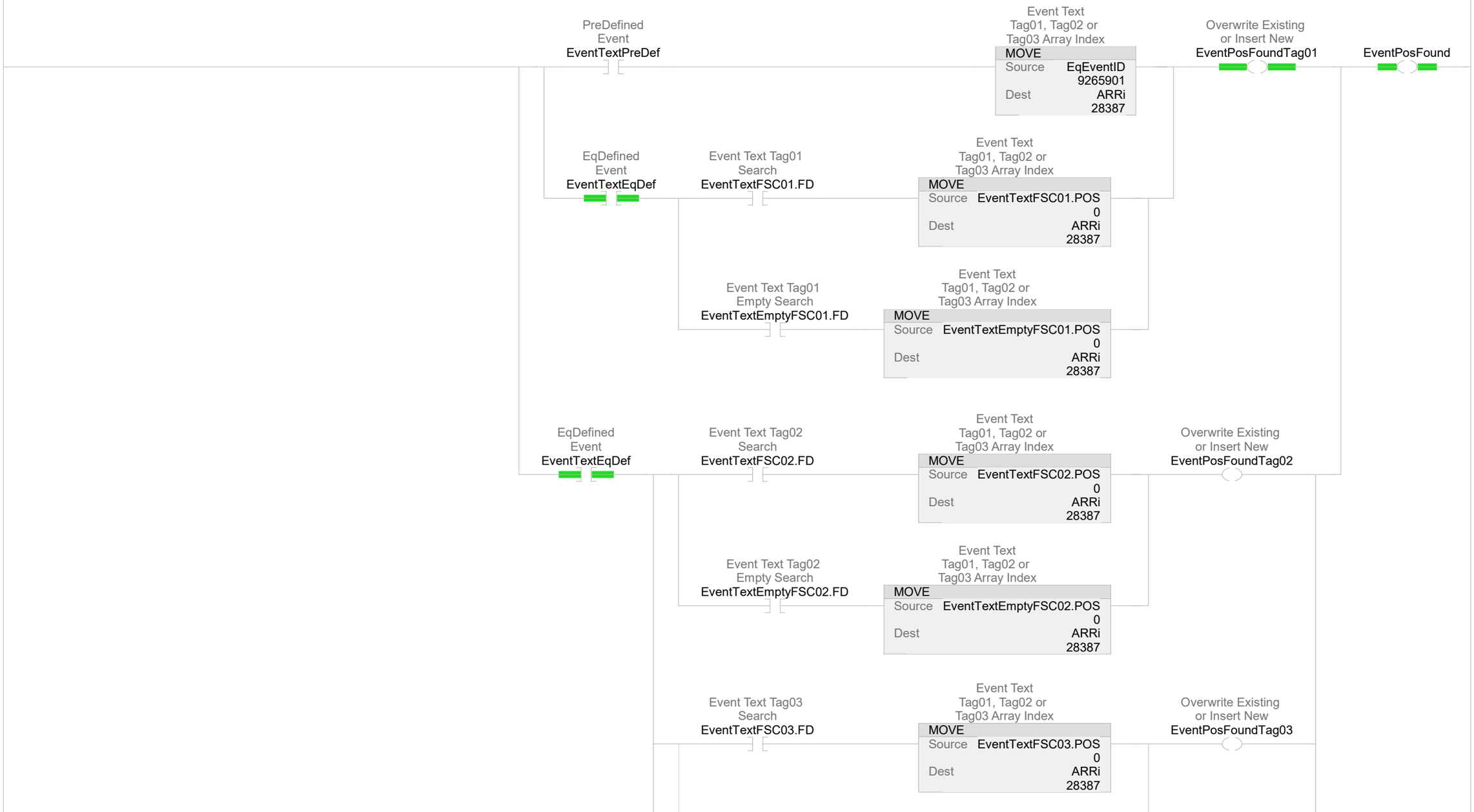
Event Text Tag06 Empty Search EventTextEmptyFSC06.FD

ADD
Source A EventTextEmptyFSC06.POS 0
Source B EDC_HMI.CFG.EvTxt06.si 162500
Dest EventEmptyLastFD 45063

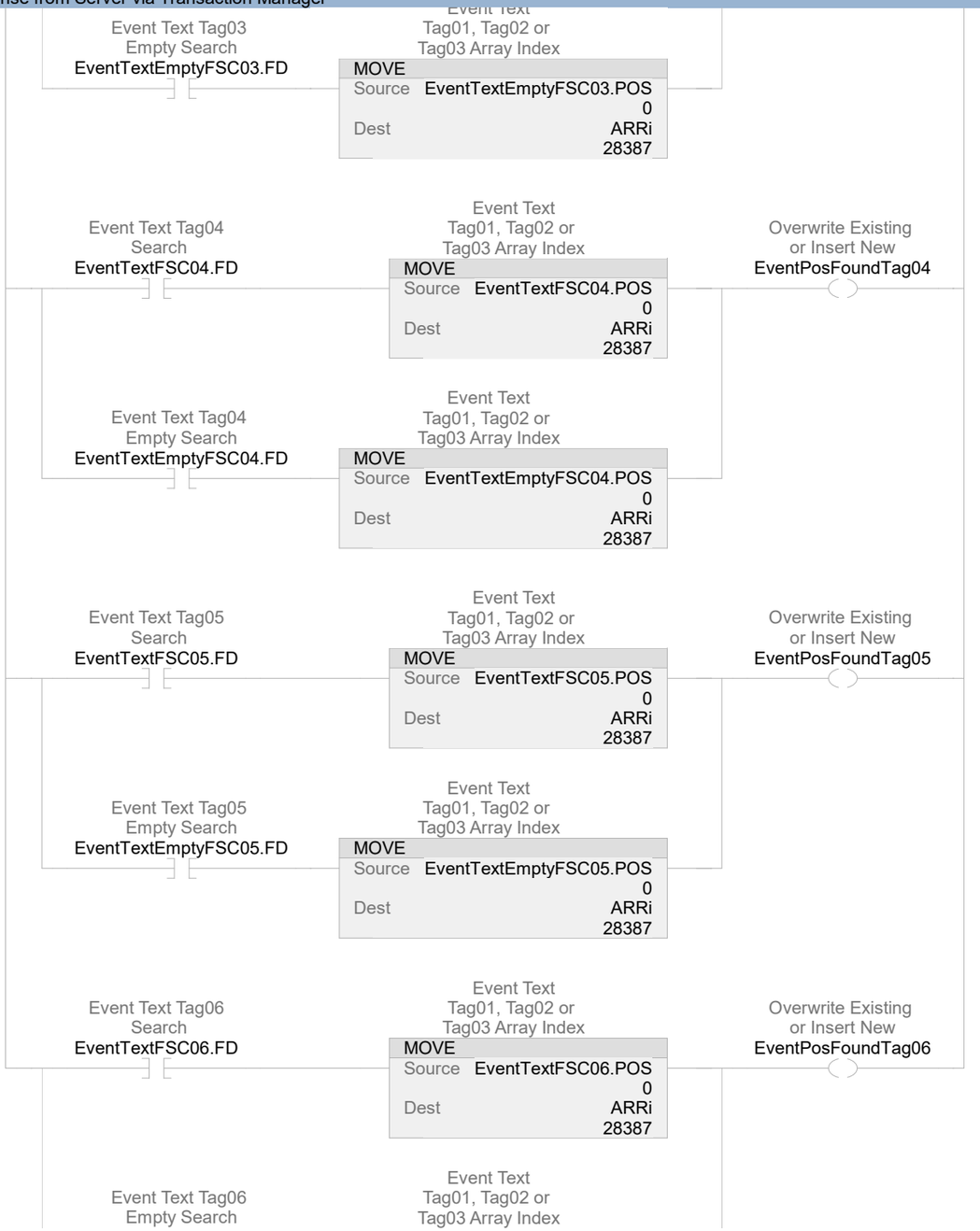
Configuration Response from Server via Transaction Manager

Load Event Text array Index

31



Configuration Response from Server via Transaction Manager



Configuration Response from Server via Transaction Manager

EventTextEmptyFSC06.FD

MOVE
Source EventTextEmptyFSC06.POS
0
Dest ARRi
28387

Equipment Event Text Array Full
(shouldn't happen - requires Manual Intervention and Manual Reset of Error)

32

Overwrite Existing or Insert New EventPosFoundTag01 /
Overwrite Existing or Insert New EventPosFoundTag02 /
Overwrite Existing or Insert New EventPosFoundTag03 /
Overwrite Existing or Insert New EventPosFoundTag04 /
Overwrite Existing or Insert New EventPosFoundTag05 /
Overwrite Existing or Insert New EventPosFoundTag06 /

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4024
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Event Array Full ER

EDC_Status01.ServerCfg.Update.EquipEventArrayFull
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store
Log Transaction

33

Overwrite Existing or Insert New EventPosFoundTag01

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt01[ARRi].EqEventID 9262317

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt01[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Overwrite Existing or Insert New EventPosFoundTag02

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt02[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt02[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Overwrite Existing or Insert New EventPosFoundTag03

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt03[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt03[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Overwrite Existing or Insert New EventPosFoundTag04

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt04[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt04[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

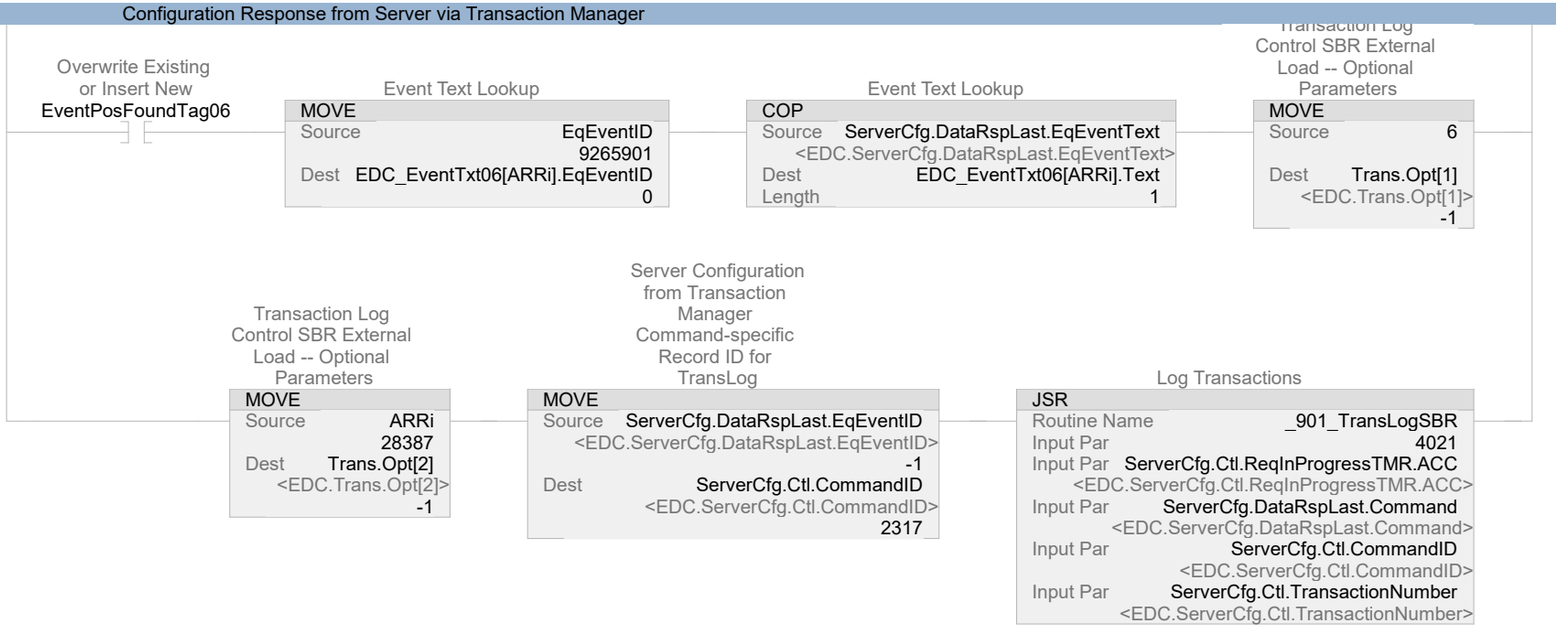
Overwrite Existing or Insert New EventPosFoundTag05

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt05[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt05[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 5
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Configuration Response from Server via Transaction Manager



Configuration Response from Server via Transaction Manager

Update Equipment-Defined array search limits for EQID existing or new Event

34

EqDefined Event EventTextEqDef

Event Text Tag01 Search EventTextFSC01.FD

Event Text Tag01 Empty Search EventTextEmptyFSC01.FD

MOVE
Source ARRi 28387
Dest TXTi 28387

Event Text Tag02 Search EventTextFSC02.FD

Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt02.si 32500
Dest TXTi 28387

Event Text Tag03 Search EventTextFSC03.FD

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt03.si 65000
Dest TXTi 28387

Event Text Tag04 Search EventTextFSC04.FD

Event Text Tag04 Empty Search EventTextEmptyFSC04.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt04.si 97500
Dest TXTi 28387

Event Text Tag05 Search EventTextFSC05.FD

Event Text Tag05 Empty Search EventTextEmptyFSC05.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt05.si 130000
Dest TXTi 28387

Event Text Tag06 Search EventTextFSC06.FD

Event Text Tag06 Empty Search

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt06.si

LT
Source A TXTi 28387
Source B TXTsi 28256

Equipment Data Collection Lines and Equipment Array

MOVE
Source TXTi 28387
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTsi 23544

Event Text Start Index

EQ
Source A TXTsi 28256
Source B 0

GT
Source A TXTi 28387
Source B TXTei 28387

Equipment Data Collection Lines and Equipment Array

MOVE
Source TXTi 28387
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTei 23779

Configuration Response from Server via Transaction Manager

EventTextEmptyFSC06.FD

Dest 162500 TXTi 28387

Data OK
DN009 May 5, 2020 -- Ack data for any EQID 1-9999

35

lbl_DataOk
LBL

Server Interface Status State = Red - Equip Event Rx Data OK
EDC_Status01.ServerCfg.Update.EquipEventRxDataOK
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Update Event Text Fragment Statistics
(for New Event Text only)

36

EqDefined Event
EventTextEqDef

EventPosFound

Event Text Tag01 Empty Search
EventTextEmptyFSC01.FD

Event DtType Ignore
EventDtIgnore

Event DtType Info
EventDtInfo

Event NoDisplay
EventNoDisplay

Update Event Text Statistics
UpdateEventTextStat

Event Text Tag02 Empty Search
EventTextEmptyFSC02.FD

Event Text Tag03 Empty Search
EventTextEmptyFSC03.FD

Event Text Tag04 Empty Search
EventTextEmptyFSC04.FD

Event Text Tag05 Empty Search
EventTextEmptyFSC05.FD

Event Text Tag06 Empty Search
EventTextEmptyFSC06.FD

Configuration Response from Server via Transaction Manager

Update Event Controls
Event Statistics and Masks

37

Decommissioned Equipment
DecomEQi
/

EqDefined Event
EventTextEqDef

Configuration Response from Server via Transaction Manager
JSR
Routine Name _837_CfgResponseEquipmentEventCtlEq

EqDefined Event
EventTextEqDef
/

Configuration Response from Server via Transaction Manager
JSR
Routine Name _837_CfgResponseEquipmentEventCtlPre

Exit Routine

38

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 17 - Equipment-Defined Event Text
Update Event Text Fragment Statistics and Masks
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 837
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check for good Event Looklup Index for EDC_Event EvLKi

2

Equipment-Event Index for EDC_Event Lookup

CLR
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array

LIMIT
Low Limit 0
Test EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
High Limit EDC_Event.CFG.EQ.mi 2999

Equipment-Event Index for EDC_Event Lookup

MOVE
Source EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
Dest EvLKi 164

Event Lookup Index within array limits
EvLKi_IndexOK

Equipment-Event Index for EDC_Event matches

3

Event Lookup Index EQID Match
EvLKi_EQIDMatch
U

Event Lookup Index within array limits
EvLKi_IndexOK

Event Controls Equipment Fragment Statistics Array

EQ
Source A EDC_Event.CTL.Equip[EvLKi].EQID 926
Source B EQID 531

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Configuration Response from Server via Transaction Manager

Search for EQID Index if not found
Store result to avoid future searches

(redundant to CfgResponseEquipment - should always exist from Equipment Download)

4

Event Lookup Index EQID Match
EvLKi_EQIDMatch
/

Diagnostic
ADD
Source A EventEqFscCount 16
Source B 1
Dest EventEqFscCount 16

Fragment Statistics Equipment Search
EventEqFSC
RES

Fragment Statistics Equipment Search
MOVE
Source 0
Dest EventEqFSC.POS 0

Fragment Statistics Equipment Search
MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEqFSC.LEN 3000

Fragment Statistics Equipment Search
FSC
Control EventEqFSC
Length 3000
Position 0
Mode ALL
Expression EDC_Event.CTL.Equip[EventEqFSC.POS].EQID=EQID
EN
DN
ER

Fragment Statistics Equipment Search
EventEqFSC.FD

Equipment-Event Index for EDC_Event Lookup
MOVE
Source EventEqFSC.POS 0
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array
MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Event Lookup Index Found
EvLKi_Search

Configuration Response from Server via Transaction Manager

Search for Empty Index if new Equipment
Store result to avoid future searches

(redundant to CfgResponseEquipment - should always exist from Equipment Download)

5

Event Lookup Index EQID Match
EvLKi_EQIDMatch
/

Diagnostic
ADD
Source A EventEmptyFscCount 1
Source B 1
Dest EventEmptyFscCount 1

Fragment Statistics Empty Search
EventEmptyFSC
RES

Fragment Statistics Empty Search
MOVE
Source 0
Dest EventEmptyFSC.POS 271

Fragment Statistics Empty Search
MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEmptyFSC.LEN 3000

Fragment Statistics Empty Search
FSC
Control EventEmptyFSC
Length 3000
Position 271
Mode ALL
Expression EDC_Event.CTL.Equip[EventEmptyFSC.POS].EQID = 0
EN
DN
ER

Fragment Statistics Empty Search
EventEmptyFSC.FD

Equipment-Event Index for EDC_Event Lookup
MOVE
Source EventEmptyFSC.POS 271
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array
MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Controls Equipment Fragment Statistics Array
MOVE
Source EQID 531
Dest EDC_Event.CTL.Equip[EvLKi].EQID 926

Event Controls Equipment with Event Text Qty
ADD
Source A EDC_Event.CTL.EqQty 272
Source B 1
Dest EDC_Event.CTL.EqQty 272

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Event Lookup Index New
EvLKi_New

Configuration Response from Server via Transaction Manager

Equipment Not Found and Statistics Array is Full
(shouldn't happen - requires Manual Intervention and Manual Reset of Error)

Event Lookup Index EQID Match
EvLKi_EQIDMatch

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4025
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Event Fragmentation Statistics Array Full ER
EDC_Status01.ServerCfg.Update.EventStatsArrayFull
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Skip Event Text Statistics Update if new Event was not added

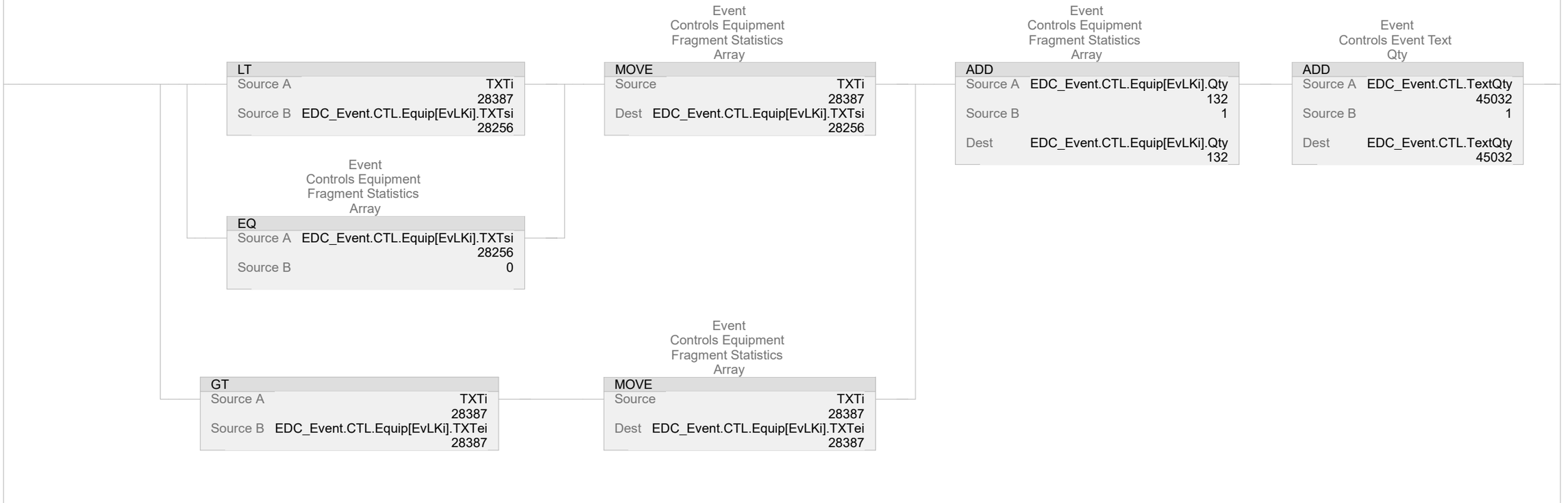
Update Event Text Statistics
UpdateEventTextStat

lbl_UpdateMasks
JMP

Configuration Response from Server via Transaction Manager

If new Event Text, Update Index Range and Increment EQ, Total Counts

8



Configuration Response from Server via Transaction Manager

Update Fragmentation for Previous Equipment
Only allow positive Fragment counts
Don't allow Previous Frag if Previous Event quantity is 0

Fragmentation = This ei - Prev si - 1
Change Fragmentation Statistics from Event% to Equip% (more meaningful - Event% overstated fragmentation)

9

Equipment-Event Index for EDC_Event Lookup

NE
Source A EvLKi 164
Source B 0

SUB
Source A EvLKi 164
Source B 1
Dest LKprv 233

MOVE
Source EDC_Event.CTL.Equip[LKprv].Frag 0
Dest PrevFrag 0

Event Controls Equipment Fragment Statistics Array

CPT
Dest EDC_Event.CTL.Equip[LKprv].Frag 0
Expression EDC_Event.CTL.Equip[EvLKi].TXTsi - EDC_Event.CTL.Equip[LKprv].TXTei - 1

Event Controls Equipment Fragment Statistics Array

LT
Source A EDC_Event.CTL.Equip[LKprv].Frag 0
Source B 0

Event Controls Equipment Fragment Statistics Array

CLR
Dest EDC_Event.CTL.Equip[LKprv].Frag 0

Event Controls Equipment Fragment Statistics Array

EQ
Source A EDC_Event.CTL.Equip[LKprv].Qty 192
Source B 0

AFI

Event Controls Event Fragment Qty

CPT
Dest EDC_Event.CTL.FragQty 4
Expression EDC_Event.CTL.FragQty + (EDC_Event.CTL.Equip[LKprv].Frag - PrevFrag)

EQ
Source A PrevFrag

Event Controls Equipment Fragment Statistics Array

NE
Source A EDC_Event.CTL.Equip[LKprv].Frag

Event Controls Event Fragment Qty

ADD
Source A EDC_Event.CTL.FragQty

Configuration Response from Server via Transaction Manager

	0
Source B	0

	0
Source B	0

	4
Source B	1
Dest	EDC_Event.CTL.FragQty 4

Configuration Response from Server via Transaction Manager

Update Fragmentation for This Equipment
Only allow positive Fragment counts

Fragmentation = This ei - Next si + 1
Change Fragmentation Statistics from Event% to Equip% (more meaningful - Event% overstated fragmentation)

10

Equipment-Event Index for EDC_Event Lookup

NE
Source A EvLKi 164
Source B EDC_Event.CFG.EQ.mi 2999

ADD
Source A EvLKi 164
Source B 1
Dest LKnxt 235

Event Controls Equipment Fragment Statistics Array

NE
Source A EDC_Event.CTL.Equip[LKnxt].TXTsi 37062
Source B 0

MOVE
Source EDC_Event.CTL.Equip[EvLKi].Frag 0
Dest PrevFrag 0

Event Controls Equipment Fragment Statistics Array

CPT
Dest EDC_Event.CTL.Equip[EvLKi].Frag 0
Expression EDC_Event.CTL.Equip[EvLKi].TXTei - EDC_Event.CTL.Equip[LKnxt].TXTsi + 1

Event Controls Equipment Fragment Statistics Array

LT
Source A EDC_Event.CTL.Equip[EvLKi].Frag 0
Source B 0

Event Controls Equipment Fragment Statistics Array

CLR
Dest EDC_Event.CTL.Equip[EvLKi].Frag 0

Event Controls Event Fragment Qty

CPT
Dest EDC_Event.CTL.FragQty 4
Expression EDC_Event.CTL.FragQty + (EDC_Event.CTL.Equip[EvLKi].Frag - PrevFrag)

AFI

Event Controls Equipment Fragment Statistics Array

EQ
Source A PrevFrag 0

Event Controls Equipment Fragment Statistics Array

NE
Source A EDC_Event.CTL.Equip[EvLKi].Frag 0

Event Controls Event Fragment Qty

ADD
Source A EDC_Event.CTL.FragQty 4

Configuration Response from Server via Transaction Manager

Source B	0

Source B	0

Source B	1
Dest	EDC_Event.CTL.FragQty 4

Update Fragmentation Percentage and Last EQ
Change Fragmentation Statistics from Event% to Equip% (more meaningful - Event% overstated fragmentation)

11

AFI

Event
Controls Event
Fragment Percentage

CPT

Dest	EDC_Event.CTL.FragPerc 1.4705882
Expression	EDC_Event.CTL.FragQty / (EDC_Event.CTL.FragQty + EDC_Event.CTL.TextQty) *100

Event
Controls Event
Fragment Percentage

CPT

Dest	EDC_Event.CTL.FragPerc 1.4705882
Expression	(EDC_Event.CTL.FragQty * 100.0) / EDC_Event.CTL.EqQty

Update Event Masks for DtOwner and DtType

12

lbl_UpdateMasks

LBL

NOP

Configuration Response from Server via Transaction Manager

New EQID - Initialize Event Masks to DtOwner = Maintenance, DtType = Fault and Ignore = pass data
Initialize Mask Config words (new data for EqDriver)

*** Ignore Mask is inverted from Configuration -- when OFF, the Mask bit is 1 to pass data thru MEQ ***
v3.02 Data Refresh Request Command

13

Event Lookup Index New
EvLKi_New

Temp Result for Array Size
MUL
Source A EDC_Event.CFG.PKT.n 8
Source B EDC_Event.CFG.DAT.n 8
Dest LEN 64

DT Owner = Maintenance Event Mask Equipment Mask Equip Array
FLL
Source -1
Dest EDC_EventMaskMaint.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Fault Event Mask Equipment Mask Equip Array
FLL
Source -1
Dest EDC_EventMaskFault.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Alarm Event Mask Equipment Mask Equip Array
FLL
Source 0
Dest EDC_EventMaskAlarm.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Alert Event Mask Equipment Mask Equip Array
FLL
Source 0
Dest EDC_EventMaskAlert.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Ignore Event Mask Equipment Mask Equip Array
FLL
Source -1
Dest EDC_EventMaskIgnore.Equip[EvLKi].Packet[0]
Length LEN

No Display Event Mask Equipment Mask Equip Array
FLL
Source 0
Dest EDC_EventMaskNoDisp.Equip[EvLKi].Packet[0]
Length LEN

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].MaintMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].FaultMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlertMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].DataRefreshReq 2#0000_0000

Configuration Response from Server via Transaction Manager

Validate Packet Indexes
PKTi, DATi and BITi

14

Event Mask
Packet Index
(0-7)

LIMIT
Low Limit 0
Test PKTi
7
High Limit EDC_Event.CFG.PKT.mi
7

Event Mask
Data Index
(0-7)

LIMIT
Low Limit 0
Test DATi
3
High Limit EDC_Event.CFG.DAT.mi
7

Event Mask
Bit Index
(0-31)

LIMIT
Low Limit 0
Test BITi
1
High Limit EDC_Event.CFG.BIT.mi
31

PacketIndexOK

PacketIndexOK
/

lbl_Exit
JMP

Update Event Mask for Downtime Owner = Maintenance
Update Mask Cfg for Download to EqDriver

May 2, 2023 - change Maint Mask to Maint-Fault Mask
(only charge Maintenance when DtOwner=Maint and DtType = Fault)

15

Event DtOwner
Maintenance
EventDtMaint

Event DtType
Fault
EventDtFault

DT Owner =
Maintenance
Event Mask Equipment
Mask Equip Array
EDC_EventMaskMaint.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
Controls Equipment
Fragment Statistics
Array
EDC_Event.CTL.Equip[EvLKi].MaintMaskSend.[PKTi]
L

Event DtOwner
Maintenance
EventDtMaint
/

DT Owner =
Maintenance
Event Mask Equipment
Mask Equip Array
EDC_EventMaskMaint.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

Event DtType
Fault
EventDtFault
/

Configuration Response from Server via Transaction Manager

Update Event Mask for Downtime Type = Fault
Update Mask Cfg for Download to EqDriver

16

Event DtType Fault EventDtFault

DT Type = Fault Event Mask Equipment Mask Equip Array EDC_EventMaskFault.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] L

Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].FaultMaskSend.[PKTi] L

Event DtType Fault EventDtFault /

DT Type = Fault Event Mask Equipment Mask Equip Array EDC_EventMaskFault.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] U

Update Event Mask for Downtime Type = Alarm
Update Mask Cfg for Download to EqDriver

17

Event DtType Alarm EventDtAlarm

DT Type = Alarm Event Mask Equipment Mask Equip Array EDC_EventMaskAlarm.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] L

Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend.[PKTi] L

Event DtType Alarm EventDtAlarm /

DT Type = Alarm Event Mask Equipment Mask Equip Array EDC_EventMaskAlarm.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] U

Update Event Mask for Downtime Type = Alert
Update Mask Cfg for Download to EqDriver

18

Event DtType Alert EventDtAlert

DT Type = Alert Event Mask Equipment Mask Equip Array EDC_EventMaskAlert.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] L

Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].AlertMaskSend.[PKTi] L

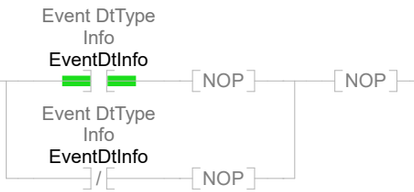
Event DtType Alert EventDtAlert /

DT Type = Alert Event Mask Equipment Mask Equip Array EDC_EventMaskAlert.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] U

Configuration Response from Server via Transaction Manager

Downtime Type = Info
(no Mask for Info -- it's the default status if not Fault, Alarm, Alert or Ignore)

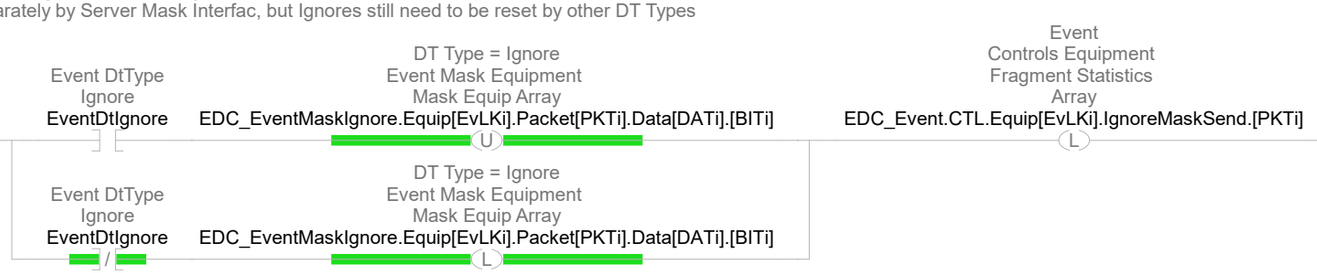
19



Update Event Mask for Downtime Type = Ignore
Update Mask Cfg for Download to EqDriver

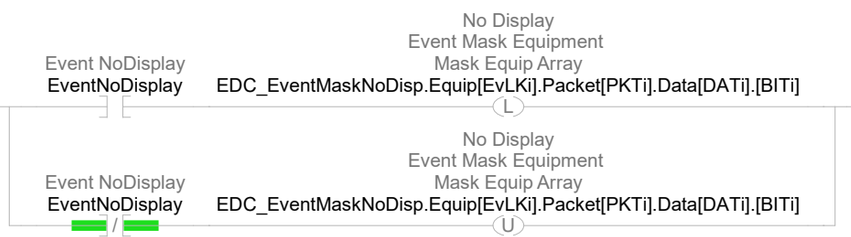
*** Ignore Mask is inverted from Configuration -- when Ignore is OFF, the Mask bit is 1 to pass data thru MEQ ***
Ignore Events are downloaded separately by Server Mask Interfac, but Ignores still need to be reset by other DT Types

20



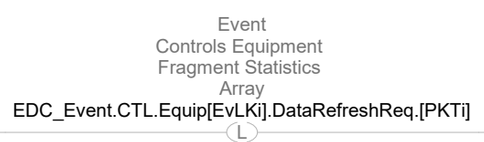
Update Event Mask for NoDisplay

21



v3.02 Data Refresh Request Command

22



Configuration Response from Server via Transaction Manager

Set Summary Data KPI Option ID 1-5 Event Mask
and Equipment Packet Config State

23

KPI Option 1
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 2
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 3
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 4
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 5
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 1

KPI Option 1
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]
L

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 2

KPI Option 2
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi]
L

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 3

KPI Option 3
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi]
L

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 4

KPI Option 4
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi]
L

0 = not defined

KPI Option 5
Event Mask Equipment

Event
KPI Options
Equipment Event KPI

Configuration Response from Server via Transaction Manager

1-5 = Option 1-5

EQ	
Source A	KpiOptionID
	0
Source B	5

Mask Equip Array
EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Options
EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi]
L

Reset Equipment Packet Config State for KPI Option 1
if no Events Configured

24

0 = not defined
1-5 = Option 1-5

NE	
Source A	KpiOptionID
	0
Source B	1

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[0]
	16#0000_0000
Source B	0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[1]
	16#0000_0000
Source B	0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[2]
	16#0000_0000
Source B	0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[3]
	16#0000_0000
Source B	0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[4]
	16#0000_0000
Source B	0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[5]
	16#0000_0000
Source B	0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[6]
	16#0000_0000
Source B	0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ	
Source A	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[7]
	16#0000_0000
Source B	0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 2
if no Events Configured

25

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 2

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[3]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[4]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[5]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[6]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[7]
16#0000_0000
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 3
if no Events Configured

26

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 3

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[3]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[4]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[5]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[6]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[7]
16#0000_0000
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 4
if no Events Configured

27

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 4

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[3]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[4]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[5]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[6]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[7]
16#0000_0000
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 5
if no Events Configured

28

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 5

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[3]
16#ffff_fffc
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[4]
16#ffff_ffff
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[5]
16#ffff_ffff
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[6]
16#ffff_ffff
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[7]
16#ffff_ffff
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

v3.5 Refresh Equipment Events if not Full Download
(full Download already has a periodic refresh)

29

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.Update
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.UpdateOview
L

Equipment Data Collection Lines and Equipment Array
FLL
Source -1
Dest EDC.Line[LNi].Equip[EQi].EventCTL.UpdateZone
Length 1

Exit Routine

30

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 16 - PreDefined Event Text
Update Event Text Masks

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 837
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Update Event Mask for Downtime Owner = Maintenance

May 2, 2023 - change Maint Mask to Maint-Fault Mask
(only charge Maintenance when DtOwner=Maint and DtType = Fault)

2

Event DtOwner Maintenance EventDtMaint

Event DtType Fault EventDtFault

PreDefined Event Mask DtOwner Maintenance Mask EDC_EventMaskPreDef.Maint.[BITi] (L)

Event DtOwner Maintenance EventDtMaint (/)

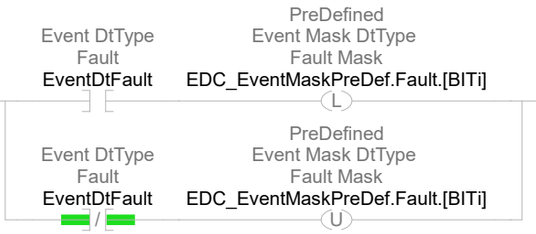
Event DtType Fault EventDtFault (/)

PreDefined Event Mask DtOwner Maintenance Mask EDC_EventMaskPreDef.Maint.[BITi] (U)

Configuration Response from Server via Transaction Manager

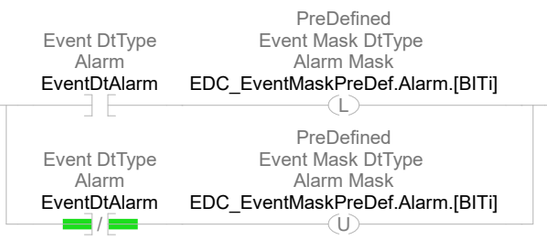
Update Event Mask for Downtime Type = Fault

3



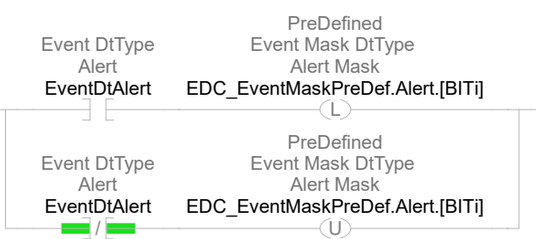
Update Event Mask for Downtime Type = Alarm

4



Update Event Mask for Downtime Type = Alert

5



Configuration Response from Server via Transaction Manager

Downtime Type = Info
(no Mask for Info -- it's the default status if not Fault, Alarm, Alert or Ignore)

6

Event DtType Info EventDtInfo — NOP — NOP
Event DtType Info EventDtInfo (/) — NOP

Update Event Mask for Downtime Type = Ignore

*** Ignore Mask is inverted from Configuration -- when Ignore is OFF, the Mask bit is 1 to pass data thru MEQ ***
Ignore Events are downloaded separately by Server Mask Interfac, but Ignores still need to be reset by other DT Types

7

Event DtType Ignore EventDtIgnore — PreDefined Event Mask DtType Ignore Mask EDC_EventMaskPreDef.Ignore.[BITi] (U)
Event DtType Ignore EventDtIgnore (/) — PreDefined Event Mask DtType Ignore Mask EDC_EventMaskPreDef.Ignore.[BITi] (L)

Update Event Mask for NoDisplay

8

Event NoDisplay EventNoDisplay — PreDefined Event Mask No Display Option EDC_EventMaskPreDef.NoDisp.[BITi] (L)
Event NoDisplay EventNoDisplay (/) — PreDefined Event Mask No Display Option EDC_EventMaskPreDef.NoDisp.[BITi] (U)

v3.5 Refresh All Events if not Full Download
(full Download already has a periodic refresh)

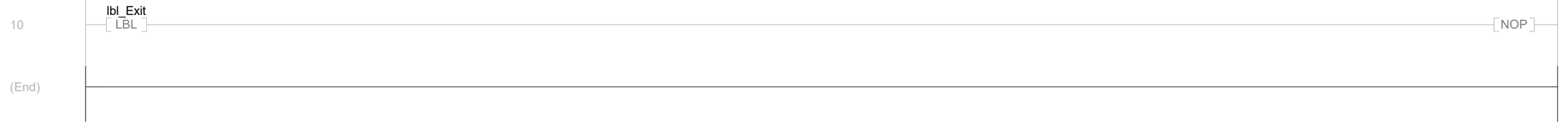
9

Server Configuration from Transaction Manager Response Download All Started ServerCfg.Ctl.DownloadAllStarted <EDC.ServerCfg.Ctl.DownloadAllStarted> (/)

HMI Refresh HMI Events after Config All EDC_HMI.CTL.DownloadAllRefresh (L)

Configuration Response from Server via Transaction Manager

Exit Routine



Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 8 - Shop
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 838
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0] <EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Short Name

NE
Source A ServerCfg.DataRspLast.ShortName.DATA[0] <EDC.ServerCfg.DataRspLast.ShortName.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4018
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Shop Rx Data ER
EDC_Status01.ServerCfg.Update.ShopRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Full Download Request -- Reset ALL existing Configuration
(Shop Configuration command forces Reset and Download ALL configuration)

EDC2 WL31 Equipment Complete -- once download reaches the Events, all Equipment should have a good configuration
EDC2 WL31 Block EqDriver -- don't service EqDriver Messages until Equipment Configuration is complete and EQ Disable status can be re-established

4

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>
L

Server Configuration from Transaction Manager Response Download All Complete
ServerCfg.Ctl.DownloadAllComplete
<EDC.ServerCfg.Ctl.DownloadAllComplete>
U

Server Configuration from Transaction Manager Download Started - Equipment Config Complete
ServerCfg.Ctl.DownloadEquipComplete
<EDC.ServerCfg.Ctl.DownloadEquipComplete>
U

Server Configuration from Transaction Manager Download Started - Block Equipment Drivers until Equipment Config Complete
ServerCfg.Ctl.DownloadBlockEqDriver
<EDC.ServerCfg.Ctl.DownloadBlockEqDriver>
L

JSR
Routine Name _810_Reset

Configuration Response from Server via Transaction Manager

Process Data and Store

Equipment Data Collection Gateway Name

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.CFG.Name
Length 1

Equipment Data Collection Shop Site ID

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.CFG.SiteID
Length 1

HMI Shop Name

COP
Source EDC.CFG.Name
Dest EDC_HMI.Name
Length 1

5

PdM Full Download Initialize

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Server PdM Configuration from Transaction Manager Response Download All Started
ServerPdm.Ctl.DownloadAllStarted
<EDC.ServerPdm.Ctl.DownloadAllStarted>
L

Server PdM Configuration from Transaction Manager Response Download All Complete
ServerPdm.Ctl.DownloadAllComplete
<EDC.ServerPdm.Ctl.DownloadAllComplete>
U

Configuration Initialization before full download

JSR
Routine Name _858_PdmDownloadInit

6

Mask Full Download Initialize

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Server Ignore Mask Configuration from Transaction Manager Response Download All Started
ServerMask.Ctl.DownloadAllStarted
<EDC.ServerMask.Ctl.DownloadAllStarted>
L

Server Ignore Mask Configuration from Transaction Manager Response Download All Complete
ServerMask.Ctl.DownloadAllComplete
<EDC.ServerMask.Ctl.DownloadAllComplete>
U

7

Exit Routine

lbl_Exit
LBL

NOP

8

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 9 - Quality Tool Carrier Scanner
LineID - used for Conv ID
PitchID- used for Quality Tool ID
PosID - used for Quality Tool Data ID
ZoneDispID- used for Carrier Scanner Type
ConfigStatus - used for Enable, Emtpy bits

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 839
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Convert EQID to LNi and EQi

2

Equipment Index

MOD
Source A ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1
Source B 100
Dest EQi 31

Line Index

CPT
Dest LNi 5
Expression (ServerCfg.DataRspLast.EqID - EQi) / 100

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

3

Line Index
LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

Equipment Index
LIMIT
Low Limit 0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

Server Configuration from Transaction Manager Line ID (0=Offline)
NE
Source A ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Source B -1

Server Configuration from Transaction Manager Pitch ID (0-350)
NE
Source A ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
Source B -1

Server Configuration from Transaction Manager Position ID (0-5)
NE
Source A ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
Source B -1

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)
NE
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Source B -1

Server Configuration from Transaction Manager Name
NE
Source A ServerCfg.DataRspLast.Name.DATA[0]
<EDC.ServerCfg.DataRspLast.Name.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

4

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4019
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-ER
Carrier Scanner Data Read/Write
EDC_Status01.ServerCfg.Update.CarrScanDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Initialize Carrier Scanner Index

5

CLR
Dest i
300

Carrier Scanner Empty Array Position Found
ScanEmptyFD
U

Carrier Scanner Matching Array Position Found
ScanMatchFD
U

Configuration Response from Server via Transaction Manager

Search for matching Scanner Record to update
(and search for first unused array position)

6

lbl_CheckPos
LBL

EQ
Source A: EDC_Track.ScanCfg[i].EQID ??
Source B: 0

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

Carrier Scanner Empty Array Position Found
ScanEmptyFD
/

MOVE
Source: i 300
Dest: ScanPOS 4

Carrier Scanner Empty Array Position Found
ScanEmptyFD
L

lbl_NextPos
JMP

Check Carrier Scanner Position for match of EQID, ToolID, DataID

7

EQ
Source A: EDC_Track.ScanCfg[i].EQID ??
Source B: ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

EQ
Source A: EDC_Track.ScanCfg[i].ToolID ??
Source B: ServerCfg.DataRspLast.PitchID <EDC.ServerCfg.DataRspLast.PitchID> -1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

EQ
Source A: EDC_Track.ScanCfg[i].DataID ??
Source B: ServerCfg.DataRspLast.PositionID <EDC.ServerCfg.DataRspLast.PositionID> -1

Carrier Scanner Empty Array Position

MOVE
Source: i 300
Dest: ScanPOS 4

Carrier Scanner Matching Array Position Found
ScanMatchFD
L

Increment Carrier Scanner Index

8

lbl_NextPos
LBL

Carrier Scanner Matching Array Position Found
ScanMatchFD
/

ADD
Source A: i 300
Source B: 1
Dest: i 300

LE
Source A: i 300
Source B: EDC_Track.Cfg.SCN.mi 299

lbl_CheckPos
JMP

Configuration Response from Server via Transaction Manager

Carrier Scanner Array Full

9

Carrier Scanner Empty Array Position Found
ScanEmptyFD
/

Carrier Scanner Matching Array Position Found
ScanMatchFD
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4026
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-Carrier Scanner Array Full ER Read/Write
EDC_Status01.ServerCfg.Update.CarrScanArrayFull
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

LineID - used for Conv ID
PitchID- used for Quality Tool ID
PosID - used for Quality Tool Data ID
ZoneDispID- used for Carrier Scanner Type
ConfigStatus - used for Enable, Emtpy bits

10

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.QualityMsgTrackEN (L)

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Enable (U)

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Empty (U)

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Enable (L)

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Empty (L)

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID> -1
Dest EDC_Track.ScanCfg[ScanPOS].EQID 114

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID> -1
Dest EDC_Track.ScanCfg[ScanPOS].ToolID 1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID> -1
Dest EDC_Track.ScanCfg[ScanPOS].DataID 1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID> -1
Dest EDC_Track.ScanCfg[ScanPOS].ConvID 25

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
Dest EDC_Track.ScanCfg[ScanPOS].Type 4

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

CPT
Dest EDC_Track.ScanCfg[ScanPOS].Srch 11401
Expression (ServerCfg.DataRspLast.EqID *100) + ServerCfg.DataRspLast.PitchID

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_Track.Conv[ServerCfg.DataRspLast.LineID].Cfg.Name
Length 1

Configuration Response from Server via Transaction Manager

Data OK

11

Server Interface Status State = Green- Carrier Scanner Data OK
EDC_Status01.ServerCfg.Update.CarrScanDataOK
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Exit Routine

12

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 6 - Schedule
LineID - used for Array Index
EQID - used for ShiftID
PitchID- used for Quarter ID
PosID - used for Hour ID
ZoneDispID- used for Hour Start
EqEventID - used for Shift Reset

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 840
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.ShopKey
<EDC.ServerCfg.DataRspLast.ShopKey>
-1
Source B 1

Server Configuration from Transaction Manager Line ID (0=Offline)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
High Limit CfgSched.mi
99

Server Configuration from Transaction Manager Equipment ID

NE
Source A ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Source B -1

Server Configuration from Transaction Manager Pitch ID (0-350)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
High Limit 5

Server Configuration from Transaction Manager Position ID (0-5)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
High Limit 10

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
High Limit 240000

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
High Limit 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance
(share Shop status - need to expand Status to support new downloads)

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4019
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Shop Rx Data ER
EDC_Status01.ServerCfg.Update.ShopRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store
if HourStart = 240000, this is the last record -- initialize remaining array positions to -1

LineID - used for array Index
EqID - used for Shift
PitchID- used for Quarter 0-5
PosID - used for Hour 0-10
ZoneDispID- used for HourStart (0-240000)
EqEventID - used for Shift Reset (0-1)

4

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID> -1
Dest index
22

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID> -1
Dest ScheduleLookup[index].Shift
2

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID> -1
Dest ScheduleLookup[index].Qtr
2

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID> -1
Dest ScheduleLookup[index].Hour
4

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
Dest ScheduleLookup[index].HourStart
240000

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID> -1
Dest ScheduleLookup[index].Reset
0

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

EQ
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
Source B 240000

LT
Source A index
22
Source B CfgSched.mi
99

Temp Result for Array Size

SUB
Source A CfgSched.mi
99
Source B index
22
Dest LEN
64

Schedule Hours lookup array

FLL
Source -1
Dest ScheduleLookup[index+1]
Length LEN

Configuration Response from Server via Transaction Manager

adjust array end index
used for FSC search length

Schedule Array
Config Array End
Index

5

GT
Source A index
22
Source B CfgSched.ei
22

LE
Source A index
22
Source B CfgSched.mi
99

MOVE
Source index
22
Dest CfgSched.ei
22

Server Configuration
from Transaction
Manager Maintenance
Zone Display ID
(1-8)

EQ
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Source B 240000

Data OK
(share Shop status - need to expand Status to support new downloads)

Server Interface
Status State = Red-
Shop Rx Data OK
EDC_Status01.ServerCfg.Update.ShopRxDataOK

6

L

Update Status State
for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Exit Routine

lbl_Exit

7

LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 7 - Gateway
Zone - used for TimeZoneID
EqEvent
RankNum - used for DstAdjust
Name - used for SntpServerAddress
EQID - used for LdtEQi
ConfigStatus - used for 3 config bits

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 841
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

Logix Wallclock allows Time Zone selection from -12 to +14

2

Server Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.ShopKey
<EDC.ServerCfg.DataRspLast.ShopKey>
-1
Source B 1

Server Configuration from Transaction Manager Equipment Event Text

NE
Source A ServerCfg.DataRspLast.EqEventText.DATA[0]
<EDC.ServerCfg.DataRspLast.EqEventText.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

NE
Source A ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Source B -1

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0]
<EDC.ServerCfg.DataRspLast.Name.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Equipment ID

NE
Source A ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Source B -1

Server Configuration from Transaction Manager Response has Good Data Record

ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance
(share Shop status - need to expand Status to support new downloads)

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
[/]

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.ShopKey
<EDC.ServerCfg.DataRspLast.ShopKey>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4019
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Shop Rx Data ER
EDC_Status01.ServerCfg.Update.ShopRxDataER
(L)

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig (...)
Status EDC_Status01.ServerCfg

lbl_Exit
(JMP)

Configuration Response from Server via Transaction Manager

EqEventTxt- used for TimeZone (59-char max - expanded EqEventTxt from 40 to 82)
RankNum - used for DstAdjust
Name - used for SntpServerAddress
EQID - used for LdtEQi
ConfigStatus - used for 3 config bits

4

Gateway Config Download Wall Clock Time Zone String

COP
Source ServerCfg.DataRspLast.EqEventText
<EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_CfgGateway.TimeZoneString
Length 1

Gateway Config Download Wall Clock Daylight Savings Time Adjustment (minutes)

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Dest EDC_CfgGateway.DstAdjustment
60

Gateway Config Download Wall Clock Time Sync SNTP Server IP Address

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_CfgGateway.SntpServerIP
Length 1

Gateway Config Download Line Downtime (KPI) Equip Index when Rollup not enabled (default 0)

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest EDC_CfgGateway.LdtEQi
0

Gateway Config Download HMI Common Line Equipment - AVI Equip Index (default 0)

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Dest EDC_CfgGateway.HmiComAviEQi
9

Gateway Config Download HMI Common Line Equipment - Andon Equip Index (default 0)

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
Dest EDC_CfgGateway.HmiComAndonEQi
1

Gateway Config Download HMI Common Line Equipment - Conveyor Equip Index (default 2)

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
Dest EDC_CfgGateway.HmiComConvEQi
2

Gateway Config Download HMI Common Line Equipment - Entrance Equip Index (default 3)

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Dest EDC_CfgGateway.HmiComEntEQi
3

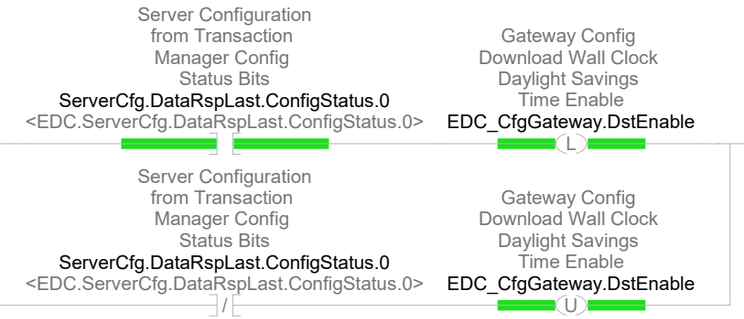
Gateway Config Download HMI Common Line Equipment - Exit Equip Index (default 4)

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest EDC_CfgGateway.HmiComExitEQi
4

Configuration Response from Server via Transaction Manager

Bit 0 - Enable Daylight Savings Time (DST)
if enabled, applies the DST offset during DST

5



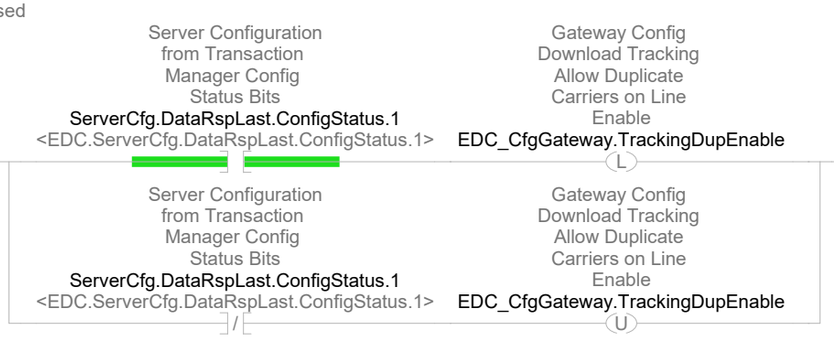
Bit 1 - Enable Tracking Duplicates

Tracking - allow duplicate Die/Mold on Line

Assembly, Paint and BodyWeld do not allow duplicate Carriers on a Line
Plastics and Stamping do allow duplicate Die/Mold on a Line

this may need to be applied at the Line level, but for now it is Shop-based

6

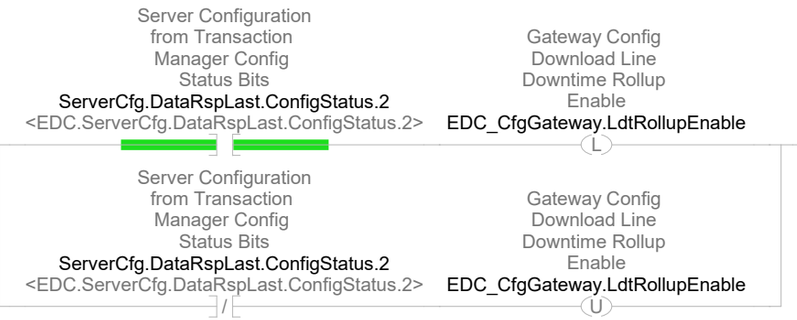


Configuration Response from Server via Transaction Manager

Bit 2 - Enable Downtime Rollup
Rollup of All Equipment worst case or one Equipment (i.e. Andon)
(rarely used - typically the Line DT = Andon Equipment DT)

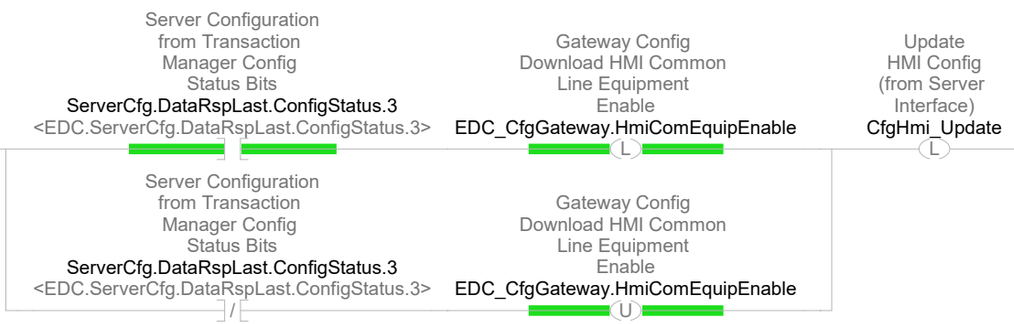
Line Downtime Configuration

7



Bit 3 - Disable HMI Overview Common Line Equipment
(used in shops with EQID xx00-xx05 assigned for a common display of AVI, Andon, Conveyor, Entrance LIfter and Exit Lifter)

8



Configuration Response from Server via Transaction Manager

Bit 4 - Enable HMI Overview for Health Alarm/Fault Event Display

9

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.4
<EDC.ServerCfg.DataRspLast.ConfigStatus.4>

Gateway Config Download HMI Display Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.4
<EDC.ServerCfg.DataRspLast.ConfigStatus.4>
/

Gateway Config Download HMI Display Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable
U

Data OK
(share Shop status - need to expand Status to support new downloads)

10

Server Interface Status State = Red- Shop Rx Data OK
EDC_Status01.ServerCfg.Update.ShopRxDataOK
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Configuration Request to Server via Transaction Manager

Configuration Data Request to Server via Transaction Manager

- Logix Requests Data (-1 Command)
- Server Responds with Data (101-103 Command), No Data (0 Command) or No Response (-1)

- if Data, Logix processes Data and Sends Ack (Echo Server Command)
- Server uses Ack to set Download Timestamp for Config Record
- Server Responds with next Data/NoData

- if NoData, Logix waits and then Requests Data (-1 Command)

0

NOP

Request Commands to TM/SQL

-1 = New Data Request (delay between requests)
101-103 = Ack Data Response and New Data Request (no delay between requests)

1

NOP

Response Commands from Server

-1 = No Data Error
0 = No Data to Download
101 = Quality Tool Configuration
102 = Trend Tool Configuration
103 = Network Configuration

2

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

3

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 845
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data

4

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Duplicate Error
ServerPdm.Ctl.RspDupER
<EDC.ServerPdm.Ctl.RspDupER>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerPdm.Ctl.ReqFastTime
<EDC.ServerPdm.Ctl.ReqFastTime>
0
Dest ServerPdm.Ctl.ReqNextTMR.PRE
<EDC.ServerPdm.Ctl.ReqNextTMR.PRE>
10000

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Duplicate Error
ServerPdm.Ctl.RspDupER
<EDC.ServerPdm.Ctl.RspDupER>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerPdm.Ctl.ReqSlowTime
<EDC.ServerPdm.Ctl.ReqSlowTime>
10000
Dest ServerPdm.Ctl.ReqNextTMR.PRE
<EDC.ServerPdm.Ctl.ReqNextTMR.PRE>
10000

Server PdM Configuration from Transaction Manager Request Next Record Timer

TON
Timer ServerPdm.Ctl.ReqNextTMR
<EDC.ServerPdm.Ctl.ReqNextTMR>
Preset 10000
Accum 9505
EN
DN

Configuration Request to Server via Transaction Manager

Request Next
Ready for next Transaction Manager Trigger

Server PdM Configuration from Transaction Manager Request Next Record Timer
ServerPdm.Ctl.ReqNextTMR.DN
<EDC.ServerPdm.Ctl.ReqNextTMR.DN>

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>
L

Server PdM Configuration from Transaction Manager Request Record Complete
ServerPdm.Ctl.ReqComplete
<EDC.ServerPdm.Ctl.ReqComplete>
U

5

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager - Data Init

JSR
Routine Name _846_PdmRequestInit

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
/

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

6

Configuration Request to Server via Transaction Manager

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

7

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B 1
Dest ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B 1

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210

Configuration Request to Server via Transaction Manager

Request Next
Log Transaction & Performance

8

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4300
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataReq.Command
<EDC.ServerPdm.DataReq.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4302
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataReq.Command
<EDC.ServerPdm.DataReq.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerPdm.Msg.EN

Configuration Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

9

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B ServerPdm.Ctl.Interface
<EDC.ServerPdm.Ctl.Interface>
3210

Server PdM Configuration from Transaction Manager Request Record Complete
ServerPdm.Ctl.ReqComplete
<EDC.ServerPdm.Ctl.ReqComplete>
L

Configuration Request to Server via Transaction Manager

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
Also problems with asynchronous data from TM - handshake can update before data values.
2-Check method is fast and avoids errors for occasional slow update of data.
Wait 1 scan before 1st check to eliminate 99% of missing data issues.
2nd check after 2000msec if Check 1 fails.
(anything >2 seconds will generate a TM Transaction error based on Tag Retrieval rate)

10

Server PdM Configuration from Transaction Manager Request Record Complete
ServerPdm.Ctl.ReqComplete
<EDC.ServerPdm.Ctl.ReqComplete>

Server PdM Configuration from Transaction Manager Request Record First Check Timer

MOVE
Source 10
Dest ServerPdm.Ctl.ReqCompleteCheckTMR.PRE
<EDC.ServerPdm.Ctl.ReqCompleteCheckTMR.PRE>
10

Server PdM Configuration from Transaction Manager Request Record First Check Timer

TON
Timer ServerPdm.Ctl.ReqCompleteCheckTMR
<EDC.ServerPdm.Ctl.ReqCompleteCheckTMR>
Preset 10
Accum 14
EN
DN

Server PdM Configuration from Transaction Manager Request Record First Check Timer
ServerPdm.Ctl.ReqCompleteCheckTMR.DN
<EDC.ServerPdm.Ctl.ReqCompleteCheckTMR.DN>

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Complete Delay Timer

MOVE
Source 2000
Dest ServerPdm.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerPdm.Ctl.ReqCompleteDelayTMR.PRE>
2000

Server PdM Configuration from Transaction Manager Request Complete Delay Timer

TON
Timer ServerPdm.Ctl.ReqCompleteDelayTMR
<EDC.ServerPdm.Ctl.ReqCompleteDelayTMR>
Preset 2000
Accum 2006
EN
DN

Server PdM Configuration from Transaction Manager Request Complete Delay Timer
ServerPdm.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerPdm.Ctl.ReqCompleteDelayTMR.DN>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Configuration Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforced in data update.)

11

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Record First Check 1-shot
ServerPdm.Ctl.ReqCompleteCheckONS
<EDC.ServerPdm.Ctl.ReqCompleteCheckONS>
ONS

Server PdM Configuration from Transaction Manager Request Complete 1st Check Count
ADD
Source A ServerPdm.Ctl.ReqComp1stChkCnt
<EDC.ServerPdm.Ctl.ReqComp1stChkCnt>
663124
Source B 1
Dest ServerPdm.Ctl.ReqComp1stChkCnt
<EDC.ServerPdm.Ctl.ReqComp1stChkCnt>
663124

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Request Record Complete 1-shot
ServerPdm.Ctl.ReqCompleteONS
<EDC.ServerPdm.Ctl.ReqCompleteONS>
ONS

Server PdM Configuration from Transaction Manager Request Complete 2nd Check Count
ADD
Source A ServerPdm.Ctl.ReqComp2ndChkCnt
<EDC.ServerPdm.Ctl.ReqComp2ndChkCnt>
9982
Source B 1
Dest ServerPdm.Ctl.ReqComp2ndChkCnt
<EDC.ServerPdm.Ctl.ReqComp2ndChkCnt>
9982

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4301
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataReq.Command
<EDC.ServerPdm.DataReq.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerPdm.Update.MsgRxOK
L

Update PdM State for Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...
Status EDC_Status01.ServerPdm

Configuration Request to Server via Transaction Manager

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerPdm.Msg.DN

Configuration Response from Server via Transaction Manager

JSR
Routine Name _850_PdmResponse

Request Complete Delay Reset Request

12

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>
U

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

13

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Request in Progress Timer

MOVE
Source 62000
Dest ServerPdm.Ctl.ReqInProgressTMR.PRE
<EDC.ServerPdm.Ctl.ReqInProgressTMR.PRE>
62000

Server PdM Configuration from Transaction Manager Request in Progress Timer

TON
Timer ServerPdm.Ctl.ReqInProgressTMR
<EDC.ServerPdm.Ctl.ReqInProgressTMR>
Preset 62000
Accum 0
EN
DN

Server PdM Configuration from Transaction Manager Request in Progress Timer
ServerPdm.Ctl.ReqInProgressTMR.DN
<EDC.ServerPdm.Ctl.ReqInProgressTMR.DN>

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

14

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4304
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataReq.Command
<EDC.ServerPdm.DataReq.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red-Msg Rx ER
EDC_Status01.ServerPdm.Update.MsgRxER
(L)

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm (...)
Status EDC_Status01.ServerPdm

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans (...)
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerPdm.Msg.ER

Server PdM Configuration from Transaction Manager Request in Progress Timer
ServerPdm.Ctl.ReqInProgressTMR
<EDC.ServerPdm.Ctl.ReqInProgressTMR>
(RES)

(End)

Configuration Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Server to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 846
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

DN011 Separate Req & Rsp Tags
Create Data Init Packet

2

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

/

Server PdM Configuration from Transaction Manager - Data

FLL
Source -1
Dest ServerPdmInit
Length 1

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Clear Response to ensure new data on next DB Response
DHF017 Jan 30, 2018 -- Clear Server Request if no Ack for Good Message Response
Copy Response to Request for Good Data Ack to DB

3

Server PdM Configuration from Transaction Manager Transaction Response Data

COP	
Source	ServerPdmInit
Dest	ServerPdm.DataRsp <EDC.ServerPdm.DataRsp>
Length	1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Transaction Request Data

COP	
Source	ServerPdmInit
Dest	ServerPdm.DataReq <EDC.ServerPdm.DataReq>
Length	1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Transaction Request Data

COP	
Source	ServerPdm.DataRspLast <EDC.ServerPdm.DataRspLast>
Dest	ServerPdm.DataReq <EDC.ServerPdm.DataReq>
Length	1

Store Last Request

4

Server PdM Configuration from Transaction Manager Transaction Last Request Data

COP	
Source	ServerPdm.DataReq <EDC.ServerPdm.DataReq>
Dest	ServerPdm.DataReqLast <EDC.ServerPdm.DataReqLast>
Length	1

Configuration Request to Server via Transaction Manager - Data Init

(End)

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 850
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Detect Duplicate Config Data Response
(this indicates the database is stuck on a bad config item)

2

Server PdM Configuration from Transaction Manager Request/Response Command

NE
Source A ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command> 0
Source B 0

Server PdM Configuration from Transaction Manager Request/Response Command

NE
Source A ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command> 0
Source B -1

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command> 0
Source B ServerPdm.DataRsp.Command <EDC.ServerPdm.DataRsp.Command> 0

Server PdM Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.ShopKey <EDC.ServerPdm.DataRspLast.ShopKey> -1
Source B ServerPdm.DataRsp.ShopKey <EDC.ServerPdm.DataRsp.ShopKey> -1

Server PdM Configuration from Transaction Manager Line Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.LineKey <EDC.ServerPdm.DataRspLast.LineKey> -1
Source B ServerPdm.DataRsp.LineKey <EDC.ServerPdm.DataRsp.LineKey> -1

Server PdM Configuration from Transaction Manager Equip Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.EqKey <EDC.ServerPdm.DataRspLast.EqKey> -1
Source B ServerPdm.DataRsp.EqKey <EDC.ServerPdm.DataRsp.EqKey> -1

Server PdM Configuration from Transaction Manager Tool Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.ToolKey <EDC.ServerPdm.DataRspLast.ToolKey> -1
Source B ServerPdm.DataRsp.ToolKey <EDC.ServerPdm.DataRsp.ToolKey> -1

Server PdM Configuration from Transaction Manager PdM Tool Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.PdmToolKey <EDC.ServerPdm.DataRspLast.PdmToolKey> -1
Source B ServerPdm.DataRsp.PdmToolKey <EDC.ServerPdm.DataRsp.PdmToolKey> -1

Server PdM Configuration from Transaction Manager PdM Trend Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.PdmTrendKey <EDC.ServerPdm.DataRspLast.PdmTrendKey> -1
Source B ServerPdm.DataRsp.PdmTrendKey <EDC.ServerPdm.DataRsp.PdmTrendKey> -1

Server PdM Configuration from Transaction Manager EquipmentID (0001-9999) & ToolID (001-999) combined for fast searches

EQ
Source A ServerPdm.DataRspLast.EqToolID <EDC.ServerPdm.DataRspLast.EqToolID> -1
Source B ServerPdm.DataRsp.EqToolID <EDC.ServerPdm.DataRsp.EqToolID> -1

Server PdM Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerPdm.DataRspLast.EqID <EDC.ServerPdm.DataRspLast.EqID> -1
Source B ServerPdm.DataRsp.EqID <EDC.ServerPdm.DataRsp.EqID> -1

Server PdM Configuration from Transaction Manager Tool ID (001-999)

EQ
Source A ServerPdm.DataRspLast.ToolID <EDC.ServerPdm.DataRspLast.ToolID> -1
Source B ServerPdm.DataRsp.ToolID <EDC.ServerPdm.DataRsp.ToolID> -1

Server PdM Configuration from Transaction Manager Network Index

EQ
Source A ServerPdm.DataRspLast.Data[0].NETi

Server PdM Configuration from Transaction Manager Response Duplicate
ServerPdm.Ctl.RspDup
<EDC.ServerPdm.Ctl.RspDup>

Configuration Response from Server via Transaction Manager

<EDC.ServerPdm.DataRspLast.Data[0].NETi>
-1
Source B ServerPdm.DataRsp.Data[0].NETi
<EDC.ServerPdm.DataRsp.Data[0].NETi>
-1

Duplicate Message Count and Detection
(this indicates the database is stuck on a bad config item)

3

Server PdM Configuration from Transaction Manager Response Duplicate
ServerPdm.Ctl.RspDup
<EDC.ServerPdm.Ctl.RspDup>

ADD
Source A PdmRspDuplicateCount
0
Source B
1
Dest PdmRspDuplicateCount
0

GT
Source A PdmRspDuplicateCount
0
Source B PdmRspDuplicateMax
1

MOVE
Source PdmRspDuplicateCount
0
Dest PdmRspDuplicateMax
1

GT
Source A PdmRspDuplicateCount
0
Source B
5

Server PdM Configuration from Transaction Manager Response Duplicate Error
ServerPdm.Ctl.RspDupER
<EDC.ServerPdm.Ctl.RspDupER>

Server PdM Configuration from Transaction Manager Response Duplicate
ServerPdm.Ctl.RspDup
<EDC.ServerPdm.Ctl.RspDup>

CLR
Dest PdmRspDuplicateCount
0

Move Response to working tag and Initialize Response Good Data
DN011 Separate Req & Rsp Tags

Moving DataRsp to DataLastRsp prevents SQL/TM from changing data while Logix is processing.

4

Server PdM Configuration from Transaction Manager Transaction Last Response Data

COP
Source ServerPdm.DataRsp
<EDC.ServerPdm.DataRsp>
Dest ServerPdm.DataRspLast
<EDC.ServerPdm.DataRspLast>
Length 1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Configuration Response from Server via Transaction Manager

Check Response Command for Bad Data
Valid Commands 0, 101-103

5

Server PdM Configuration from Transaction Manager Request/Response Command

NE
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 0

Server PdM Configuration from Transaction Manager Request/Response Command

LT
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 101

Server PdM Configuration from Transaction Manager Request/Response Command

GT
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 103

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4312
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE

Performance Transaction

aoiEDC_PerfTrans

Configuration Response from Server via Transaction Manager

Dest TempPerfMsg.TransLast
110

Source ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerPdm.Msg.KA

lbl_RspComp
JMP

If Response Command received, process Response

6

PdmRspCmdRcvd

Server has no new Configuration data to download

7

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 0

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
L

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

ADD
Source A PdmRspCmd[0]
698142
Source B 1

Dest PdmRspCmd[0]
698142

Configuration Response from Server via Transaction Manager

Process Response Command

8

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 101

Configuration Response from Server via Transaction Manager

JSR
Routine Name _851_PdmResponseTool

ADD
Source A PdmRspCmd[1]
18929
Source B 1
Dest PdmRspCmd[1]
18929

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 102

Configuration Response from Server via Transaction Manager

JSR
Routine Name _852_PdmResponseTrend

ADD
Source A PdmRspCmd[2]
133
Source B 1
Dest PdmRspCmd[2]
133

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 103

Configuration Response from Server via Transaction Manager

JSR
Routine Name _853_PdmResponseNetwork

ADD
Source A PdmRspCmd[3]
7146
Source B 1
Dest PdmRspCmd[3]
7146

Command Response Complete

lbl_RspComp

9

LBL

NOP

Configuration Response from Server via Transaction Manager

Store Last Good, Bad or Incomplete Response w/ Diagnostic Counts
Incomplete - Command Num received with incomplete data packet on on 1st Check
Bad - Incomplete data packet on 2nd Check

10

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Transaction Last Good Response Data

COP
Source ServerPdm.DataRspLast
<EDC.ServerPdm.DataRspLast>
Dest ServerPdm.DataRspLastGood
<EDC.ServerPdm.DataRspLastGood>
Length 1

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Response Good on 1st Check

ADD
Source A ServerPdm.Ctl.RspGood1stChkCnt
<EDC.ServerPdm.Ctl.RspGood1stChkCnt>
653142
Source B 1
Dest ServerPdm.Ctl.RspGood1stChkCnt
<EDC.ServerPdm.Ctl.RspGood1stChkCnt>
653142

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Response Good on 2nd Check

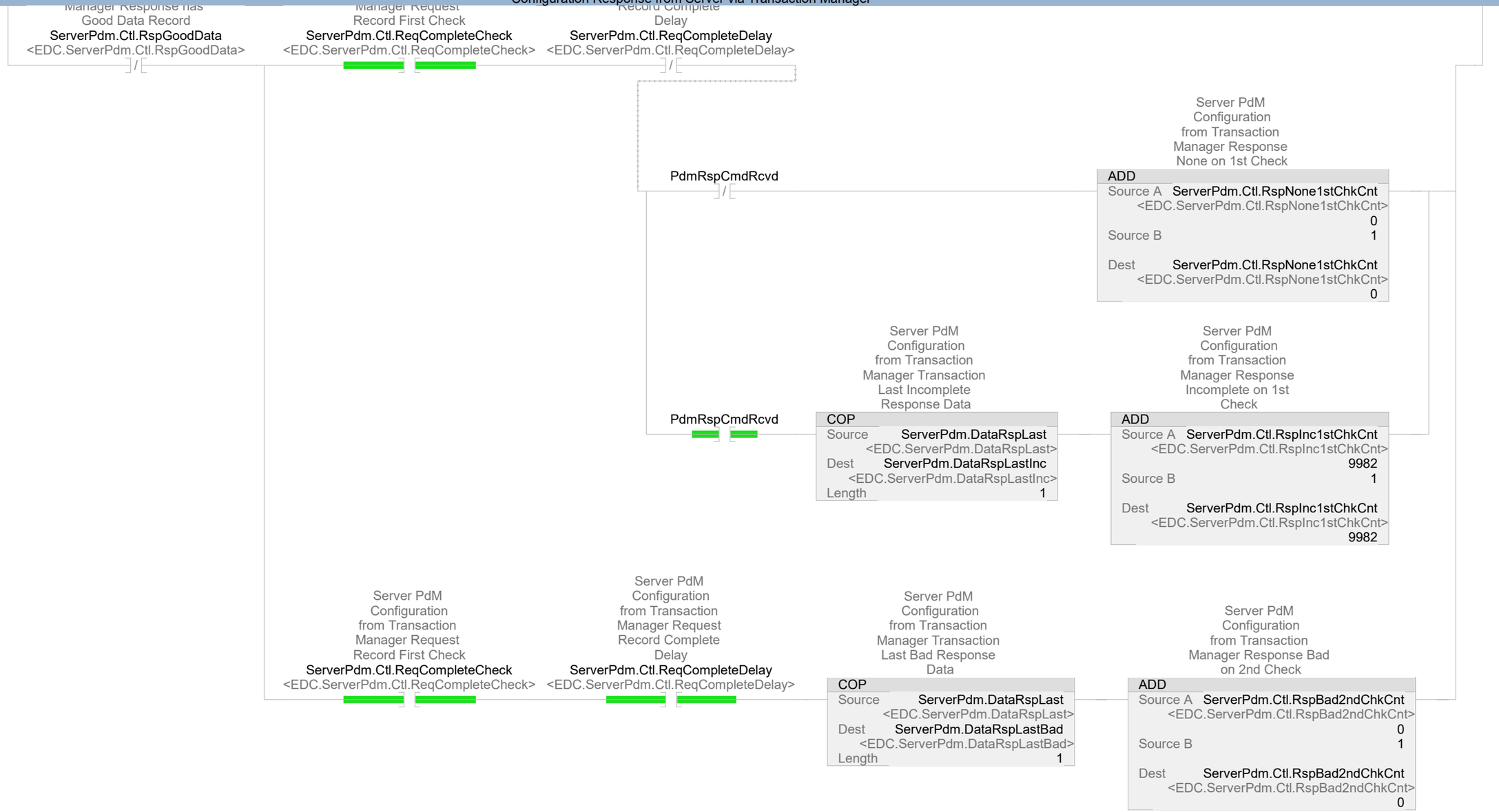
ADD
Source A ServerPdm.Ctl.RspGood2ndChkCnt
<EDC.ServerPdm.Ctl.RspGood2ndChkCnt>
9982
Source B 1
Dest ServerPdm.Ctl.RspGood2ndChkCnt
<EDC.ServerPdm.Ctl.RspGood2ndChkCnt>
9982

Server PdM Configuration from Transaction Manager Response has

Server PdM Configuration from Transaction Manager Request

Server PdM Configuration from Transaction Manager Request Record Complete

Configuration Response from Server via Transaction Manager



Configuration Response from Server via Transaction Manager

Download ALL configuration is complete

11

Server PdM Configuration from Transaction Manager Response Download All Started
ServerPdm.Ctl.DownloadAllStarted
<EDC.ServerPdm.Ctl.DownloadAllStarted>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Response Download All Started
ServerPdm.Ctl.DownloadAllStarted
<EDC.ServerPdm.Ctl.DownloadAllStarted>
U

Server PdM Configuration from Transaction Manager Response Download All Complete
ServerPdm.Ctl.DownloadAllComplete
<EDC.ServerPdm.Ctl.DownloadAllComplete>
L

Server PdM Configuration from Transaction Manager Response Download All Complete 1-shot
ServerPdm.Ctl.DownloadAllCompleteONS
<EDC.ServerPdm.Ctl.DownloadAllCompleteONS>

Configuration Udpate after full download

JSR
Routine Name _859_PdmDownloadComplete

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 101 - Tool
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 851
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data
-- allow Tool w/out Data Configuration (Data is typically created after Tool is created and will generate another download to update config)

2

Server PdM Configuration from Transaction Manager EquipmentID (0001-9999) & ToolID (001-999) combined for fast searches

GT
Source A ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID> -1
Source B 0

Server PdM Configuration from Transaction Manager Equipment ID (0001-9999)

NE
Source A ServerPdm.DataRspLast.EqID
<EDC.ServerPdm.DataRspLast.EqID> -1
Source B -1

Server PdM Configuration from Transaction Manager Tool ID (001-999)

GT
Source A ServerPdm.DataRspLast.ToolID
<EDC.ServerPdm.DataRspLast.ToolID> -1
Source B 0

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)

NE
Source A ServerPdm.DataRspLast.ToolEnable
<EDC.ServerPdm.DataRspLast.ToolEnable> 0
Source B -1

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)

NE
Source A ServerPdm.DataRspLast.TriggerEnable
<EDC.ServerPdm.DataRspLast.TriggerEnable> 0
Source B -1

Server PdM Configuration from Transaction Manager Sample Period ID

NE
Source A ServerPdm.DataRspLast.SamplePeriodID
<EDC.ServerPdm.DataRspLast.SamplePeriodID> -1
Source B -1

Server PdM Configuration from Transaction Manager Data Start Index

NE
Source A ServerPdm.DataRspLast.DATsi
<EDC.ServerPdm.DataRspLast.DATsi> -1
Source B -1

Server PdM Configuration from Transaction Manager Data End Index

NE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei> -1
Source B -1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Validate Data 0 config

3

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 0

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[0].NETi
<EDC.ServerPdm.DataRspLast.Data[0].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[0].DATi
<EDC.ServerPdm.DataRspLast.Data[0].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[0].TypeID
<EDC.ServerPdm.DataRspLast.Data[0].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 1 config

4

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 1

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[1].NETi
<EDC.ServerPdm.DataRspLast.Data[1].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[1].DATi
<EDC.ServerPdm.DataRspLast.Data[1].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[1].TypeID
<EDC.ServerPdm.DataRspLast.Data[1].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[1].Enable
<EDC.ServerPdm.DataRspLast.Data[1].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 2 config

5

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 2

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[2].NETi
<EDC.ServerPdm.DataRspLast.Data[2].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[2].DATi
<EDC.ServerPdm.DataRspLast.Data[2].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[2].TypeID
<EDC.ServerPdm.DataRspLast.Data[2].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[2].Enable
<EDC.ServerPdm.DataRspLast.Data[2].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 3 config

6

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 3

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[3].NETi
<EDC.ServerPdm.DataRspLast.Data[3].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[3].DATi
<EDC.ServerPdm.DataRspLast.Data[3].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[3].TypeID
<EDC.ServerPdm.DataRspLast.Data[3].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[3].Enable
<EDC.ServerPdm.DataRspLast.Data[3].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 4 config

7

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 4

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[4].NETi
<EDC.ServerPdm.DataRspLast.Data[4].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[4].DATi
<EDC.ServerPdm.DataRspLast.Data[4].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[4].TypeID
<EDC.ServerPdm.DataRspLast.Data[4].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[4].Enable
<EDC.ServerPdm.DataRspLast.Data[4].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 5 config

8

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 5

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[5].NETi
<EDC.ServerPdm.DataRspLast.Data[5].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[5].DATi
<EDC.ServerPdm.DataRspLast.Data[5].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[5].TypeID
<EDC.ServerPdm.DataRspLast.Data[5].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[5].Enable
<EDC.ServerPdm.DataRspLast.Data[5].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 6 config

9

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 6

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[6].NETi
<EDC.ServerPdm.DataRspLast.Data[6].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[6].DATi
<EDC.ServerPdm.DataRspLast.Data[6].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[6].TypeID
<EDC.ServerPdm.DataRspLast.Data[6].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[6].Enable
<EDC.ServerPdm.DataRspLast.Data[6].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 7 config

10

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 7

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[7].NETi
<EDC.ServerPdm.DataRspLast.Data[7].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[7].DATi
<EDC.ServerPdm.DataRspLast.Data[7].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[7].TypeID
<EDC.ServerPdm.DataRspLast.Data[7].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[7].Enable
<EDC.ServerPdm.DataRspLast.Data[7].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 8 config

11

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 8

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[8].NETi
<EDC.ServerPdm.DataRspLast.Data[8].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[8].DATi
<EDC.ServerPdm.DataRspLast.Data[8].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[8].TypeID
<EDC.ServerPdm.DataRspLast.Data[8].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[8].Enable
<EDC.ServerPdm.DataRspLast.Data[8].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 9 config

12

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 9

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[9].NETi
<EDC.ServerPdm.DataRspLast.Data[9].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[9].DATi
<EDC.ServerPdm.DataRspLast.Data[9].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[9].TypeID
<EDC.ServerPdm.DataRspLast.Data[9].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[9].Enable
<EDC.ServerPdm.DataRspLast.Data[9].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate unused Data 0 not Enabled

13

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 0

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

PdmGoodData

Validate unused Data 1 not Enabled

14

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[1].Enable
<EDC.ServerPdm.DataRspLast.Data[1].Enable>

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 2 not Enabled

15

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 2

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[2].Enable
<EDC.ServerPdm.DataRspLast.Data[2].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 3 not Enabled

16

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 3

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[3].Enable
<EDC.ServerPdm.DataRspLast.Data[3].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 4 not Enabled

17

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 4

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[4].Enable
<EDC.ServerPdm.DataRspLast.Data[4].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 5 not Enabled

18

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 5

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[5].Enable
<EDC.ServerPdm.DataRspLast.Data[5].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 6 not Enabled

19

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 6

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[6].Enable
<EDC.ServerPdm.DataRspLast.Data[6].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 7 not Enabled

20

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 7

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[7].Enable
<EDC.ServerPdm.DataRspLast.Data[7].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 8 not Enabled

21

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 8

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[8].Enable
<EDC.ServerPdm.DataRspLast.Data[8].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 9 not Enabled

22

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 9

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[9].Enable
<EDC.ServerPdm.DataRspLast.Data[9].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

23

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
/

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID>
-1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4313
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red-Tool Rx Data ER
EDC_Status01.ServerPdm.Update.ToolRxDataER
L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

lbl_Exit
JMP

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

24

Pdm Tool Array Search
PdmToolFSC
RES

PdM Config Array End Index

GT
Source A PdM_Cfg.TL.ei 914
Source B 0

Pdm Tool Array Search

ADD
Source A PdM_Cfg.TL.ei 914
Source B 1
Dest PdmToolFSC.LEN 915

Pdm Tool Array Search

MOVE
Source 0
Dest PdmToolFSC.POS 839

Pdm Tool Array Search

FSC
Control PdmToolFSC
Length 915
Position 839
Mode ALL
Expression PdM_Tool[PdmToolFSC.POS].Cfg.EqToolID=ServerPdm.DataRspLast.EqToolID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Search Pdm Config for Empty Position

25

Pdm Tool Array Search: PdmToolFSC.FD [/]

Pdm Tool Array Empty Search: PdmToolEmptyFSC [RES]

Pdm Tool Array Empty Search — MOVE
Source: PdM_Cfg.TL.n 3000
Dest: PdmToolEmptyFSC.LEN 3000

Pdm Tool Array Empty Search — MOVE
Source: 0
Dest: PdmToolEmptyFSC.POS 1

Pdm Tool Array Empty Search — FSC
Control: PdmToolEmptyFSC
Length: 3000
Position: 1
Mode: ALL
Expression: PdM_Tool[PdmToolEmptyFSC.POS].Cfg.EqToolID = 0
EN DN ER

Store Array Position for Update/Insert

26

Pdm Tool Array Search: PdmToolFSC.FD

MOVE
Source: PdmToolFSC.POS 839
Dest: TLi 839

Pdm Tool Array Search: PdmToolFSC.FD [/]

Pdm Tool Array Empty Search: PdmToolEmptyFSC.FD

MOVE
Source: PdmToolEmptyFSC.POS 1
Dest: TLi 839

ADD
Source A: Pdm_ToolCount 756
Source B: 1
Dest: Pdm_ToolCount 756

GT
Source A: TLi 839
Source B: PdM_Cfg.TL.ei 914

PdM Config Array End Index

MOVE
Source: TLi 839
Dest: PdM_Cfg.TL.ei 914

Tool Array is Full

27

Pdm Tool Array Search: PdmToolFSC.FD [/]

Pdm Tool Array Empty Search: PdmToolEmptyFSC.FD [/]

PdM Config Tool Configuration Array is Full: PdM_Cfg.ToolArrayFull [L]

lbl_Exit [JMP]

Configuration Response from Server via Transaction Manager

28

Process Data and Store

Save but don't enable Tool if no Data values have been configured

PdM Tool Array
PdM_Tool[TLi].Cfg.DownloadPend
U

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID>
-1
Dest PdM_Tool[TLi].Cfg.EqToolID
1004032

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.EqID
<EDC.ServerPdm.DataRspLast.EqID>
-1
Dest PdM_Tool[TLi].Cfg.EQID
1004

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.ToolID
<EDC.ServerPdm.DataRspLast.ToolID>
-1
Dest PdM_Tool[TLi].Cfg.ToolID
32

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.SamplePeriodID
<EDC.ServerPdm.DataRspLast.SamplePeriodID>
-1
Dest PdM_Tool[TLi].Cfg.SamplePeriodID
5

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.DATsi
<EDC.ServerPdm.DataRspLast.DATsi>
-1
Dest PdM_Tool[TLi].Cfg.DATsi
0

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Dest PdM_Tool[TLi].Cfg.DATei
3

PdM Tool Array
COP
Source ServerPdm.DataRspLast.Data[0]
<EDC.ServerPdm.DataRspLast.Data[0]>
Dest PdM_Tool[TLi].Cfg.Data[0]
Length 10

PdM Tool Array
PdM_Tool[TLi].Cfg.Enable
U

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)
EQ
Source A ServerPdm.DataRspLast.ToolEnable
<EDC.ServerPdm.DataRspLast.ToolEnable>
0
Source B 1

Server PdM Configuration from Transaction Manager Data Start Index
NE
Source A ServerPdm.DataRspLast.DATsi
<EDC.ServerPdm.DataRspLast.DATsi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data End Index
NE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B -1

PdM Tool Array
PdM_Tool[TLi].Cfg.Enable
L

PdM Tool Array
PdM_Tool[TLi].Cfg.TriggerEnable
U

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)
EQ
Source A ServerPdm.DataRspLast.TriggerEnable
<EDC.ServerPdm.DataRspLast.TriggerEnable>
0
Source B 1

PdM Tool Array
PdM_Tool[TLi].Cfg.TriggerEnable
L

Configuration Response from Server via Transaction Manager

Initialize Tool's Last Sample value to -1 to prevent a sample when the rate is changed
(PdM code will synch newly configured tool to the current Sample Period without sampling.)

Initialize Tool's Start signal to off to force Init/Reset of PdM Data/Array
(Config change to Data NETi/DATi wasn't clearing previously collected data)

Sunday Configuration Download will reset all Sampled tools with some data loss for the current sample.

29

PdM Tool Array
PdM_Tool[TLi].Cfg.TriggerEnable
/

PdM Tool Array
MOVE
Source -1
Dest PdM_Tool[TLi].Ctl.PeriodLast
185

PdM Tool Array
PdM_Tool[TLi].Ctl.Start
U

Data OK

30

Server Interface
Status State = Green
- Tool Rx Data OK
EDC_Status01.ServerPdm.Update.ToolRxDataOK
L

Update PdM State for
Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

Exit Routine

31

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 102 - Trend

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 852
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

2

Server PdM Configuration from Transaction Manager EquipmentID (0001-9999) & ToolID (001-999) combined for fast searches

GT	
Source A	ServerPdm.DataRspLast.EqToolID
	<EDC.ServerPdm.DataRspLast.EqToolID>
	-1
Source B	0

Server PdM Configuration from Transaction Manager Equipment ID (0001-9999)

NE	
Source A	ServerPdm.DataRspLast.EqID
	<EDC.ServerPdm.DataRspLast.EqID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Tool ID (001-999)

GT	
Source A	ServerPdm.DataRspLast.ToolID
	<EDC.ServerPdm.DataRspLast.ToolID>
	-1
Source B	0

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)

NE	
Source A	ServerPdm.DataRspLast.ToolEnable
	<EDC.ServerPdm.DataRspLast.ToolEnable>
	0
Source B	-1

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)

NE	
Source A	ServerPdm.DataRspLast.TriggerEnable
	<EDC.ServerPdm.DataRspLast.TriggerEnable>
	0
Source B	-1

Server PdM Configuration from Transaction Manager Sample Period ID

NE	
Source A	ServerPdm.DataRspLast.SamplePeriodID
	<EDC.ServerPdm.DataRspLast.SamplePeriodID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Sample Rate ID

NE	
Source A	ServerPdm.DataRspLast.SampleRateID
	<EDC.ServerPdm.DataRspLast.SampleRateID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data Start Index

NE	
Source A	ServerPdm.DataRspLast.DATsi
	<EDC.ServerPdm.DataRspLast.DATsi>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data End Index

NE	
Source A	ServerPdm.DataRspLast.DATei
	<EDC.ServerPdm.DataRspLast.DATei>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Network Index

NE	
Source A	ServerPdm.DataRspLast.Data[0].NETi
	<EDC.ServerPdm.DataRspLast.Data[0].NETi>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data Index

NE	
Source A	ServerPdm.DataRspLast.Data[0].DATi
	<EDC.ServerPdm.DataRspLast.Data[0].DATi>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE	
Source A	ServerPdm.DataRspLast.Data[0].TypeID
	<EDC.ServerPdm.DataRspLast.Data[0].TypeID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Response has Good Data Record

ServerPdm.Ctl.RspGoodData

<EDC.ServerPdm.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
[/]

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID>
-1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4314
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red-Trend Rx Data ER
EDC_Status01.ServerPdm.Update.TrendRxDataER
(L)

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm (...)
Status EDC_Status01.ServerPdm

lbl_Exit
(JMP)

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

4

Pdm Trend Array Search
PdmTrendFSC
(RES)

PdM Config Array End Index

GT
Source A PdM_Cfg.TRN.ei 3
Source B 0

Pdm Trend Array Search

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest PdmTrendFSC.LEN 4

Pdm Trend Array Search

MOVE
Source 0
Dest PdmTrendFSC.POS 3

Pdm Trend Array Search

FSC
Control PdmTrendFSC
Length 4
Position 3
Mode ALL
Expression PdM_Trend[PdmTrendFSC.POS].Cfg.EqToolID=ServerPdm.DataRspLast.EqToolID
(EN) (DN) (ER)

Configuration Response from Server via Transaction Manager

Search Pdm Config for Empty Position

5

Pdm Trend Array Search PdmTrendFSC.FD

PdmTrendEmptyFSC (RES)

MOVE
Source PdM_Cfg.TRN.n 1000
Dest PdmTrendEmptyFSC.LEN 1000

MOVE
Source 0
Dest PdmTrendEmptyFSC.POS 3

FSC
Control PdmTrendEmptyFSC
Length 1000
Position 3
Mode ALL
Expression PdM_Trend[PdmTrendEmptyFSC.POS].Cfg.EqToolID=0
EN
DN
ER

Store Array Position for Update/Insert

6

Pdm Trend Array Search PdmTrendFSC.FD

MOVE
Source PdmTrendFSC.POS 3
Dest TRNi 3

Pdm Trend Array Search PdmTrendFSC.FD

PdmTrendEmptyFSC.FD

MOVE
Source PdmTrendEmptyFSC.POS 3
Dest TRNi 3

ADD
Source A Pdm_TrendCount 3
Source B 1
Dest Pdm_TrendCount 3

GT
Source A TRNi 3
Source B PdM_Cfg.TRN.ei 3

PdM Config Array End Index

MOVE
Source TRNi 3
Dest PdM_Cfg.TRN.ei 3

Trend Array is Full

7

Pdm Trend Array Search PdmTrendFSC.FD

PdmTrendEmptyFSC.FD

PdM Config Trend Configuration Array is Full
PdM_Cfg.TrendArrayFull (L)

lbl_Exit (JMP)

Configuration Response from Server via Transaction Manager

8

Process Data and Store

PdM Trend Array
PdM_Trend[TRNi].Cfg.DownloadPend
U

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID> -1
Dest PdM_Trend[TRNi].Cfg.EqToolID
967001

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.EqID
<EDC.ServerPdm.DataRspLast.EqID> -1
Dest PdM_Trend[TRNi].Cfg.EQID
967

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.ToolID
<EDC.ServerPdm.DataRspLast.ToolID> -1
Dest PdM_Trend[TRNi].Cfg.ToolID
1

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.SamplePeriodID
<EDC.ServerPdm.DataRspLast.SamplePeriodID> -1
Dest PdM_Trend[TRNi].Cfg.SamplePeriodID
0

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.SampleRateID
<EDC.ServerPdm.DataRspLast.SampleRateID> -1
Dest PdM_Trend[TRNi].Cfg.SampleRateID
14

PdM Trend Array
COP
Source ServerPdm.DataRspLast.Data[0]
<EDC.ServerPdm.DataRspLast.Data[0]>
Dest PdM_Trend[TRNi].Cfg.Data
Length 1

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)

PdM Trend Array
PdM_Trend[TRNi].Cfg.Enable
U

EQ
Source A ServerPdm.DataRspLast.ToolEnable
<EDC.ServerPdm.DataRspLast.ToolEnable> 0
Source B 1

PdM Trend Array
PdM_Trend[TRNi].Cfg.Enable
L

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)

PdM Trend Array
PdM_Trend[TRNi].Cfg.TriggerEnable
U

EQ
Source A ServerPdm.DataRspLast.TriggerEnable
<EDC.ServerPdm.DataRspLast.TriggerEnable> 0
Source B 1

PdM Trend Array
PdM_Trend[TRNi].Cfg.TriggerEnable
L

Configuration Response from Server via Transaction Manager

Initialize Trend's Last Sample value to -1
to prevent a sample when the rate is changed
(PdM code will sync newly configured trend to the current Sample Period without sampling.)

9

PdM Trend Array
PdM_Trend[TRNi].Cfg.TriggerEnable
/

PdM Trend Array
MOVE
Source -1
Dest PdM_Trend[TRNi].Ctl.PeriodLast
0

PdM Trend Array
PdM_Trend[TRNi].Ctl.Start
U

Data OK

10

Server Interface
Status State = Green
- Trend Rx Data OK
EDC_Status01.ServerPdm.Update.TrendRxDataOK
L

Update PdM State for
Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 103 - Network

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 853
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server PdM Configuration from Transaction Manager Shop Key - for DB Ack only

NE
Source A ServerPdm.DataRspLast.ShopKey <EDC.ServerPdm.DataRspLast.ShopKey> -1
Source B -1

Server PdM Configuration from Transaction Manager Network Index

LIMIT
Low Limit PdM_Cfg.NET.si 0
Test ServerPdm.DataRspLast.Data[0].NETi <EDC.ServerPdm.DataRspLast.Data[0].NETi> -1
High Limit PdM_Cfg.NET.mi 299

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[0].TypeID <EDC.ServerPdm.DataRspLast.Data[0].TypeID> -1
Source B -1

Diagnostic

ADD
Source A NetDataGoodCount 74724
Source B 1
Dest NetDataGoodCount 74724

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData <EDC.ServerPdm.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Diagnostic

ADD
Source A: NetDataBadCount 3293
Source B: 1
Dest: NetDataBadCount 3293

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source: ServerPdm.DataRspLast.Data[0].NETi <EDC.ServerPdm.DataRspLast.Data[0].NETi> -1
Dest: ServerPdm.Ctl.CommandID <EDC.ServerPdm.Ctl.CommandID> 298

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4315
Input Par: ServerPdm.Ctl.ReqInProgressTMR.ACC <EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par: ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command>
Input Par: ServerPdm.Ctl.CommandID <EDC.ServerPdm.Ctl.CommandID>
Input Par: ServerPdm.Ctl.TransactionNumber <EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red- Network Rx Data ER
EDC_Status01.ServerPdm.Update.NetworkRxDataER
L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...
Status: EDC_Status01.ServerPdm

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

4

MOVE
Source ServerPdm.DataRspLast.Data[0].NETi
<EDC.ServerPdm.DataRspLast.Data[0].NETi>
-1
Dest NETi
299

Network Modules

MOVE
Source ServerPdm.DataRspLast.Data[0].TypeID
<EDC.ServerPdm.DataRspLast.Data[0].TypeID>
-1
Dest PdM_NetMod[NETi].Cfg.TypeID
2

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>

Network Modules
PdM_NetMod[NETi].Cfg.Enable
L

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>

Network Modules
PdM_NetMod[NETi].Cfg.Enable
U

Data OK

5

Server Interface Status State = Green - Reset Rx Data OK
EDC_Status01.ServerPdm.Update.ResetRxDataOK
L

Update PdM State for Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Configuration Initialization before full download

PdM Interface Initialization
(called before Full Configuration Download)

Tools, Trends and Network are not actually reset before full download because this would cause dropouts in the Sampled PdM data
Existing Networks are updated by new configurations
Existing Tools and Trends are marked for Download Pending -- and unmarked when updated by new configuration.
Pending when the Full Configuration Download is complete, any Tools and Trends still marked Download Pending are disabled.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 858
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Tool Loop

2

CLR
Dest TOOLi 915

Tool Loop
mark for Download Pending

3

lbl_ToolLoop
LBL

PdM Tool Array
PdM_Tool[TOOLi].Cfg.DownloadPend
L

ADD
Source A TOOLi 915
Source B 1
Dest TOOLi 915

LE
Source A TOOLi 915
Source B PdM_Cfg.TL.ei 914

lbl_ToolLoop
JMP

Configuration Initialization before full download

Initialize Trend Loop

4

CLR
Dest TRNDi 4

Trend Loop
mark for Download Pending

5

lbl_TrendLoop
LBL

PdM Trend Array
PdM_Trend[TRNDi].Cfg.DownloadPend
L

ADD
Source A TRNDi 4
Source B 1
Dest TRNDi 4

LE
Source A TRNDi 4
Source B PdM_Cfg.TRN.ei 3

lbl_TrendLoop
JMP

Reset Pdm Config Errors

6

PdM Config Tool Configuration Array is Full
PdM_Cfg.ToolArrayFull
U

PdM Config Trend Configuration Array is Full
PdM_Cfg.TrendArrayFull
U

PdM Config Tool Processing Error - Invalid EQID
PdM_Cfg.ToolInvalidEQID
U

PdM Config Tool Processing Error - Invalid Data Index
PdM_Cfg.ToolInvalidDataIndex
U

PdM Config Tool Processing Error - Invalid Sample Period
PdM_Cfg.ToolInvalidSamplePeriod
U

PdM Config Tool Processing Error - Invalid Network Data
PdM_Cfg.ToolInvalidNetworkData
U

PdM Config Trend Processing Error - Invalid EQID
PdM_Cfg.TrendInvalidEQID
U

PdM Config Trend Processing Error - Invalid Sample Period
PdM_Cfg.TrendInvalidSamplePeriod
U

PdM Config Trend Processing Error - Invalid Sample Rate
PdM_Cfg.TrendInvalidSampleRate
U

PdM Config Trend Processing Error - Invalid Network Data
PdM_Cfg.TrendInvalidNetworkData
U

Exit Routine

7

lbl_Exit
LBL

NOP

(End)

Configuration Udpate after full download

PdM Interface Initialization
(called after Full Configuration Download)

Tools, Trends and Network are not actually reset before full download because this would cause dropouts in the Sampled PdM data
Existing Networks are updated by new configurations
Existing Tools and Trends are marked for Download Pending -- and unmarked when updated by new configuration.
Pending when the Full Configuration Download is complete, any Tools and Trends still marked Download Pending are disabled.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 859
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Tool Loop

2

CLR
Dest Pdm_ToolCount 756

CLR
Dest TOOLi 915

Configuration Udpate after full download

Tool Loop
Clear Tool Config, Control and Data if still marked for Download Pending

3

lbl_ToolLoop
LBL

PdM Tool Array
PdM_Tool[TOOLi].Cfg.DownloadPend

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Cfg
Length 1

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Ctl
Length 1

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Data[0]
Length PdM_Cfg.DAT.n

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Array[0]
Length PdM_Cfg.DAT.n

PdM Tool Array
PdM_Tool[TOOLi].Cfg.Enable

ADD
Source A Pdm_ToolCount 756
Source B 1
Dest Pdm_ToolCount 756

ADD
Source A TOOLi 915
Source B 1
Dest TOOLi 915

LE
Source A TOOLi 915
Source B PdM_Cfg.TL.ei 914

lbl_ToolLoop
JMP

Initialize Trend Loop

4

CLR
Dest Pdm_TrendCount 3

CLR
Dest TRNDi 4

Configuration Udpate after full download

Trend Loop
Clear Tool Config, Control and Data if still marked for Download Pending

5

lbl_TrendLoop
LBL

PdM Trend Array
PdM_Trend[TRNDi].Cfg.DownloadPend

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Cfg
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Ctl
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Data
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Array
Length 1

PdM Trend Array
PdM_Trend[TRNDi].Cfg.Enable

ADD
Source A Pdm_TrendCount 3
Source B 1
Dest Pdm_TrendCount 3

ADD
Source A TRNDi 4
Source B 1
Dest TRNDi 4

LE
Source A TRNDi 4
Source B PdM_Cfg.TRN.ei 3

lbl_TrendLoop
JMP

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Status Request to Server via Transaction Manager

Status Data Request to Server via Transaction Manager

- Logix Requests Data (1-13 Command for each row of Status table)
- Server Responds with Data (1-13 Command) or NoData (-1 Command)

- if Data, Logix stores Data for HMI Overview display
- if NoData, Logix waits and then Requests Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 860
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Status Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data & not Last Table Row

SQL Job runs every 20 seconds to update the Status data
Gateway refreshes Status data every 10 seconds (2x update rate)

2

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>
[/]

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>
[ ]

Server Status from Transaction Manager Response has Last Row Record
ServerStat.Ctl.RspLastRow
<EDC.ServerStat.Ctl.RspLastRow>
[/]

Server Status from Transaction Manager Request Next Record Timer

MOVE
Source ServerStat.Ctl.ReqFastTime
<EDC.ServerStat.Ctl.ReqFastTime>
0
Dest ServerStat.Ctl.ReqNextTMR.PRE
<EDC.ServerStat.Ctl.ReqNextTMR.PRE>
10000

Server Status from Transaction Manager Request Next Record Timer

TON
Timer ServerStat.Ctl.ReqNextTMR
<EDC.ServerStat.Ctl.ReqNextTMR>
Preset 10000
Accum 1945
EN
DN

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>
[/]

Server Status from Transaction Manager Response has Last Row Record
ServerStat.Ctl.RspLastRow
<EDC.ServerStat.Ctl.RspLastRow>
[ ]

Server Status from Transaction Manager Request Next Record Timer

MOVE
Source ServerStat.Ctl.ReqSlowTime
<EDC.ServerStat.Ctl.ReqSlowTime>
10000
Dest ServerStat.Ctl.ReqNextTMR.PRE
<EDC.ServerStat.Ctl.ReqNextTMR.PRE>
10000

Request Next
Ready for next Transaction Manager Trigger

3

Server Status from Transaction Manager Request Next Record Timer
ServerStat.Ctl.ReqNextTMR.DN
<EDC.ServerStat.Ctl.ReqNextTMR.DN>
[ ]

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>
( )

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>
(L)

Server Status from Transaction Manager Request Record Complete
ServerStat.Ctl.ReqComplete
<EDC.ServerStat.Ctl.ReqComplete>
(U)

Status Request to Server via Transaction Manager

Request Next Initialize
if Response Good Data, request next Row of Data in Response to Server
if not Response Good Data, send new Data Request after wait interval

4

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Status Request to Server via Transaction Manager - Data Init

JSR
Routine Name _861_StatRequestInit

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

5

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B 1
Dest ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B 1

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319

Status Request to Server via Transaction Manager

6

Request Next
Log Transaction & Performance

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4100
Input Par	ServerStat.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerStat.Data.Command
	<EDC.ServerStat.Data.Command>
Input Par	ServerStat.Ctl.CommandID
	<EDC.ServerStat.Ctl.CommandID>
Input Par	ServerStat.Ctl.TransactionNumber
	<EDC.ServerStat.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR	
Dest	TempPerfMsg.TransTMR.ACC
	110

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4102
Input Par	ServerStat.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerStat.Data.Command
	<EDC.ServerStat.Data.Command>
Input Par	ServerStat.Ctl.CommandID
	<EDC.ServerStat.Ctl.CommandID>
Input Par	ServerStat.Ctl.TransactionNumber
	<EDC.ServerStat.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerStat.Msg.EN

Status Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

7

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B ServerStat.Ctl.Interface
<EDC.ServerStat.Ctl.Interface>
7319

Server Status from Transaction Manager Request Record Complete
ServerStat.Ctl.ReqComplete
<EDC.ServerStat.Ctl.ReqComplete>
L

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify

8

Server Status from Transaction Manager Request Record Complete
ServerStat.Ctl.ReqComplete
<EDC.ServerStat.Ctl.ReqComplete>

Server Status from Transaction Manager Request Complete Delay Timer

MOVE
Source 50
Dest ServerStat.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerStat.Ctl.ReqCompleteDelayTMR.PRE>
50

Server Status from Transaction Manager Request Complete Delay Timer

TON
Timer ServerStat.Ctl.ReqCompleteDelayTMR
<EDC.ServerStat.Ctl.ReqCompleteDelayTMR>
Preset 50
Accum 55
EN
DN

Server Status from Transaction Manager Request Complete Delay Timer
ServerStat.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerStat.Ctl.ReqCompleteDelayTMR.DN>

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>
U

Status Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforeced in data update.)

9

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

Server Status from Transaction Manager Request Record Complete 1-shot
ServerStat.Ctl.ReqCompleteONS
<EDC.ServerStat.Ctl.ReqCompleteONS>
ONS

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4101
Input Par ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerStat.Update.MsgRxOK
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...
Status EDC_Status01.ServerStat

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerStat.Msg.DN

Status Response from Server via Transaction Manager
JSR
Routine Name _865_StatResponse

Status Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

Dec 31, 2025 - reduced from 62 to 12 seconds to capture failure before next TM service restart

10

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>
/

Server Status from Transaction Manager Request in Progress Timer

MOVE
Source 12000
Dest ServerStat.Ctl.ReqInProgressTMR.PRE
<EDC.ServerStat.Ctl.ReqInProgressTMR.PRE>
12000

Server Status from Transaction Manager Request in Progress Timer

TON
Timer ServerStat.Ctl.ReqInProgressTMR
<EDC.ServerStat.Ctl.ReqInProgressTMR>
Preset 12000
Accum 0
EN
DN

Server Status from Transaction Manager Request in Progress Timer
ServerStat.Ctl.ReqInProgressTMR.DN
<EDC.ServerStat.Ctl.ReqInProgressTMR.DN>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Status Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

11

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4104
Input Par ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Server Interface Status State = Red-Msg Rx ER
EDC_Status01.ServerStat.Update.MsgRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...
Status EDC_Status01.ServerStat

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerStat.Msg.ER

Server Status from Transaction Manager Request in Progress Timer
ServerStat.Ctl.ReqInProgressTMR
<EDC.ServerStat.Ctl.ReqInProgressTMR>
RES

Status Request to Server via Transaction Manager

Enable RB high-speed data transfer
use the Server Status Message to monitor TM status

If TM communications stop, throttle RB speed until it has recovered for 1 minute.
This provides time for TM to restart properly if the applicaiton was restarted.
(was seeing TM Log Error = The Worker thread could not signal the Scheduler thread)

12

s:fs

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.TT

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

Server Status TM Transaction OK Enable High Speed RB

MOVE
Source 60000
Dest ServerStat_TMokTMR.PRE 60000

Server Status TM Transaction OK Enable High Speed RB

TON
Timer ServerStat_TMokTMR
Preset 60000
Accum 60008
EN
DN

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

StatCompleteONS
ONS

ADD
Source A ServerStat_CompleteCount 6917818
Source B 1
Dest ServerStat_CompleteCount 6917818

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

StatTimeoutONS
ONS

ADD
Source A ServerStat_TimeoutCount 135
Source B 1
Dest ServerStat_TimeoutCount 135

Status Request to Server via Transaction Manager

(End)

Status Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, request next Row of Data in Response to Server
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 861
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Status Request to Server via Transaction Manager - Data Init

Initialize Data Values
(FLL -1 corrupts FLOAT values and causes TM Transaction Failure - overwrite with MOV -1)

2

Server Status from Transaction Manager Status Transaction Data

FLL
Source -1
Dest ServerStat.Data
<EDC.ServerStat.Data>
Length 1

Server Status from Transaction Manager Status Description

FLL
Source 0
Dest ServerStat.Data.Desc
<EDC.ServerStat.Data.Desc>
Length 1

Server Status from Transaction Manager Last Process Time

FLL
Source 0
Dest ServerStat.Data.LastProcTime
<EDC.ServerStat.Data.LastProcTime>
Length 1

Server Status from Transaction Manager Status Description

MOVE
Source 1
Dest ServerStat.Data.Desc.DATA[0]
<EDC.ServerStat.Data.Desc.DATA[0]>
'T'

Server Status from Transaction Manager Last Process Time

MOVE
Source 1
Dest ServerStat.Data.LastProcTime.DATA[0]
<EDC.ServerStat.Data.LastProcTime.DATA[0]>
'2'

Server Status from Transaction Manager Backlog Time (minutes)

MOVE
Source -1
Dest ServerStat.Data.BacklogTime
<EDC.ServerStat.Data.BacklogTime>
0.3

Status Request to Server via Transaction Manager - Data Init

Increment Command
(Table Row Index)
DN005 Health Status Alarm Callout

3

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

Server Status from Transaction Manager Command-specific Record ID for TransLog

ADD
Source A ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Source B 1
Dest ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13

Server Status from Transaction Manager Command-specific Record ID for TransLog

LT
Source A ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Source B 1

Server Status from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source 1
Dest ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13

Server Status from Transaction Manager Command-specific Record ID for TransLog

GT
Source A ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Source B 13

Status Request to Server via Transaction Manager - Data Init

Load Command

Server Status from Transaction Manager Request/Response Command

MOVE
Source ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Dest ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13

4

Store Last Request

Server Status from Transaction Manager Transaction Data Last Request

COP
Source ServerStat.Data
<EDC.ServerStat.Data>
Dest ServerStat.DataLastReq
<EDC.ServerStat.DataLastReq>
Length 1

5

(End)

Status Response from Server via Transaction Manager

Status Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 865
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Response Good Data

2

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

U

Status Response from Server via Transaction Manager

Check Response Command for No Data
DN005 Health Status Alarm Callout

3

Server Status from Transaction Manager Request/Response Command

LT
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B 1

Server Status from Transaction Manager Request/Response Command

GT
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B 15

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4113
Input Par ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

lbl_Exit
JMP

Process Response Command

4

Status Response from Server via Transaction Manager

JSR
Routine Name _866_StatResponseRow

Exit Routine

lbl_Exit
5 LBL

NOP

Status Response from Server via Transaction Manager

Store Last Response and Last Good Response

6

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

Server Status from Transaction Manager Transaction Data Last Response

COP
Source ServerStat.Data
<EDC.ServerStat.Data>
Dest ServerStat.DataLastResp
<EDC.ServerStat.DataLastResp>
Length 1

Server Status from Transaction Manager Transaction Data Last Good Response

COP
Source ServerStat.Data
<EDC.ServerStat.Data>
Dest ServerStat.DataLastRespGood
<EDC.ServerStat.DataLastRespGood>
Length 1

(End)

Status Response from Server via Transaction Manager

Process Status Row from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 866
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Status Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data
DN005 Health Status Alarm Callout

2

Server Status from Transaction Manager Request/Response Command

EQ
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13

Server Status from Transaction Manager Request/Response Command

EQ
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B ServerStat.Data.StatusID
<EDC.ServerStat.Data.StatusID>
13

Server Status from Transaction Manager Status ID

LIMIT
Low Limit 1
Test ServerStat.Data.StatusID
<EDC.ServerStat.Data.StatusID>
13
High Limit EDC_HMI.CFG.Stat.n
15

Server Status from Transaction Manager Status Description

NE
Source A ServerStat.Data.Desc.DATA[0]
<EDC.ServerStat.Data.Desc.DATA[0]>
'T'
Source B 1

Server Status from Transaction Manager Last Process Time

NE
Source A ServerStat.Data.LastProcTime.DATA[0]
<EDC.ServerStat.Data.LastProcTime.DATA[0]>
'2'
Source B 1

Server Status from Transaction Manager Backlog Time (minutes)

NE
Source A ServerStat.Data.BacklogTime
<EDC.ServerStat.Data.BacklogTime>
0.3
Source B -1

Server Status from Transaction Manager Backlog Count (events)

NE
Source A ServerStat.Data.BacklogCount
<EDC.ServerStat.Data.BacklogCount>
3
Source B -1

Server Status from Transaction Manager Shift Record Count (events)

NE
Source A ServerStat.Data.ShiftTotal
<EDC.ServerStat.Data.ShiftTotal>
96603
Source B -1

Server Status from Transaction Manager Shift Error Count (events)

GE
Source A ServerStat.Data.ErrorCount
<EDC.ServerStat.Data.ErrorCount>
0
Source B 0

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>
L

Status Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>
/

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4114
Input Par ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerStat.Update.RxDataER
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...
Status EDC_Status01.ServerStat

lbl_Exit
JMP

Data Good - Reset Data ER

4

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerStat.State.RxDataER

Server Interface Status Reset Errors
EDC_Status01.ServerStat.Update.ResetER
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...
Status EDC_Status01.ServerStat

Status Response from Server via Transaction Manager

Store Data
(Server Rows 1-15 stored in HMI Array 0-14)

5

Row Index

SUB
Source A ServerStat.Data.StatusID
<EDC.ServerStat.Data.StatusID>
13
Source B 1
Dest Ri
12

HMI Server Status Block

COP
Source ServerStat.Data.Desc
<EDC.ServerStat.Data.Desc>
Dest EDC_HMI.Oview.Status[Ri].Desc
Length 1

HMI Server Status Block

COP
Source ServerStat.Data.LastProcTime
<EDC.ServerStat.Data.LastProcTime>
Dest EDC_HMI.Oview.Status[Ri].LastProcTime
Length 1

HMI Server Status Block

MOVE
Source ServerStat.Data.BacklogTime
<EDC.ServerStat.Data.BacklogTime>
0.3
Dest EDC_HMI.Oview.Status[Ri].BacklogTime
0.3

HMI Server Status Block

MOVE
Source ServerStat.Data.BacklogCount
<EDC.ServerStat.Data.BacklogCount>
3
Dest EDC_HMI.Oview.Status[Ri].BacklogCount
3

HMI Server Status Block

MOVE
Source ServerStat.Data.ShiftTotal
<EDC.ServerStat.Data.ShiftTotal>
96603
Dest EDC_HMI.Oview.Status[Ri].ShiftTotal
96603

HMI Server Status Block

MOVE
Source ServerStat.Data.ErrorCount
<EDC.ServerStat.Data.ErrorCount>
0
Dest EDC_HMI.Oview.Status[Ri].ErrCount
0

Status Response from Server via Transaction Manager

Update HMI Overview Status Color

6

Server Status from Transaction Manager Backlog Time (minutes)

LT
Source A ServerStat.Data.BacklogTime <EDC.ServerStat.Data.BacklogTime> 0.3
Source B 5

HMI Server Status Block

MOVE
Source 1
Dest EDC_HMI.Oview.Status[Ri].BackColor 1

Server Status from Transaction Manager Backlog Time (minutes)

LIMIT
Low Limit 5
Test ServerStat.Data.BacklogTime <EDC.ServerStat.Data.BacklogTime> 0.3
High Limit 10

HMI Server Status Block

MOVE
Source 2
Dest EDC_HMI.Oview.Status[Ri].BackColor 1

Server Status from Transaction Manager Backlog Time (minutes)

GE
Source A ServerStat.Data.BacklogTime <EDC.ServerStat.Data.BacklogTime> 0.3
Source B 10

HMI Server Status Block

MOVE
Source 3
Dest EDC_HMI.Oview.Status[Ri].BackColor 1

Last Row of Table received (delay before next table update)

7

Row Index

GE
Source A Ri 12
Source B EDC_HMI.CFG.Stat.ei 12

Server Status from Transaction Manager Response has Last Row Record ServerStat.Ctl.RspLastRow <EDC.ServerStat.Ctl.RspLastRow>

Status Response from Server via Transaction Manager

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

Detect and Callout CIS Message Errors

Detect and Callout CIS Message Errors
- Each Error increment latches SendCallout
- When SendCallout and 1-minute dwell since last – set Callout bit for 30 seconds
(this provides 30 seconds on/off for each event)
(also guarantees first & last events will be annunciated regardless of quantity during a 'burst' of events)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 870
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Server Status Table
Load CIS Row Index
(SQL Status ID 6 = Gateway Status Row 5)

2

MOVE
Source 5
Dest CisServerStatRow 5

Detect new CIS Error

3

HMI Server Status Block

NE
Source A EDC_HMI.Oview.Status[CisServerStatRow].ErrCount 0
Source B CisLastErrCount 0

MOVE
Source EDC_HMI.Oview.Status[CisServerStatRow].ErrCount 0
Dest CisLastErrCount 0

NE
Source A CisLastErrCount 0
Source B 0

CisNewError
(L)

Detect and Callout CIS Message Errors

Send CIS Error to Callout for 30 seconds
Block new Errors until this Error sent and Delay is done
(might not send Callout for each Error, but at minimum will indicate First and Last in a burst without bugging the heck out of the recipient)

4

CisNewError CisErrorSendDelay CisNewError CisSendErrorTMR.DN

CisErrorSend

MOVE
Source 30000
Dest CisSendErrorTMR.PRE 30000

TON
Timer CisSendErrorTMR
Preset 30000
Accum 0
EN
DN

CisErrorSend CisErrorSendDelay

Server Status from Transaction Manager Response has CIS Error - Send Callout Bit
ServerStat.Ctl.CisErrorSendCallout
<EDC.ServerStat.Ctl.CisErrorSendCallout>

Delay 30 seconds before sending any new Errors

5

CisErrorSend CisErrorSendDelay

MOVE
Source 30000
Dest CisSendDelayTMR.PRE 30000

TON
Timer CisSendDelayTMR
Preset 30000
Accum 0
EN
DN

CisSendDelayTMR.DN

CisErrorSendDelay

(End)

Detect and Callout Health Status Errors

DN005 Health Status Alarm Callout
Detect and Callout Health Status Errors
- Each Error increment latches SendCallout
- When SendCallout and 1-minute dwell since last – set Callout bit for 30 seconds
(this provides 30 seconds on/off for each event)
(also guarantees first & last events will be annunciated regardless of quantity during a 'burst' of events)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 871
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Server Status Table
Load Health Status Row Index
(SQL Status ID 7 = Gateway Status Row 6)

2

MOVE
Source 6
Dest HealthServStatRow 6

Detect new Health Status Error

3

HMI Server Status Block

NE
Source A EDC_HMI.Oview.Status[HealthServStatRow].ErrCount 0
Source B HealthLastErrCount 0

MOVE
Source EDC_HMI.Oview.Status[HealthServStatRow].ErrCount 0
Dest HealthLastErrCount 0

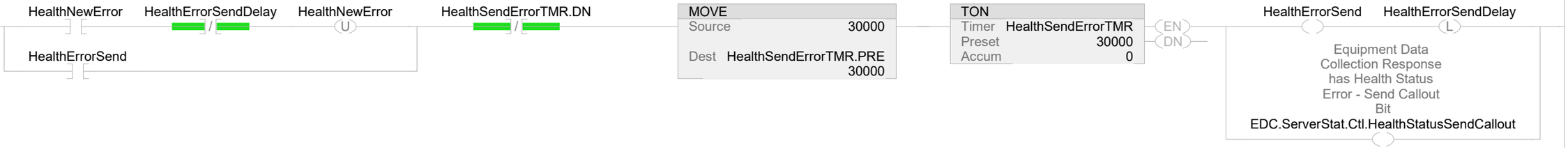
NE
Source A HealthLastErrCount 0
Source B 0

HealthNewError
(L)

Detect and Callout Health Status Errors

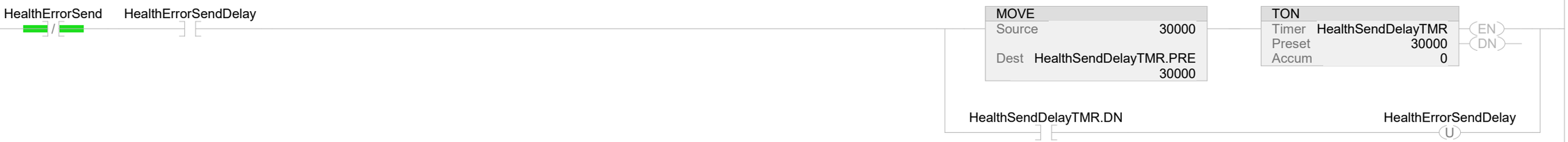
Send Health Status Error to Callout for 30 seconds
Block new Errors until this Error sent and Delay is done
(might not send Callout for each Error, but at minimum will indicate First and Last in a burst without bugging the heck out of the recipient)

4



Delay 30 seconds before sending any new Errors

5



(End)

EQ Health Request to Server via Transaction Manager

EQ Health Data Request to Server via Transaction Manager
DN014 Health Displays

- Logix Requests Data
- Server Responds with Data (EQID and Status) or NoData (0 Command)

- if Data, Logix stores Data for Callout and Acks
- if NoData, Logix waits and then Requests Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 880
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

EQ Health Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had good response with new data

2

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

Server EQ Health from Transaction Manager Request Next Record Timer

MOVE
Source ServerHealth.Ctl.ReqFastTime
<EDC.ServerHealth.Ctl.ReqFastTime>
0
Dest ServerHealth.Ctl.ReqNextTMR.PRE
<EDC.ServerHealth.Ctl.ReqNextTMR.PRE>
1000

Server EQ Health from Transaction Manager Request Next Record Timer

TON
Timer ServerHealth.Ctl.ReqNextTMR
<EDC.ServerHealth.Ctl.ReqNextTMR>
Preset 1000
Accum 678
EN
DN

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

Server EQ Health from Transaction Manager Request Next Record Timer

MOVE
Source ServerHealth.Ctl.ReqSlowTime
<EDC.ServerHealth.Ctl.ReqSlowTime>
1000
Dest ServerHealth.Ctl.ReqNextTMR.PRE
<EDC.ServerHealth.Ctl.ReqNextTMR.PRE>
1000

Request Next
Ready for next Transaction Manager Trigger

3

Server EQ Health from Transaction Manager Request Next Record Timer
ServerHealth.Ctl.ReqNextTMR.DN
<EDC.ServerHealth.Ctl.ReqNextTMR.DN>

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>
L

Server EQ Health from Transaction Manager Request Record Complete
ServerHealth.Ctl.ReqComplete
<EDC.ServerHealth.Ctl.ReqComplete>
U

EQ Health Request to Server via Transaction Manager

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

4

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

EQ Helath Request to Server via Transaction Manager - Data Init

JSR
Routine Name _881_HealthRequestInit

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

5

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B 1
Dest ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B 1

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750

EQ Health Request to Server via Transaction Manager

6

Request Next
Log Transaction & Performance

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4200
Input Par	ServerHealth.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par	0
Input Par	ServerHealth.Data.EQID
	<EDC.ServerHealth.Data.EQID>
Input Par	ServerHealth.Ctl.TransactionNumber
	<EDC.ServerHealth.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR

Dest	TempPerfMsg.TransTMR.ACC
	110

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4202
Input Par	ServerHealth.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par	0
Input Par	ServerHealth.Data.EQID
	<EDC.ServerHealth.Data.EQID>
Input Par	ServerHealth.Ctl.TransactionNumber
	<EDC.ServerHealth.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans

aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerHealth.Msg.EN

EQ Health Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

7

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B ServerHealth.Ctl.Interface
<EDC.ServerHealth.Ctl.Interface>
3750

Server EQ Health from Transaction Manager Request Record Complete
ServerHealth.Ctl.ReqComplete
<EDC.ServerHealth.Ctl.ReqComplete>
L

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify

8

Server EQ Health from Transaction Manager Request Record Complete
ServerHealth.Ctl.ReqComplete
<EDC.ServerHealth.Ctl.ReqComplete>

Server EQ Health from Transaction Manager Request Complete Delay Timer

MOVE
Source 50
Dest ServerHealth.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerHealth.Ctl.ReqCompleteDelayTMR.PRE>
50

Server EQ Health from Transaction Manager Request Complete Delay Timer

TON
Timer ServerHealth.Ctl.ReqCompleteDelayTMR
<EDC.ServerHealth.Ctl.ReqCompleteDelayTMR>
Preset 50
Accum 62
EN
DN

Server EQ Health from Transaction Manager Request Complete Delay Timer
ServerHealth.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerHealth.Ctl.ReqCompleteDelayTMR.DN>

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>
U

EQ Health Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake Before Data - order of TM variables not enforeced in data update.)

9

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request Record Complete 1-shot
ServerHealth.Ctl.ReqCompleteONS
<EDC.ServerHealth.Ctl.ReqCompleteONS>
ONS

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4201
Input Par ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par 0
Input Par ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
Input Par ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerHealth.Update.MsgRxOK
L

Update Health State for Transactions
aoiEDC_StatusUpdateServerHealth
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerHealth ...
Status EDC_Status01.ServerHealth

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerHealth.Msg.DN

EQ Health Response from Server via Transaction Manager
JSR
Routine Name _885_HealthResponse

EQ Health Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

Dec 31, 2025 - reduced from 62 to 12 seconds to capture failure before next TM service restart

10

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request in Progress Timer

MOVE
Source 12000
Dest ServerHealth.Ctl.ReqInProgressTMR.PRE
<EDC.ServerHealth.Ctl.ReqInProgressTMR.PRE>
12000

Server EQ Health from Transaction Manager Request in Progress Timer

TON
Timer ServerHealth.Ctl.ReqInProgressTMR
<EDC.ServerHealth.Ctl.ReqInProgressTMR>
Preset 12000
Accum 0
EN
DN

Server EQ Health from Transaction Manager Request in Progress Timer
ServerHealth.Ctl.ReqInProgressTMR.DN
<EDC.ServerHealth.Ctl.ReqInProgressTMR.DN>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

EQ Health Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

11

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4204
Input Par: ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par: 0
Input Par: ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
Input Par: ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>

Server Interface Status State = Red- Msg Rx ER
EDC_Status01.ServerHealth.Update.MsgRxER
L

Update Health State for Transactions

aoiEDC_StatusUpdateServerHealth
aoiEDC_StatusUpdateS...: aoiEDC_StatusUpdateServerHealth
Status: EDC_Status01.ServerHealth

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest: TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source: ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
0
Dest: TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: TempPerfMsg
PerfTrans: EDC_Status01.ServerHealth.Msg.ER

Server EQ Health from Transaction Manager Request in Progress Timer
ServerHealth.Ctl.ReqInProgressTMR
<EDC.ServerHealth.Ctl.ReqInProgressTMR>
RES

EQ Health Request to Server via Transaction Manager

use the Server Health Message to monitor TM status

12

S:FS

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Server Health TM Transaction OK
ServerHealth_TMokTMR.TT

Server Health TM Transaction OK
ServerHealth_TMokTMR.DN

Server Health TM Transaction OK

MOVE
Source 60000
Dest ServerHealth_TMokTMR.PRE
60000

Server Health TM Transaction OK

TON
Timer ServerHealth_TMokTMR
Preset 60000
Accum 60014
EN
DN

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

HealthCompleteONS
ONS

ADD
Source A ServerHealth_CompleteCount
5602672
Source B 1
Dest ServerHealth_CompleteCount
5602672

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

HealthTimeoutONS
ONS

ADD
Source A ServerHealth_TimeoutCount
133
Source B 1
Dest ServerHealth_TimeoutCount
133

(End)

EQ Helath Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 881
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Data Values
(FLL -1 corrupts FLOAT values and causes TM Transaction Failure - overwrite with MOV -1)

2

Server EQ Health from Transaction Manager Status Transaction Data

FLL
Source -1
Dest ServerHealth.Data <EDC.ServerHealth.Data>
Length 1

Server EQ Health from Transaction Manager Last Process Time

FLL
Source 0
Dest ServerHealth.Data.LastProcTime <EDC.ServerHealth.Data.LastProcTime>
Length 1

Server EQ Health from Transaction Manager Last Process Time

MOVE
Source 1
Dest ServerHealth.Data.LastProcTime.DATA[0] <EDC.ServerHealth.Data.LastProcTime.DATA[0]> 'O'

EQ Helath Request to Server via Transaction Manager - Data Init

Ack Command if Good

3

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Request/Response EQID

MOVE
Source ServerHealth.DataLastResp.EQID
<EDC.ServerHealth.DataLastResp.EQID>
1010
Dest ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
1010

Server EQ Health from Transaction Manager Status

MOVE
Source ServerHealth.DataLastResp.Status
<EDC.ServerHealth.DataLastResp.Status>
0
Dest ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

NOP

Store Last Request

4

Server EQ Health from Transaction Manager Transaction Data Last Request

COP
Source ServerHealth.Data
<EDC.ServerHealth.Data>
Dest ServerHealth.DataLastReq
<EDC.ServerHealth.DataLastReq>
Length 1

(End)

EQ Health Response from Server via Transaction Manager

Health Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 885
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Response Good Data

2

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

U

Move Command to EQID
(Callout Table Index)

3

MOVE
Source ServerHealth.Data.EQID <EDC.ServerHealth.Data.EQID> 1010
Dest EQID 531

Equipment Index

MOD
Source A EQID 531
Source B 100
Dest EQi 31

Line Index

CPT
Dest LNi 5
Expression (EQID - EQi) / 100

EQ Health Response from Server via Transaction Manager

Check Response Command for Legal Callout Table Index (i.e. EQID)

4

Line Index

LT	
Source A	LNi
	5
Source B	0

Line Index

GT	
Source A	LNi
	5
Source B	CFG.CalloutLN.mi
	<EDC.CFG.CalloutLN.mi>
	28

Equipment Index

LT	
Source A	EQi
	31
Source B	0

Equipment Index

GT	
Source A	EQi
	31
Source B	CFG.CalloutEQ.mi
	<EDC.CFG.CalloutEQ.mi>
	99

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4213
Input Par	ServerHealth.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par	0
Input Par	ServerHealth.Data.EQID
	<EDC.ServerHealth.Data.EQID>
Input Par	ServerHealth.Ctl.TransactionNumber
	<EDC.ServerHealth.Ctl.TransactionNumber>

lbl_Exit

JMP

Process Response Command

Allowable Data Values enforced by Data Base - no Data checking other than array indexes

5

EQ Health Response from Server via Transaction Manager

JSR	
Routine Name	_886_HealthResponseEQID

Exit Routine

lbl_Exit

6

LBL

NOP

EQ Health Response from Server via Transaction Manager

Store Last Response and Last Good Response

7

Server EQ Health from Transaction Manager Transaction Data Last Response

COP
Source ServerHealth.Data <EDC.ServerHealth.Data>
Dest ServerHealth.DataLastResp <EDC.ServerHealth.DataLastResp>
Length 1

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData <EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Transaction Data Last Good Response

COP
Source ServerHealth.Data <EDC.ServerHealth.Data>
Dest ServerHealth.DataLastRespGood <EDC.ServerHealth.DataLastRespGood>
Length 1

(End)

EQ Health Response from Server via Transaction Manager

Process EQID Callout from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 886
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Line Index

LIMIT
Low Limit 0
Test LNi 5
High Limit CFG.CalloutLN.mi <EDC.CFG.CalloutLN.mi> 28

Equipment Index

LIMIT
Low Limit 0
Test EQi 31
High Limit CFG.CalloutEQ.mi <EDC.CFG.CalloutEQ.mi> 99

Server EQ Health from Transaction Manager Status

LIMIT
Low Limit 0
Test ServerHealth.Data.Status <EDC.ServerHealth.Data.Status> 0
High Limit 4

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData <EDC.ServerHealth.Ctl.RspGoodData>

L

EQ Health Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4214
Input Par	ServerHealth.Ctl.ReqInProgressTMR.ACC <EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par	0
Input Par	ServerHealth.Data.EQID <EDC.ServerHealth.Data.EQID>
Input Par	ServerHealth.Ctl.TransactionNumber <EDC.ServerHealth.Ctl.TransactionNumber>

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerHealth.Update.RxDataER
L

Update Health State for Transactions

aoiEDC_StatusUpdateServerHealth	
aoiEDC_StatusUpdateS...	aoiEDC_StatusUpdateServerHealth ...
Status	EDC_Status01.ServerHealth

lbl_Exit
JMP

EQ Health Response from Server via Transaction Manager

Store Data
Update EQID Callout for Health Alarm (19) and Health Fault (20)

States 1 & 2 - Low & High Alarms (Yellow)
States 3 & 4 - Low & High Faults (Red)

New Data if Last Process Time is NULL or if PreDef bits Change

4

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>
U

Server EQ Health from Transaction Manager Last Process Time

EQ
Source A ServerHealth.Data.LastProcTime.LEN
<EDC.ServerHealth.Data.LastProcTime.LEN>
19
Source B 0

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>
L

Gateway Config Download HMI Display Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable

Server EQ Health from Transaction Manager Status

EQ
Source A ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0
Source B 0

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>
L

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
U

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
U

Server EQ Health from Transaction Manager Status

LIMIT
Low Limit 1
Test ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0
High Limit 2

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
/

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>
L

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
L

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
U

Server EQ Health from Transaction Manager Status

LIMIT

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

EQ Health Response from Server via Transaction Manager

Low Limit 3
Test ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0
High Limit 4

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
/

L

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
U

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
L

Oct 4, 2021 - Disabled Health Alarm/Fault bits from EDC Server

5

Gateway Config
Download HMI Display
Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable
/

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
U

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
U

If new Data, Verify LNi, EQi Limits are OK - if not exit
(next Callout Message will update the status if this fails, but that could take minutes)

v3.5 Update Equipment Events with/without NewData to force update after configuration changes

6

Server EQ Health
from Transaction
Manager Response has
New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

Line Index
LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

Equipment Index
LIMIT
Low Limit 0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

LimitsOK

LimitsOK
/

lbl_Exit
JMP

EQ Health Response from Server via Transaction Manager

Force an update of the Overview & Zone Events
(same actions taken when a new Callout Message is received)

7

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.Update
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.UpdateOview
L

Equipment Data Collection Lines and Equipment Array

FLL
Source -1
Dest EDC.Line[LNi].Equip[EQi].EventCTL.UpdateZone
Length 1

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

Configuration Request to Server via Transaction Manager

Configuration Data Request to Server via Transaction Manager

- Logix Requests Data (-1 Command)
- Server Responds with Data (201 Command) or NoData (0 Command)

- if Data, Logix processes Data and Sends Ack (Echo Server Command)
- Server uses Ack to set Download Timestamp for Config Record
- Server Responds with next Data/NoData

- if NoData, Logix waits and then Requests Data (-1 Command)

0

NOP

Request Commands to TM/SQL

-1 = New Data Request (delay between requests)
201 = Ack Data Response and New Data Request (no delay between requests)

1

NOP

Response Commands from Server

-1 = Data Error
0 = No Data to Download
201 = Event Ignore Mask

2

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

3

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 890
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data

4

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Duplicate Error
ServerMask.Ctl.RspDupER
<EDC.ServerMask.Ctl.RspDupER>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerMask.Ctl.ReqFastTime
<EDC.ServerMask.Ctl.ReqFastTime>
0
Dest ServerMask.Ctl.ReqNextTMR.PRE
<EDC.ServerMask.Ctl.ReqNextTMR.PRE>
10000

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Duplicate Error
ServerMask.Ctl.RspDupER
<EDC.ServerMask.Ctl.RspDupER>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerMask.Ctl.ReqSlowTime
<EDC.ServerMask.Ctl.ReqSlowTime>
10000
Dest ServerMask.Ctl.ReqNextTMR.PRE
<EDC.ServerMask.Ctl.ReqNextTMR.PRE>
10000

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer

TON
Timer ServerMask.Ctl.ReqNextTMR
<EDC.ServerMask.Ctl.ReqNextTMR>
EN
Preset 10000
DN
Accum 7220

Configuration Request to Server via Transaction Manager

Request Next
Ready for next Transaction Manager Trigger

5

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer
ServerMask.Ctl.ReqNextTMR.DN
<EDC.ServerMask.Ctl.ReqNextTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>
L

Server Ignore Mask Configuration from Transaction Manager Request Record Complete
ServerMask.Ctl.ReqComplete
<EDC.ServerMask.Ctl.ReqComplete>
U

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

6

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager - Data Init

JSR
Routine Name _891_MaskRequestInit

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
/

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Configuration Request to Server via Transaction Manager

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

7

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372
Source B 1
Dest ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372
Source B 1

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372

Configuration Request to Server via Transaction Manager

8

Request Next
Log Transaction & Performance

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4400
Input Par	ServerMask.Ctl.ReqInProgressTMR.ACC <EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerMask.DataReq.Command <EDC.ServerMask.DataReq.Command>
Input Par	ServerMask.Ctl.CommandID <EDC.ServerMask.Ctl.CommandID>
Input Par	ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR

Dest	TempPerfMsg.TransTMR.ACC
	110

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4402
Input Par	ServerMask.Ctl.ReqInProgressTMR.ACC <EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerMask.DataReq.Command <EDC.ServerMask.DataReq.Command>
Input Par	ServerMask.Ctl.CommandID <EDC.ServerMask.Ctl.CommandID>
Input Par	ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans

aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerMask.Msg.EN

Configuration Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

9

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>
1372
Source B ServerMask.Ctl.Interface
<EDC.ServerMask.Ctl.Interface>
1372

Server Ignore Mask Configuration from Transaction Manager Request Record Complete
ServerMask.Ctl.ReqComplete
<EDC.ServerMask.Ctl.ReqComplete>
L

Configuration Request to Server via Transaction Manager

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
Also problems with asynchronous data from TM - handshake can update before data values.
2-Check method is fast and avoids errors for occasional slow update of data.
Wait 1 scan before 1st check to eliminate 99% of missing data issues.
2nd check after 2000msec if Check 1 fails.
(anything >2 seconds will generate a TM Transaction error based on Tag Retrieval rate)

10

Server Ignore Mask Configuration from Transaction Manager Request Record Complete
ServerMask.Ctl.ReqComplete
<EDC.ServerMask.Ctl.ReqComplete>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check Timer

MOVE
Source 10
Dest ServerMask.Ctl.ReqCompleteCheckTMR.PRE
<EDC.ServerMask.Ctl.ReqCompleteCheckTMR.PRE>
10

Server Ignore Mask Configuration from Transaction Manager Request Record First Check Timer

TON
Timer ServerMask.Ctl.ReqCompleteCheckTMR
<EDC.ServerMask.Ctl.ReqCompleteCheckTMR>
Preset 10
Accum 15
EN
DN

Server Ignore Mask Configuration from Transaction Manager Request Record First Check Timer
ServerMask.Ctl.ReqCompleteCheckTMR.DN
<EDC.ServerMask.Ctl.ReqCompleteCheckTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Complete Delay Timer

MOVE
Source 2000
Dest ServerMask.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerMask.Ctl.ReqCompleteDelayTMR.PRE>
2000

Server Ignore Mask Configuration from Transaction Manager Request Complete Delay Timer

TON
Timer ServerMask.Ctl.ReqCompleteDelayTMR
<EDC.ServerMask.Ctl.ReqCompleteDelayTMR>
Preset 2000
Accum 2009
EN
DN

Server Ignore Mask Configuration from Transaction Manager Request Complete Delay Timer
ServerMask.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerMask.Ctl.ReqCompleteDelayTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Configuration Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforced in data update.)

11

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check 1-shot
ServerMask.Ctl.ReqCompleteCheckONS
<EDC.ServerMask.Ctl.ReqCompleteCheckONS>
ONS

Server Ignore Mask Configuration from Transaction Manager Request Complete 1st Check Count

ADD
Source A ServerMask.Ctl.ReqComp1stChkCnt
<EDC.ServerMask.Ctl.ReqComp1stChkCnt>
551295
Source B 1
Dest ServerMask.Ctl.ReqComp1stChkCnt
<EDC.ServerMask.Ctl.ReqComp1stChkCnt>
551295

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete 1-shot
ServerMask.Ctl.ReqCompleteONS
<EDC.ServerMask.Ctl.ReqCompleteONS>
ONS

Server Ignore Mask Configuration from Transaction Manager Request Complete 2nd Check Count

ADD
Source A ServerMask.Ctl.ReqComp2ndChkCnt
<EDC.ServerMask.Ctl.ReqComp2ndChkCnt>
537946
Source B 1
Dest ServerMask.Ctl.ReqComp2ndChkCnt
<EDC.ServerMask.Ctl.ReqComp2ndChkCnt>
537946

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4401
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataReq.Command
<EDC.ServerMask.DataReq.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerMask.Update.MsgRxOK
L

Update Mask State for Transactions

aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...
Status EDC_Status01.ServerMask

Configuration Request to Server via Transaction Manager

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerMask.Msg.DN

Configuration Response from Server via Transaction Manager

JSR
Routine Name _895_MaskResponse

Request Complete Delay Reset Request

12

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>
U

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

13

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer

MOVE
Source 62000
Dest ServerMask.Ctl.ReqInProgressTMR.PRE
<EDC.ServerMask.Ctl.ReqInProgressTMR.PRE>
62000

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer

TON
Timer ServerMask.Ctl.ReqInProgressTMR
<EDC.ServerMask.Ctl.ReqInProgressTMR>
Preset 62000
Accum 0
EN
DN

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer
ServerMask.Ctl.ReqInProgressTMR.DN
<EDC.ServerMask.Ctl.ReqInProgressTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

14

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4404
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataReq.Command
<EDC.ServerMask.DataReq.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Server Interface Status State = Red- Msg Rx ER
EDC_Status01.ServerMask.Update.MsgRxER
L

Update Mask State for Transactions

aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...
Status EDC_Status01.ServerMask

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerMask.Msg.ER

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer
ServerMask.Ctl.ReqInProgressTMR
<EDC.ServerMask.Ctl.ReqInProgressTMR>
RES

(End)

Configuration Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Server to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 891
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

DN011 Separate Req & Rsp Tags
Create Data Init Packet

2

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

/

Server Ignore Mask Configuration from Transaction Manager - Data

FLL
Source -1
Dest ServerMaskInit
Length 1

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Clear Response to ensure new data on next DB Response
DHF017 Jan 30, 2018 -- Clear Server Request if no Ack for Good Message Response
Copy Response to Request for Good Data Ack to DB

3

Server Ignore Mask Configuration from Transaction Manager Transaction Response Data

COP
Source ServerMaskInit
Dest ServerMask.DataRsp <EDC.ServerMask.DataRsp>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData <EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0 <EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Transaction Request Data

COP
Source ServerMaskInit
Dest ServerMask.DataReq <EDC.ServerMask.DataReq>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData <EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0 <EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Transaction Request Data

COP
Source ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Dest ServerMask.DataReq <EDC.ServerMask.DataReq>
Length 1

Store Last Request

4

Server Ignore Mask Configuration from Transaction Manager Transaction Last Request Data

COP
Source ServerMask.DataReq <EDC.ServerMask.DataReq>
Dest ServerMask.DataReqLast <EDC.ServerMask.DataReqLast>
Length 1

Configuration Request to Server via Transaction Manager - Data Init

(End)

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 895
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Detect Duplicate Config Data Response
(this indicates the database is stuck on a bad config item)

2

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

NE
Source A ServerMask.DataRspLast.Command <EDC.ServerMask.DataRspLast.Command> 0
Source B 0

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

NE
Source A ServerMask.DataRspLast.Command <EDC.ServerMask.DataRspLast.Command> 0
Source B -1

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerMask.DataRspLast.Command <EDC.ServerMask.DataRspLast.Command> 0
Source B ServerMask.DataRsp.Command <EDC.ServerMask.DataRsp.Command> 0

Server Ignore Mask Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerMask.DataRspLast.ShopKey <EDC.ServerMask.DataRspLast.ShopKey> -1
Source B ServerMask.DataRsp.ShopKey <EDC.ServerMask.DataRsp.ShopKey> -1

Server Ignore Mask Configuration from Transaction Manager Line Key - for DB Ack only

EQ
Source A ServerMask.DataRspLast.LineKey <EDC.ServerMask.DataRspLast.LineKey> -1
Source B ServerMask.DataRsp.LineKey <EDC.ServerMask.DataRsp.LineKey> -1

Server Ignore Mask Configuration from Transaction Manager Equip Key - for DB Ack only

EQ
Source A ServerMask.DataRspLast.EqKey <EDC.ServerMask.DataRspLast.EqKey> -1
Source B ServerMask.DataRsp.EqKey <EDC.ServerMask.DataRsp.EqKey> -1

Server Ignore Mask Configuration from Transaction Manager LineID (00-99)

EQ
Source A ServerMask.DataRspLast.LineID <EDC.ServerMask.DataRspLast.LineID> -1
Source B ServerMask.DataRsp.LineID <EDC.ServerMask.DataRsp.LineID> -1

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerMask.DataRspLast.EqID <EDC.ServerMask.DataRspLast.EqID> -1
Source B ServerMask.DataRsp.EqID <EDC.ServerMask.DataRsp.EqID> -1

Server Ignore Mask Configuration from Transaction Manager Event Packet ID (0-7)

EQ
Source A ServerMask.DataRspLast.PacketID <EDC.ServerMask.DataRspLast.PacketID> -1
Source B ServerMask.DataRsp.PacketID <EDC.ServerMask.DataRsp.PacketID> -1

Server Ignore Mask Configuration from Transaction Manager Response Duplicate
ServerMask.Ctl.RspDup <EDC.ServerMask.Ctl.RspDup>

Configuration Response from Server via Transaction Manager

Duplicate Message Count and Detection
(this indicates the database is stuck on a bad config item)

3

Server Ignore Mask Configuration from Transaction Manager Response Duplicate ServerMask.Ctl.RspDup <EDC.ServerMask.Ctl.RspDup>

ADD
Source A MaskRspDuplicateCount 0
Source B 1
Dest MaskRspDuplicateCount 0

GT
Source A MaskRspDuplicateCount 0
Source B MaskRspDuplicateMax 50

MOVE
Source MaskRspDuplicateCount 0
Dest MaskRspDuplicateMax 50

GT
Source A MaskRspDuplicateCount 0
Source B 5

Server Ignore Mask Configuration from Transaction Manager Response Duplicate ServerMask.Ctl.RspDup <EDC.ServerMask.Ctl.RspDup>

CLR
Dest MaskRspDuplicateCount 0

Server Ignore Mask Configuration from Transaction Manager Response Duplicate Error ServerMask.Ctl.RspDupER <EDC.ServerMask.Ctl.RspDupER>

Configuration Response from Server via Transaction Manager

Move Response to working tag and Initialize Response Good Data
DN011 Separate Req & Rsp Tags

Moving DataRsp to DataLastRsp prevents SQL/TM from changing data while Logix is processing.

4

Server Ignore Mask Configuration from Transaction Manager Transaction Last Response Data

COP
Source ServerMask.DataRsp <EDC.ServerMask.DataRsp>
Dest ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData <EDC.ServerMask.Ctl.RspGoodData>

U

Configuration Response from Server via Transaction Manager

Check Response Command for Bad Data
Commands 201

5

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

LT
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 201

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

GT
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 201

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4412
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerMask.Msg.KA

lbl_RspComp
JMP

If Response Command received, process Response

6

MaskRspCmdRcvd

Configuration Response from Server via Transaction Manager

Server has no new Configuration data to download

7

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 0

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
L

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

ADD
Source A MaskRspCmd[0]
0
Source B 1
Dest MaskRspCmd[0]
0

Process Response Command

8

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 201

Configuration Response from Server via Transaction Manager

JSR
Routine Name _896_MaskResponseIgnMask

ADD
Source A MaskRspCmd[1]
13555
Source B 1
Dest MaskRspCmd[1]
13555

Command Response Complete

9

lbl_RspComp
LBL

NOP

Configuration Response from Server via Transaction Manager

Store Last Good, Bad or Incomplete Response w/ Diagnostic Counts
Incomplete - Command Num received with incomplete data packet on on 1st Check
Bad - Incomplete data packet on 2nd Check

10

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Transaction Last Good Response Data

COP
Source ServerMask.DataRspLast
<EDC.ServerMask.DataRspLast>
Dest ServerMask.DataRspLastGood
<EDC.ServerMask.DataRspLastGood>
Length 1

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Response Good on 1st Check

ADD
Source A ServerMask.Ctl.RspGood1stChkCnt
<EDC.ServerMask.Ctl.RspGood1stChkCnt>
13349
Source B 1
Dest ServerMask.Ctl.RspGood1stChkCnt
<EDC.ServerMask.Ctl.RspGood1stChkCnt>
13349

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Response Good on 2nd Check

ADD
Source A ServerMask.Ctl.RspGood2ndChkCnt
<EDC.ServerMask.Ctl.RspGood2ndChkCnt>
0
Source B 1
Dest ServerMask.Ctl.RspGood2ndChkCnt
<EDC.ServerMask.Ctl.RspGood2ndChkCnt>
0

Server Ignore Mask Configuration from Transaction Manager Response has

Server Ignore Mask Configuration from Transaction Manager Request

Server Ignore Mask Configuration from Transaction Manager Request Record Complete

Configuration Response from Server via Transaction Manager

Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

MaskRspCmdRcvd

Server Ignore Mask Configuration from Transaction Manager Response None on 1st Check

ADD
Source A ServerMask.Ctl.RspNone1stChkCnt
<EDC.ServerMask.Ctl.RspNone1stChkCnt>
47
Source B 1
Dest ServerMask.Ctl.RspNone1stChkCnt
<EDC.ServerMask.Ctl.RspNone1stChkCnt>
47

MaskRspCmdRcvd

Server Ignore Mask Configuration from Transaction Manager Transaction Last Incomplete Response Data

COP
Source ServerMask.DataRspLast
<EDC.ServerMask.DataRspLast>
Dest ServerMask.DataRspLastInc
<EDC.ServerMask.DataRspLastInc>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response Incomplete on 1st Check

ADD
Source A ServerMask.Ctl.RspInc1stChkCnt
<EDC.ServerMask.Ctl.RspInc1stChkCnt>
537899
Source B 1
Dest ServerMask.Ctl.RspInc1stChkCnt
<EDC.ServerMask.Ctl.RspInc1stChkCnt>
537899

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Transaction Last Bad Response Data

COP
Source ServerMask.DataRspLast
<EDC.ServerMask.DataRspLast>
Dest ServerMask.DataRspLastBad
<EDC.ServerMask.DataRspLastBad>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response Bad on 2nd Check

ADD
Source A ServerMask.Ctl.RspBad2ndChkCnt
<EDC.ServerMask.Ctl.RspBad2ndChkCnt>
537946
Source B 1
Dest ServerMask.Ctl.RspBad2ndChkCnt
<EDC.ServerMask.Ctl.RspBad2ndChkCnt>
537946

Configuration Response from Server via Transaction Manager

Download ALL configuration is complete

11

Server Ignore Mask Configuration from Transaction Manager Response Download All Started
ServerMask.Ctl.DownloadAllStarted
<EDC.ServerMask.Ctl.DownloadAllStarted>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Response Download All Started
ServerMask.Ctl.DownloadAllStarted
<EDC.ServerMask.Ctl.DownloadAllStarted>
U

Server Ignore Mask Configuration from Transaction Manager Response Download All Complete
ServerMask.Ctl.DownloadAllComplete
<EDC.ServerMask.Ctl.DownloadAllComplete>
L

Server Ignore Mask Configuration from Transaction Manager Response Download All Complete 1-shot
ServerMask.Ctl.DownloadAllCompleteONS
<EDC.ServerMask.Ctl.DownloadAllCompleteONS>

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 201 - Event Ignore Mask
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 896
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

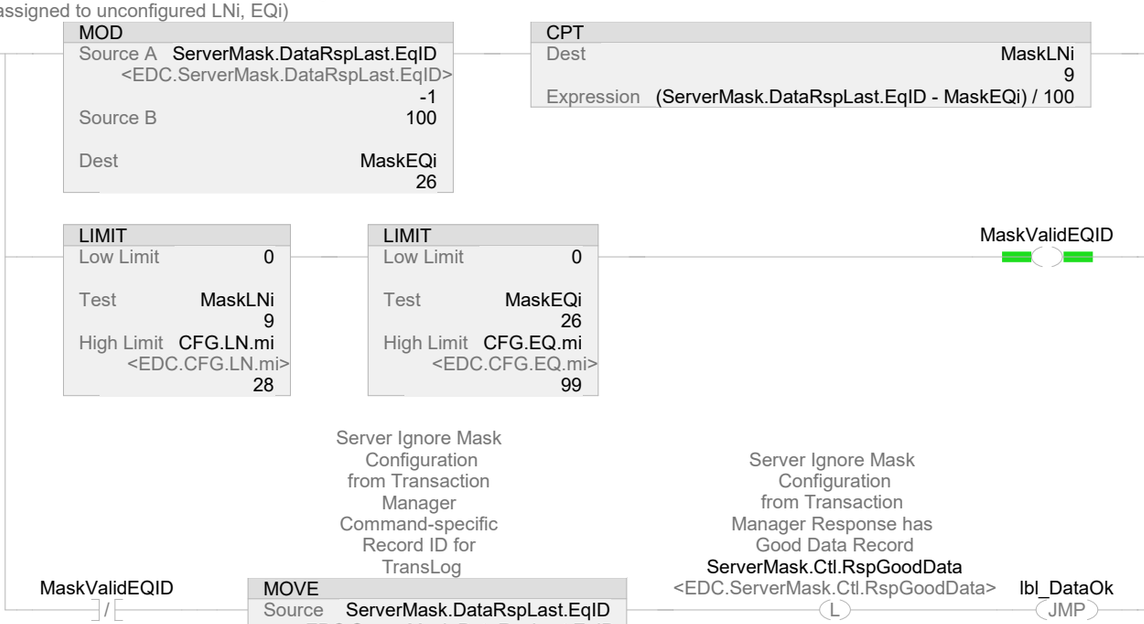
ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Convert EQID to LNi and EQi

EDC2 - ack any EQID outside configured LNi, EQi without storing config data
(unused EQID can be assigned to unconfigured LNi, EQi)

2



Configuration Response from Server via Transaction Manager

Validate Packet Index

v3.5 - ack any Packet outside array limits without storing config data
(Events be created outside limits)

3

Server Ignore Mask Configuration from Transaction Manager Event Packet ID (0-7)

LIMIT
Low Limit 0
Test ServerMask.DataRspLast.PacketID
<EDC.ServerMask.DataRspLast.PacketID>
-1
High Limit EDC_Event.CFG.PKT.mi
7

MaskValidPacketID

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MaskValidPacketID
/

MOVE
Source ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
L

lbl_DataOk
JMP

Get Eq-Event Lookup Index

4

MOVE
Source EDC.Line[MaskLNi].Equip[MaskEQi].EventCTL.EvLKi
164
Dest MaskEvLKi
164

LIMIT
Low Limit 0
Test MaskEvLKi
164
High Limit EDC_Event.CFG.EQ.mi
2999

MaskGoodLImits

MaskGoodLImits
/

lbl_SkipDataCheck
JMP

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

Mask values can be 0, -1, etc. - no way to validate and ensure full packet received
FooterID is last element and must match EqID

Verify EQID has a valid Event Lookup index

5

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

NE
Source A ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Source B -1

Server Ignore Mask Configuration from Transaction Manager Event Packet ID (0-7)

NE
Source A ServerMask.DataRspLast.PacketID
<EDC.ServerMask.DataRspLast.PacketID>
-1
Source B -1

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Source B ServerMask.DataRspLast.FooterID
<EDC.ServerMask.DataRspLast.FooterID>
-1

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Source B EDC_Event.CTL.Equip[MaskEvLKi].EQID
926

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
L

Skip Data Check if Indexes invalid

6

lbl_SkipDataCheck
LBL

NOP

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

7

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4413
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Server Interface Status State = Red-Ignore Mask Rx Data ER
EDC_Status01.ServerMask.Update.IgMaskRxDataER
L

Update Mask State for Transactions

aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...
Status EDC_Status01.ServerMask

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Store the Ignore Mask for the EQ-Event Packet
Set the Mask Send bit for updating EqDriver
v3.02 Data Refresh Request Command

8

Server Ignore Mask Configuration from Transaction Manager Packet Data Mask

SIZE
Source ServerMask.DataRspLast.Data[0]
<EDC.ServerMask.DataRspLast.Data[0]>
0
Dim. To Vary 0
Size DataLEN
8

DT Type = Ignore Event Mask Equipment Mask Equip Array

COP
Source ServerMask.DataRspLast.Data[0]
<EDC.ServerMask.DataRspLast.Data[0]>
Dest EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[0]
Length DataLEN

Event Controls Equipment Fragment Statistics Array
EDC_Event.CTL.Equip[MaskEvLKi].IgnoreMaskSend.[ServerMask.DataRspLast.PacketID]
L

Event Controls Equipment Fragment Statistics Array
EDC_Event.CTL.Equip[MaskEvLKi].DataRefreshReq.[ServerMask.DataRspLast.PacketID]
L

Initialize Loop

9

CLR
Dest MSKi
8

Configuration Response from Server via Transaction Manager

update Fault, Alarm, Alert Masks to clear Ignore items
(when set to Ignore, the Event will no longer be downloaded to update Fault, Alarm, Alert)

10

lbl_MaskLoop
LBL

DT Type = Fault
Event Mask Equipment
Mask Equip Array

AND
Source A EDC_EventMaskFault.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Source B EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Dest EDC_EventMaskFault.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??

DT Type = Alarm
Event Mask Equipment
Mask Equip Array

AND
Source A EDC_EventMaskAlarm.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Source B EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Dest EDC_EventMaskAlarm.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??

DT Type = Alert
Event Mask Equipment
Mask Equip Array

AND
Source A EDC_EventMaskAlert.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Source B EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Dest EDC_EventMaskAlert.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??

ADD
Source A MSKi 8
Source B 1
Dest MSKi 8

LT
Source A MSKi 8
Source B DataLEN 8

lbl_MaskLoop
JMP

Configuration Response from Server via Transaction Manager

Data OK

Server Interface
Status State = Green
- Ignore Mask Rx
Data OK

Update Mask State
for Transactions

11

lbl_DataOk
LBL

EDC_Status01.ServerMask.Update.IgMaskDataOK
L

aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...
Status EDC_Status01.ServerMask

Exit Routine

12

lbl_Exit
LBL

NOP

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code <EDC.Trans.Code>
Input Par Trans.Time <EDC.Trans.Time>
Input Par Trans.LNi <EDC.Trans.LNi>
Input Par Trans.EQi <EDC.Trans.EQi>
Input Par Trans.Num <EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 447423535
Source B 1
Dest Prog.RoutineScan[901] 447423535

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

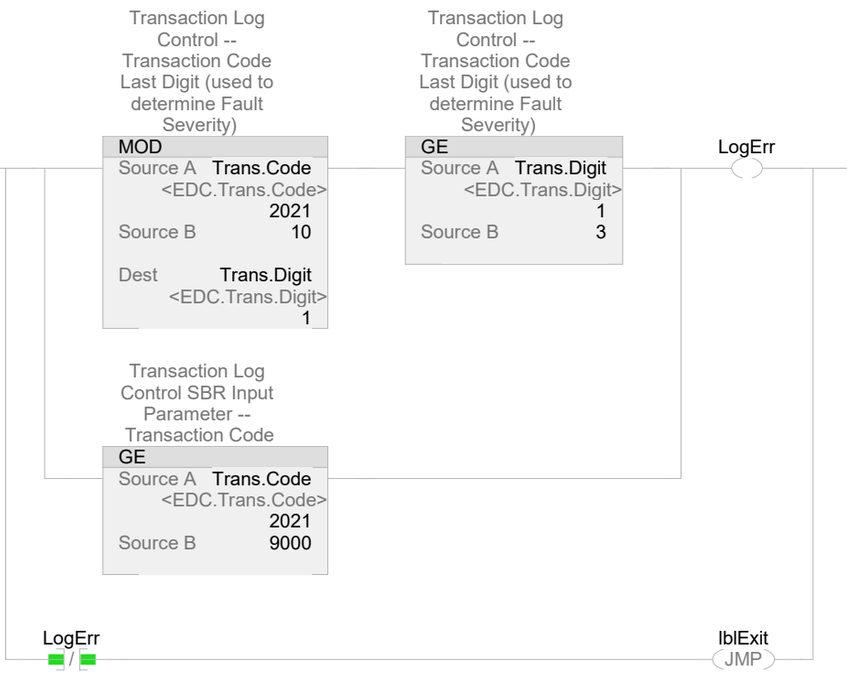
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

eAssy_EDC_Gway
MainTask
ServerInterface
_801_Main
Ladder Diagram 1
_802_Fault
Ladder Diagram 6
_810_Reset
Ladder Diagram 12
_811_Config
Ladder Diagram 17
_818_InitOview
Ladder Diagram 22
_819_InitZone
Ladder Diagram 25
_820_CfgRequest
Ladder Diagram 28
_821_CfgRequestInit
Ladder Diagram 40
_830_CfgResponse
Ladder Diagram 44
_831_CfgResponseLine
Ladder Diagram 56
_832_CfgResponseOfflineGroup
Ladder Diagram 62
_833_CfgResponseOviewDisplay
Ladder Diagram 65
_834_CfgResponseZoneDisplay
Ladder Diagram 68
_835_CfgResponseEquipment
Ladder Diagram 72
_836_CfgResponseEquipmentEventTxt
Ladder Diagram 85
_837_CfgResponseEquipmentEventCtlEq
Ladder Diagram 115
_837_CfgResponseEquipmentEventCtlPre
Ladder Diagram 135
_838_CfgResponseShop
Ladder Diagram 139
_839_CfgResponseCarrierScanner
Ladder Diagram 142
_840_CfgResponseSchedule
Ladder Diagram 149
_841_CfgResponseGateway
Ladder Diagram 154
_845_PdmRequest
Ladder Diagram 161
_846_PdmRequestInit

Ladder Diagram 172
_850_PdmResponse
Ladder Diagram 175
_851_PdmResponseTool
Ladder Diagram 184
_852_PdmResponseTrend
Ladder Diagram 205
_853_PdmResponseNetwork
Ladder Diagram 211
_858_PdmDownloadInit
Ladder Diagram 214
_859_PdmDownloadComplete
Ladder Diagram 216
_860_StatRequest
Ladder Diagram 219
_861_StatRequestInit
Ladder Diagram 229
_865_StatResponse
Ladder Diagram 233
_866_StatResponseRow
Ladder Diagram 236
_870_StatCisErrCallout
Ladder Diagram 242
_871_StatHealthErrCallout
Ladder Diagram 244
_880_HealthRequest
Ladder Diagram 246
_881_HealthRequestInit
Ladder Diagram 255
_885_HealthResponse
Ladder Diagram 257
_886_HealthResponseEQID
Ladder Diagram 260
_890_MaskRequest
Ladder Diagram 265
_891_MaskRequestInit
Ladder Diagram 276
_895_MaskResponse
Ladder Diagram 279
_896_MaskResponseIgnMask
Ladder Diagram 288
_901_TransLogSBR
Ladder Diagram 296